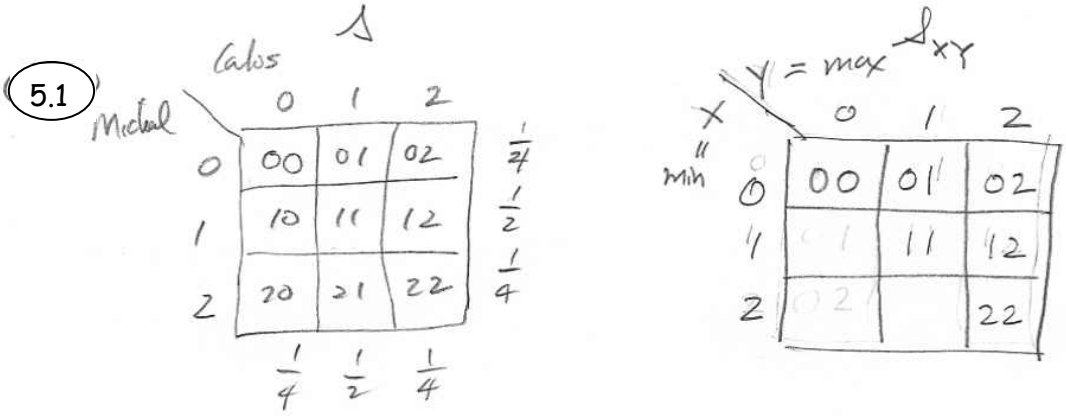
Chapter 5: Pairs of Random Variables

5.1 Two Random Variables

5.1



(b) $P[X=0, Y=0] = P[\{00\}] = \frac{1}{16}$

$P[X=0, Y=1] = P[\{01, 10\}] = \frac{1}{4}\frac{1}{2} + \frac{1}{4}\frac{1}{2} = \frac{1}{4}$

$P[X=0, Y=2] = P[\{02, 20\}] = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$

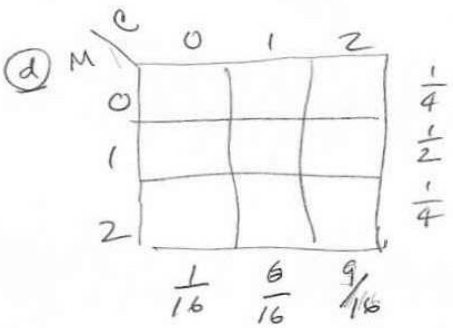
$P[X=1, Y=1] = P[\{11\}] = \frac{1}{4}$

$P[X=1, Y=2] = P[\{12, 21\}] = \frac{1}{2}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{4}$

$P[X=2, Y=2] = \frac{1}{16}$

(c) $P[X=Y] = P[(X,Y) \in \{00, 11, 22\}] = \frac{1}{16} + \frac{1}{4} + \frac{1}{16} = \frac{3}{8}$

(d)



$P[X=0, Y=0] = \frac{1}{64}$

$P[X=0, Y=1] = \frac{6}{64} + \frac{2}{64} = \frac{8}{64}$

$P[X=0, Y=2] = \frac{9}{64} + \frac{1}{64} = \frac{10}{64}$

$P[X=1, Y=1] = \frac{12}{64}$

$P[X=1, Y=2] = \frac{18}{64} + \frac{6}{64} = \frac{24}{64}$

$P[X=2, Y=2] = \frac{9}{64}$

$P[X=Y] = \frac{1}{64} + \frac{12}{64} + \frac{9}{64} = \frac{22}{64}$

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5.2

(a) (i) S

M \ N	0	1	2	
0	00	01	02	$\frac{1}{4}$
1	10	11	12	$\frac{1}{2}$
2	20	21	22	$\frac{1}{4}$
	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	

(ii) S_{XY}

X \ Y	0	1	2	3	4
-2			02		
-1		01		12	
0	00		11		22
1		10		21	
2			20		

fair coin

for biased coin:

(b) $P[X=-2, Y=2] = P[\{02\}] = \frac{1}{16}$ 9/64

$P[X=-1, Y=1] = P[\{01\}] = \frac{1}{8}$ 6/64

$P[X=-1, Y=3] = P[\{12\}] = \frac{1}{8}$ 18/64

$P[X=0, Y=0] = P[\{00\}] = \frac{1}{16}$ 1/64

$P[X=0, Y=2] = P[\{11\}] = \frac{1}{4}$ 12/64

$P[X=0, Y=4] = P[\{22\}] = \frac{1}{16}$ 9/64

$P[X=1, Y=1] = P[\{10\}] = \frac{1}{8}$ 2/64

$P[X=1, Y=3] = P[\{21\}] = \frac{1}{8}$ 6/64

$P[X=2, Y=2] = P[\{20\}] = \frac{1}{16}$ 1/64

(c) $P[X+Y=1] = 0$

$P[X+Y=2] = P[(X,Y) \in \{(-1,3), (0,2), (1,1)\}] = \frac{1}{8} + \frac{1}{4} + \frac{1}{8}$

$= \frac{1}{2}$

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5.3

(a) Sample Space: a set of outcomes where each outcome is a pair $\underline{y} = (y_1, y_2)$ where y_1 is the input and y_2 is the output.

$$S_{XY} = \{(-1,-1), (-1,0), (-1,1), (1,-1), (1,0), (1,1)\}$$

(b) $P[X=1, Y=-1]$
$= P[Y=-1 \mid X=1]P[X=1]$
$= \frac{3}{4}p$

$P[X=-1, Y=-1]$
$= \frac{1}{4}(1-p-p_e)$

$P[X=1, Y=0]$
$= P[Y=0 \mid X=1]P[X=1]$
$= \frac{3}{4}p_e$

$P[X=-1, Y=0]$
$= \frac{1}{4}p_e$

$P[X=1, Y=1]$
$= P[Y=1 \mid X=1]P[X=1]$
$= \frac{3}{4}(1-p-p_e)$

$P[X=-1, Y=1]$
$= \frac{1}{4}p$

(c) $P[X \neq Y]$
$= \frac{1}{4}p_e + \frac{1}{4}p + \frac{3}{4}p_e + \frac{3}{4}p$
$= p_e + p$

$P[Y=0]$
$= \frac{3}{4}p_e + \frac{1}{4}p_e$
$= p_e$

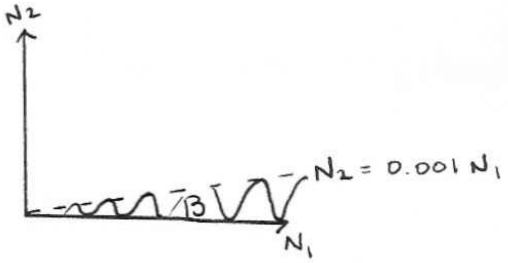
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5.4

(a) Let $\mathfrak{Z}_n$ be the number of arrivals specified by $\mathfrak{Z}$

$$0 \leq N_1 \leq \mathfrak{Z}_n \quad \text{and} \quad N_2 = \mathfrak{Z}_n - N_1$$

(b) $B = \{ 0.001\, N_1 > N_2 \}$



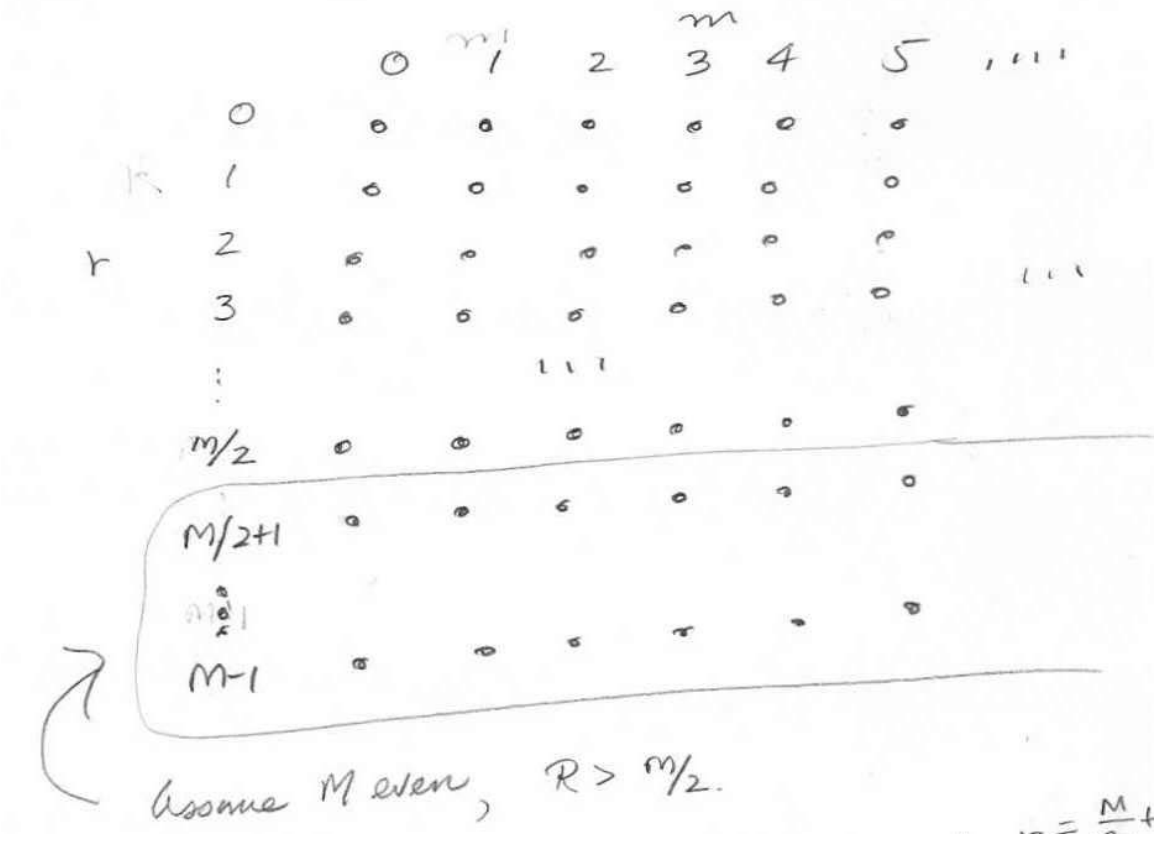
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5.5

$\xi \in \{1, 2, \ldots\}$

(a) $Q = \lfloor \frac{\xi}{M} \rfloor$ $\quad Q \in \{0, 1, 2, \ldots\}$

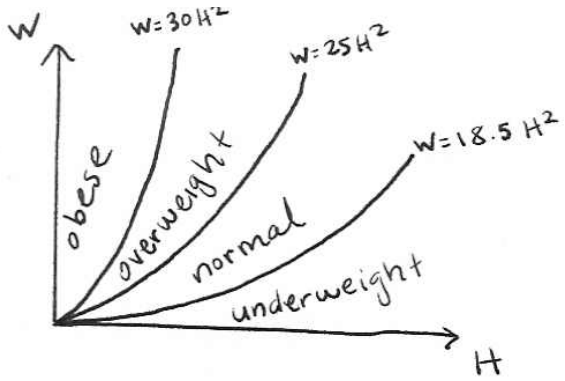
$R = \xi - \lfloor \frac{\xi}{M} \rfloor M$ $\quad R \in \{0, 1, \ldots, M-1\}$



"Last Packet More than Half Full" $= \{(q, r): q = 0, 1, \ldots \text{ and } r = \frac{M}{2}+1, \ldots, M-1\}$

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5.6

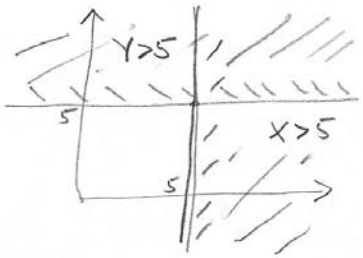


$W/H^2 < c$

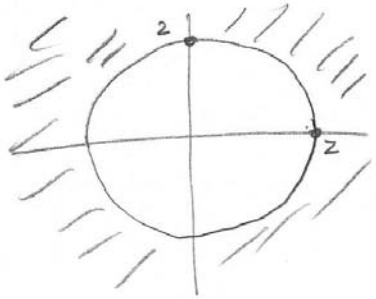
$W < cH^2$

5.7

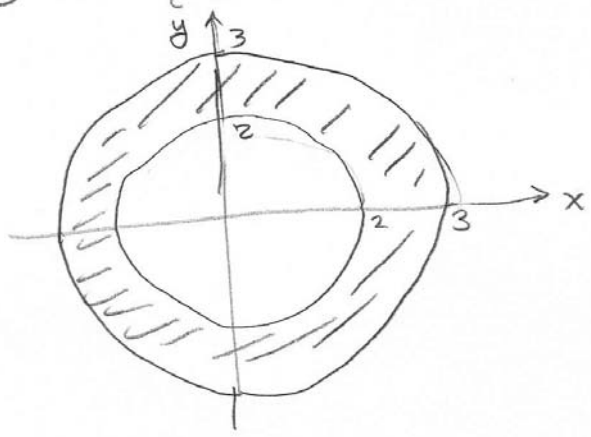
(a) $\{\max(X,Y) > 5\} = \{X > 5\} \cup \{Y > 5\}$



(b) $\{X^2 + Y^2 > 4\}$



(c) $\{4 < X^2 + Y^2 < 9\}$

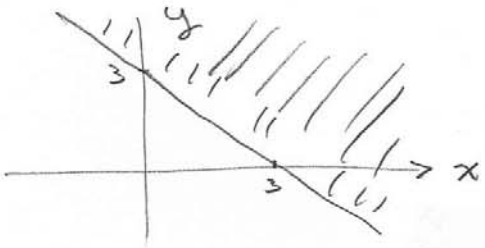


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5.8

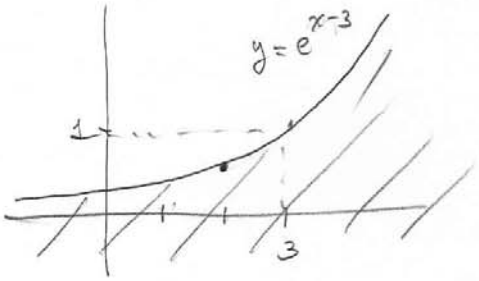
(a) $\{X+Y>3\} = \{Y>3-X\}$

Not product form



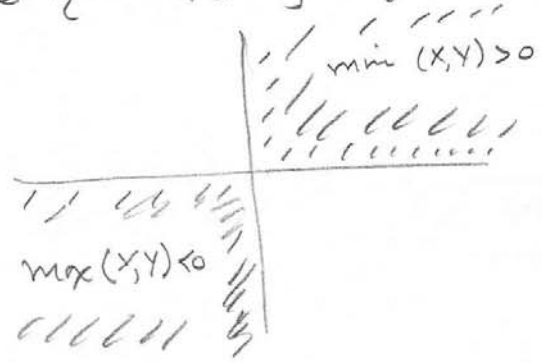
(b) $\{e^X > Ye^3\}$

$= \{Y < e^{X-3}\}$



Not product form

(c) $\{\min(X,Y)>0\} \cup \{\max(X,Y)<0\}$

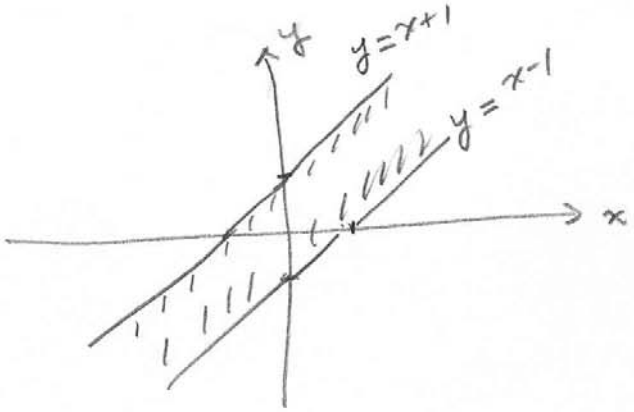


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but union of product form

(d) $\{|X-Y| \geq 1\} = \{-1 \leq X-Y \leq 1\}$

$= \{Y-1 \leq X \leq Y+1\}$



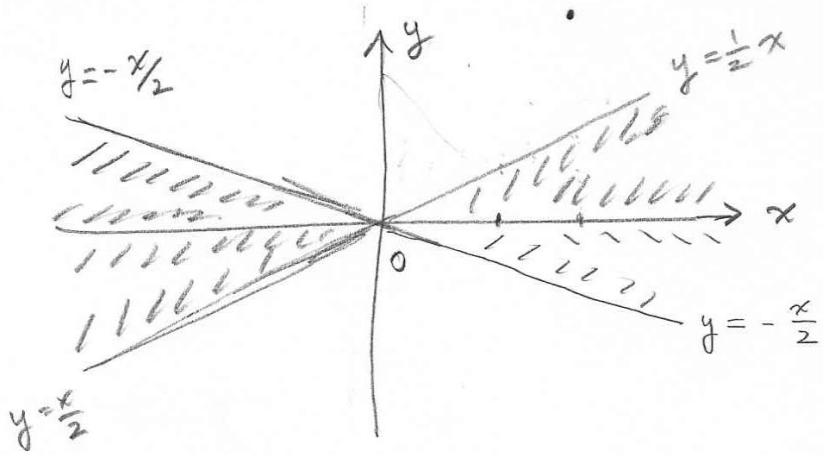
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(e) $\{|x/y| > 2\}$

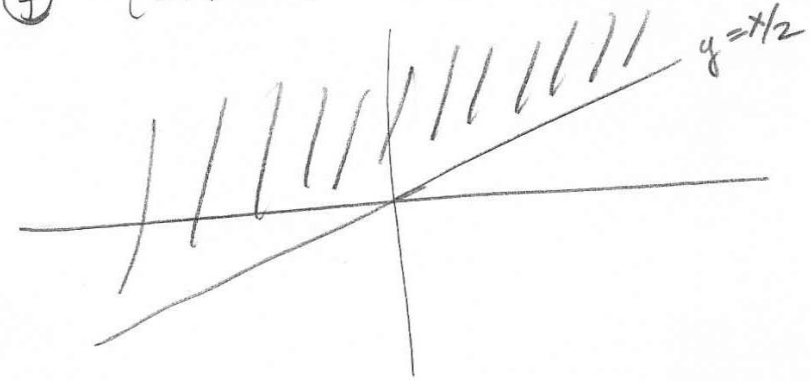
$-\frac{x}{y} > 2 \Leftrightarrow x/2 > y$ | $x/y > 2 \Leftrightarrow \frac{1}{2}x > y$

$x/y > 2 \Leftrightarrow \frac{x}{2} < y$ | $-x/y > 2 \Leftrightarrow -\frac{x}{2} < y$



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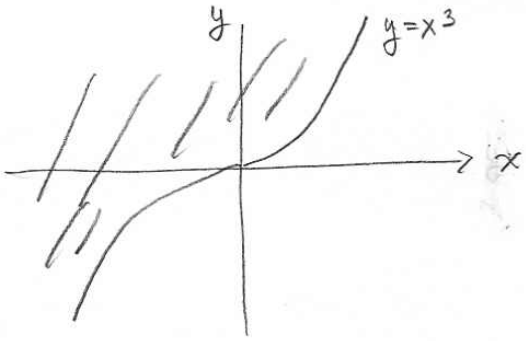
(f) $\{xy < 2\} = \{x/2 < y\}$



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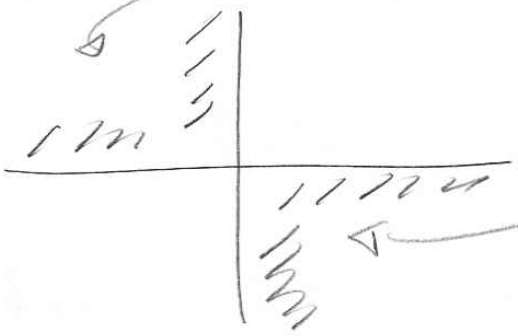
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(g) $\{X^3 > Y\}$



Not product form

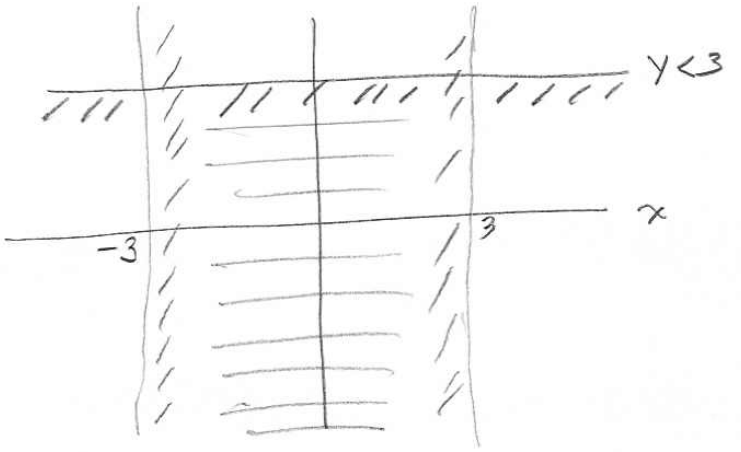
(h) $\{XY<0\} = \{X<0, Y>0\} \cup \{X>0, Y<0\}$



Not product form but union of product form

(i) $\max\{(|X|, Y) < 3\} = \max\{(-X, X, Y) < 3\}$

$= \{-X<3, X<3, Y<3\}$



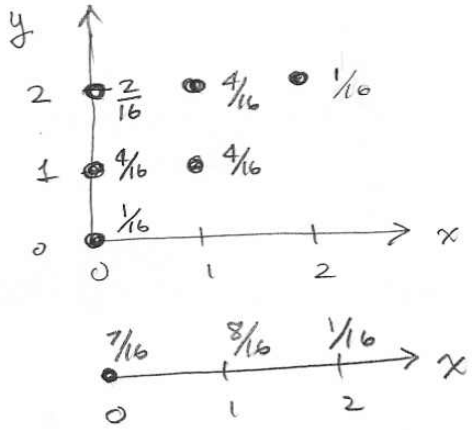
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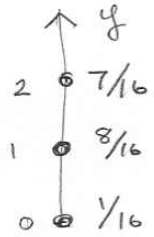
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5.2 Pairs of Discrete Random Variables

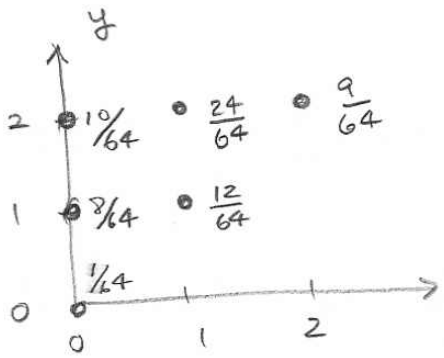
5.9

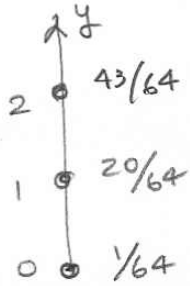
(a)



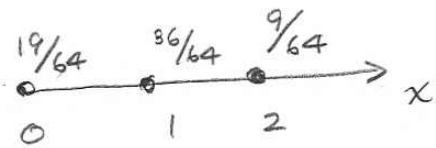


(b)





shift towards higher values of y



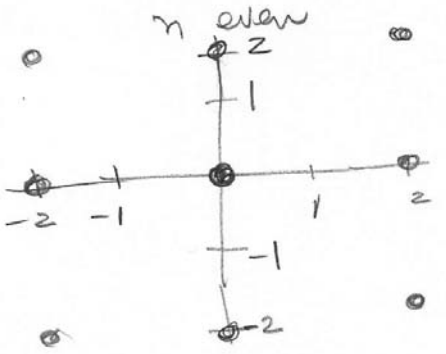
shift to higher values of x.

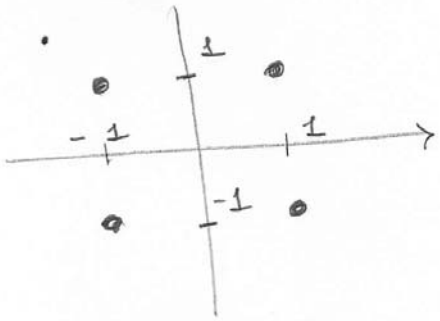
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© If n is even then X and Y can only take on even values

If n is odd then X and Y can only take odd values

$P[(X,Y)$ within $\sqrt{2}$ of origin$]$





$$P[X=0, Y=0] = P[N_1 = \tfrac{n}{2}, N_2 = \tfrac{n}{2}]$$

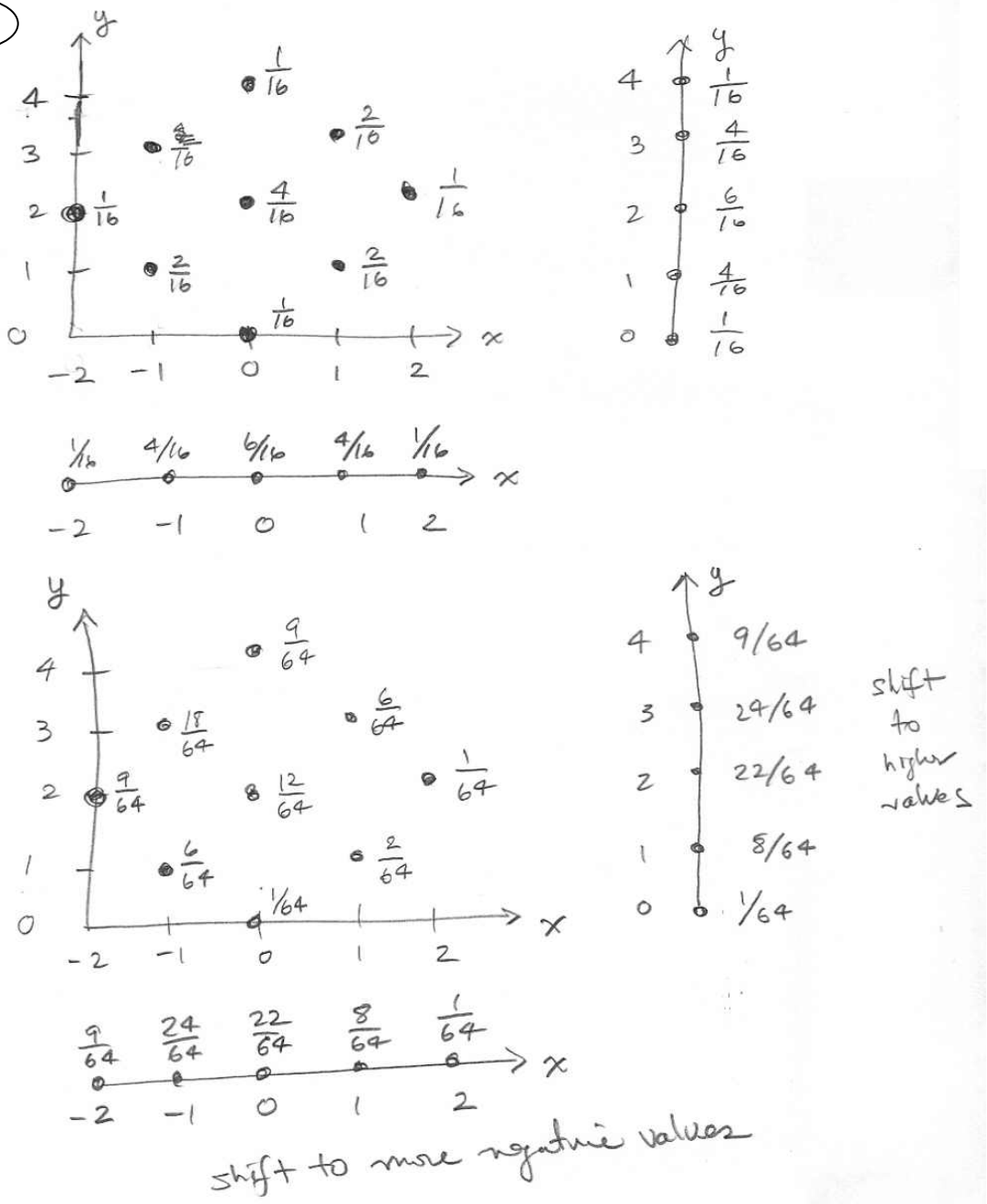
$$= \left(\binom{n}{\frac{n}{2}} p^{\frac{n}{2}} (1-p)^{\frac{n}{2}} \right)^2$$

$$\left(P[X=-1] + P[X=1]\right) \times \left(P[Y=-1] + P[Y=1]\right)$$

$$= \left[\binom{n}{l} p^{l} (1-p)^{n-l} + \binom{n}{l+1} p^{l+1} (1-p)^{n-l-1} \right]^2$$

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5.10



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5.11

(i)

X \ Y	-1	0	1	
-1	1/6	1/6	0	1/3
0	0	0	1/3	1/3
1	1/6	1/6	0	1/3
	1/3	1/3	1/3	

$P[X=i]=\frac{1}{3}$ $i\in\{-1,0,1\}$

$P[Y=i]=\frac{1}{3}$ $i\in\{-1,0,1\}$

$P[X>0]=\frac{1}{3}$

$P[X\geq Y]=\frac{1}{2}$

$P[X=-Y]=1/6$

(ii)

X \ Y	-1	0	1	
-1	1/9	1/9	1/9	1/3
0	1/9	1/9	1/9	1/3
1	1/9	1/9	1/9	1/3
	1/3	1/3	1/3	

$P[X=i]=\frac{1}{3}$ $i\in\{-1,0,1\}$

$P[Y=i]=\frac{1}{3}$ $i\in\{-1,0,1\}$

$P[X>0]=\frac{1}{3}$

$P[X\geq Y]=\frac{2}{3}$

$P[X=-Y]=\frac{1}{3}$

(iii)

X \ Y	-1	0	1	
-1	1/3	0	0	1/3
0	0	1/3	0	1/3
1	0	0	1/3	1/3
	1/3	1/3	1/3	

$P[X=i]=\frac{1}{3}$ $i\in\{-1,0,1\}$

$P[Y=i]=\frac{1}{3}$ $i\in\{-1,0,1\}$

$P[X>0]=\frac{1}{3}$

$P[X\geq Y]=1$

$P[X=-Y]=1/3$

Three different joint pmf's have the same marginal pmf's.

Events that involve joint behavior have different probabilities.

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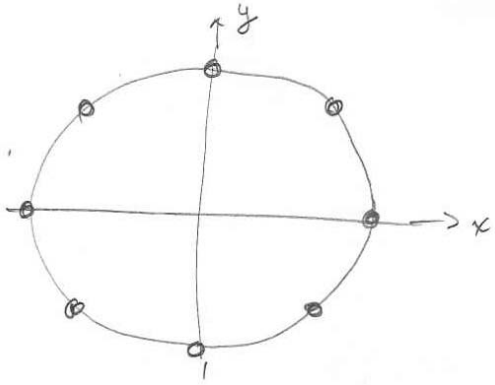
5.12

(a) Mapping S to S_{XY}

Θ	0	1	2	3	4	5	6	7
X,Y	$(r,0)$	$(\frac{r}{\sqrt{2}},\frac{r}{\sqrt{2}})$	$(0,r)$	$(\frac{-r}{\sqrt{2}},\frac{r}{\sqrt{2}})$	$(-r,0)$	$(\frac{-r}{\sqrt{2}},\frac{-r}{\sqrt{2}})$	$(0,-r)$	$(\frac{r}{\sqrt{2}},\frac{-r}{\sqrt{2}})$

(b) Joint PMF

X \ Y	$-r$	$\frac{-r}{\sqrt{2}}$	0	$\frac{r}{\sqrt{2}}$	r
$-r$	0	0	1/8	0	0
$\frac{-r}{\sqrt{2}}$	0	1/8	0	1/8	0
0	1/8	0	0	0	1/8
$\frac{r}{\sqrt{2}}$	0	1/8	0	1/8	0
r	0	0	1/8	0	0



(c) $p_X(x) = \sum_k p_{XY}(x,k)$

$p_X(r) = 1/8$ $\quad p_Y(r) = 1/8$

$p_X(\frac{r}{\sqrt{2}}) = 1/4$ $\quad p_Y(\frac{r}{\sqrt{2}}) = 1/4$

$p_X(0) = 1/4$ $\quad p_Y(0) = 1/4$

$p_X(\frac{-r}{\sqrt{2}}) = 1/4$ $\quad p_Y(\frac{-r}{\sqrt{2}}) = 1/4$

$p_X(-r) = 1/8$ $\quad p_Y(-r) = 1/8$

(d) $P[A] = p_X(0) = 1/4$

$P[B] = 1 - p_Y(1) = 7/8$

$P[C] = p_{XY}(\frac{r}{\sqrt{2}},\frac{r}{\sqrt{2}}) = 1/8$

$P[D] = p_{XY}(-r,0) = 1/8$

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5.13

(a) Sample space: 200 outcomes i representing zero or one page request for the given 1ms interval

$$S_{XY} = \{(i,j): 0 \le i \le 100,\ 0 \le j \le 100\}$$

(b) $$p_{XY}(x,y) = \binom{100}{x}(0.05)^x(0.95)^{100-x} \cdot \binom{100}{y}(0.05)^y(0.95)^{100-y}$$

(c) $$P[X=x] = \binom{100}{x}(0.05)^x(0.95)^{100-x}$$

$$P[Y=y] = \binom{100}{y}(0.05)^y(0.95)^{100-y}$$

(d) $$P[X \ge Y] = P[X>Y] + P[X=Y] = \frac{1}{2} + \frac{1}{2}P[X=Y] = \frac{1}{2} + \frac{1}{2}\sum_{j=0}^{100}(P[X=j])^2$$

$P[X>Y] = \frac{1}{2}(1 - P[X=Y])$; $\sum_{j=0}^{100}(P[X=j])^2 \approx 0.1307$; $P[X \ge Y] \approx .5654$

$$P[X=0, Y=0] = \binom{100}{0}\binom{100}{0}(0.05)^0(0.05)^0(0.95)^{100}(0.95)^{100}$$

$$= 0.95^{200}$$

$$\simeq 3.5 \times 10^{-5}$$

$$P[X>5, Y>3]$$

$$= (1 - P[X \le 5])(1 - P[Y \le 3])$$

$$= \left(1 - \sum_{x=0}^{5}\binom{100}{x}(0.05)^x(0.95)^{100-x}\right)\left(1 - \sum_{y=0}^{3}\binom{100}{y}(0.05)^y(0.95)^{100-y}\right)$$

$$= 0.285$$

(e) $P[X+Y=10]$ $\Leftarrow P[\text{"10 requests in 200 ms"}]$

$$= \binom{200}{10}(0.05)^{10}(0.95)^{190}$$

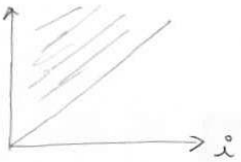
$$= 0.128$$

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5.14

(a) Sample Space: 200 outcomes representing zero or one page request for the given 1 ms interval.

$$S_{XY} = \{(i,j) : 0 \le i \le 100,\ i \le j \le 200\}$$



(b) $p_{N_1N_2}(n_1,n_2) = P[N_1 = n_1, N_2 = n_2]$

$$= \binom{100}{n_1}(0.05)^{n_1}(0.95)^{100-n_1}\binom{100}{n_2-n_1}(0.05)^{n_2-n_1}(0.95)^{100-(n_2-n_1)}$$

(c) $p_{N_1}(n_1) = \binom{100}{n_1}(0.05)^{n_1}(0.95)^{100-n_1}$

$$p_{N_2}(n_2) = \binom{200}{n_2}(0.05)^{n_2}(0.95)^{200-n_2}$$

(d) $P[A] = P[N_1 < N_2]$

$= 1 - P[N_1 = N_2]$ since $N_1 \le N_2$

$= 1 - P[\text{"0 requests in second 100 ms interval"}]$

$= 1 - \binom{100}{0}(0.05)^0(0.95)^{100}$

$= 0.994$ //

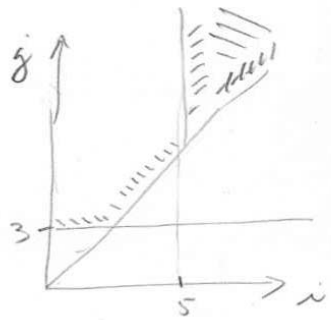
$P[B] = P[N_2 = 0]$

$= \binom{200}{0}(0.05)^0(0.95)^{200}$

$= 0.95^{200}$

$\simeq 3.5 \times 10^{-5}$

$P[C] = P[N_1 > 5,\ N_2 > 3]$



$= P[N_1 > 5]$ since $N_1 \le N_2$

$= 1 - P[N_1 \le 5]$

$= 1 - \sum_{n_1=0}^{5}\binom{100}{n_1}(0.05)^{n_1}(0.95)^{100-n_1}$

$= 0.384$ //

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5.15

(a) $S = \{(k_1, k_2) : 0 \le k_1 \le n, 0 \le k_2 \le n\}$

where $N_1 = k_1 =$ # heads in n tosses of a coin

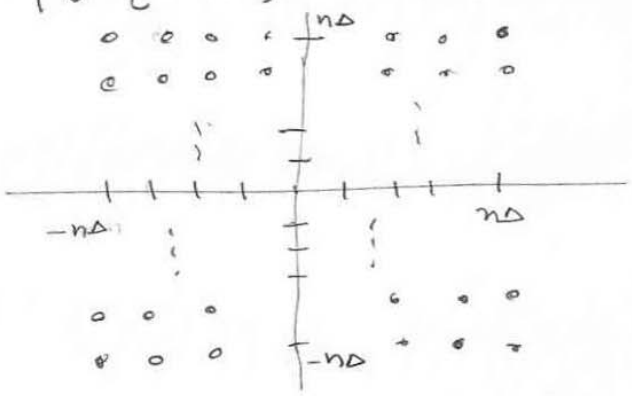
$N_2 = k_2 =$ # heads in n tosses of a coin

$$X = \Delta N_1 - (n - N_1)\Delta = 2N_1\Delta - n\Delta$$

$$Y = \Delta N_2 - (n - N_2)\Delta = 2N_2\Delta - n\Delta$$

$$X \in \{-n\Delta, (-n+2)\Delta, \ldots, n\Delta\}$$

$$Y \in \{-n\Delta, (-n+2)\Delta, \ldots, n\Delta\}$$



(b)

$$P[X = (2k-n)\Delta] = \binom{n}{k} p^k (1-p)^{n-k}$$

$$P[Y = (2j-n)\Delta] = \binom{n}{k} p^j (1-p)^{n-j}$$

$$P[X = (2k-n)\Delta, Y = (2j-n)\Delta]$$

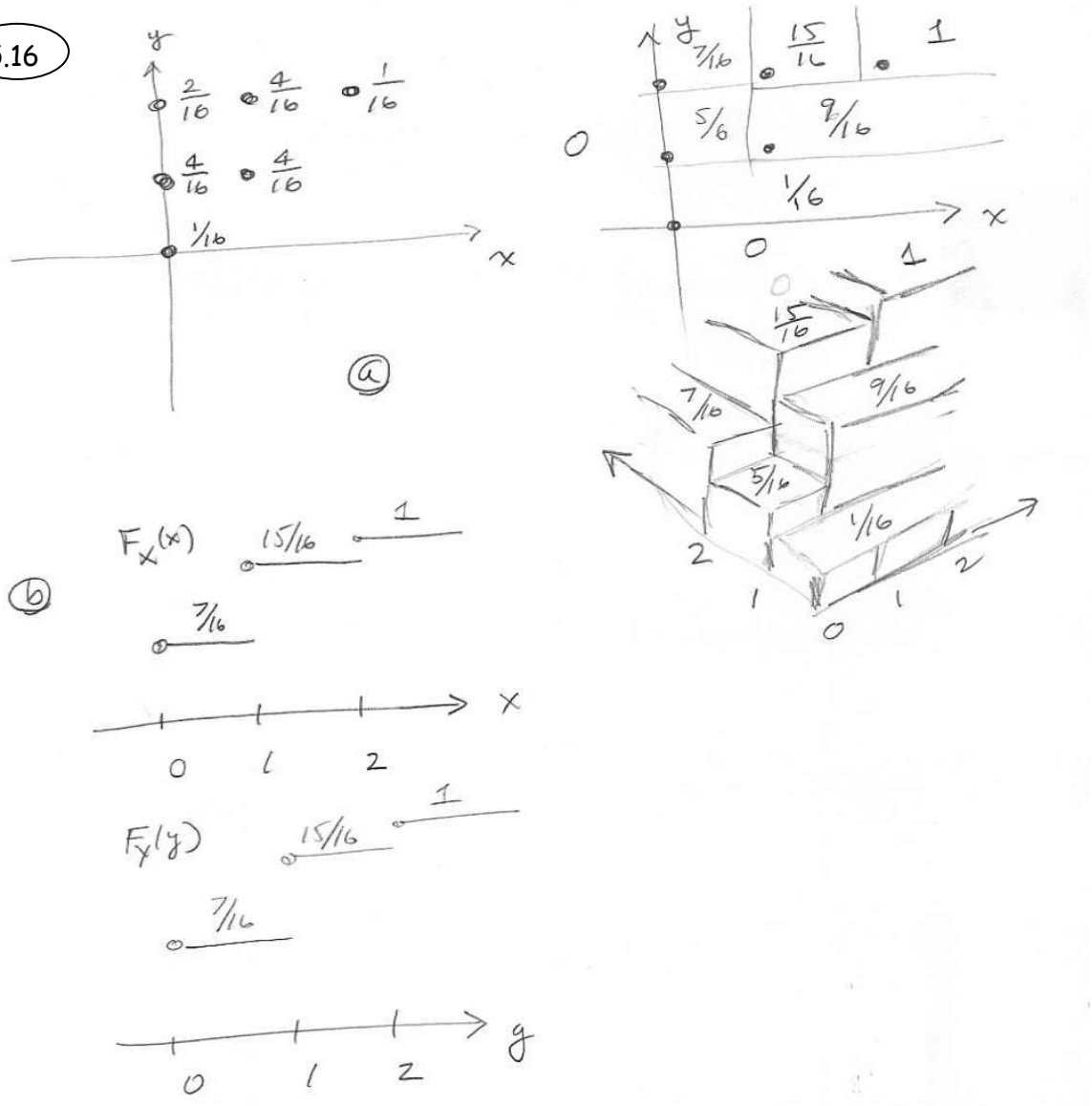
$$= \binom{n}{k} p^k (1-p)^{n-k} \binom{n}{j} p^j (1-p)^{n-j}$$

assuming same coin is used for both directions.

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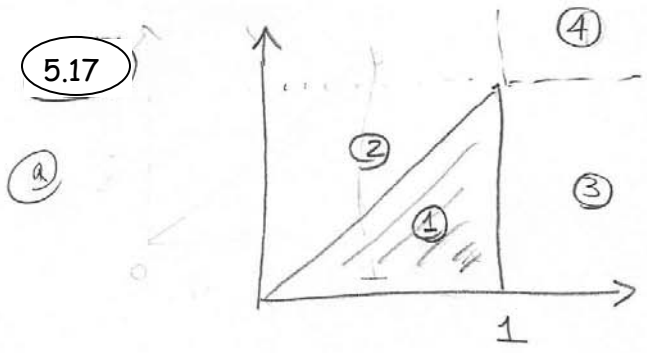
5.3 The Joint pdf of *X* and *Y*

5.16



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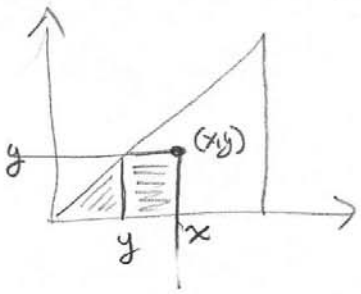
5.17



(a)

Area of Triangle = 1/2

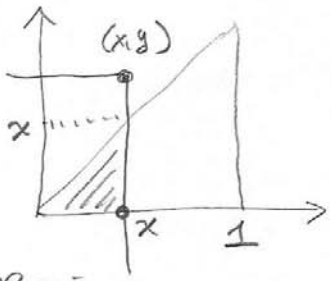
Region 1 $0 < y < x < 1$



$$P[X \le x, Y \le y] = \frac{\overbrace{\frac{y^2}{2}}^{\text{triangle}} + \overbrace{y(x-y)}^{\text{rectangle}}}{1/2}$$

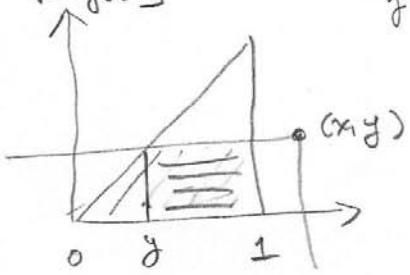
$$= 2\left(xy - \frac{y^2}{2}\right)$$

Region 2 $0 < x < y$



$$P[X \le x, Y \le y] = \frac{x^2/2}{1/2} = x^2$$

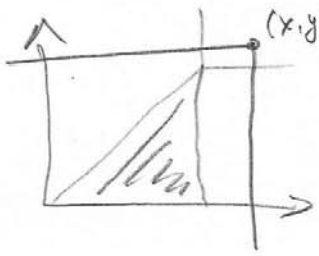
Region 3 $y < x, \; x > 1$



$$P[X \le x, Y \le y] = \frac{\frac{y^2}{2} + y(1-y)}{1/2}$$

$$= 2\left(y - \frac{y^2}{2}\right)$$

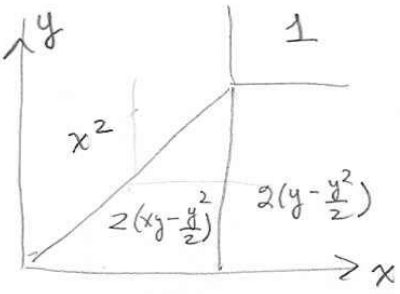
$x > 1, \; y > 1$



$$P[X \le x, Y \le y] = 1$$

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5.17



(b) $P[X \le x] = F_{XY}(x, \infty) = x^2$

$P[Y \le y] = F_{XY}(\infty, y) = 2(y - \frac{y^2}{2})$

(c) $P[X \le \frac{1}{2}, Y \le \frac{3}{4}] = (\frac{1}{2})^2 = \frac{1}{4}$ since $(\frac{1}{2}, \frac{3}{4})$ is in Region 2

$$P[\tfrac{1}{4} < X \le \tfrac{3}{4}, \tfrac{1}{4} < Y \le \tfrac{3}{4}]$$

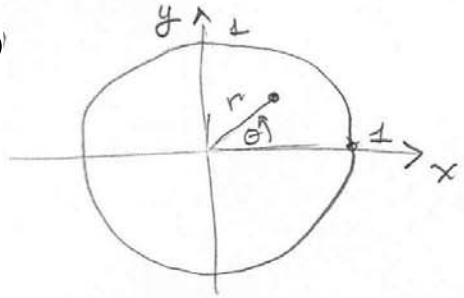
$$= F_{XY}(\tfrac{3}{4}, \tfrac{3}{4}) - F_{XY}(\tfrac{3}{4}, \tfrac{1}{4}) - F_{XY}(\tfrac{1}{4}, \tfrac{3}{4}) + F_{XY}(\tfrac{1}{4}, \tfrac{1}{4})$$

$$= (\tfrac{3}{4})^2 - 2(\tfrac{3}{4}\cdot\tfrac{1}{4} - \tfrac{1}{2}(\tfrac{1}{4})^2) - (\tfrac{1}{4})^2 + (\tfrac{1}{4})^2$$

$$= \frac{1}{4}$$

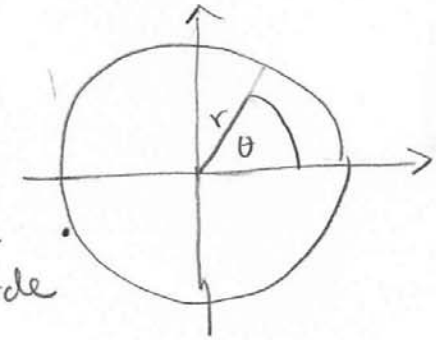
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5.18



$0 \le R \le 1$

$0 \le \Theta \le 2\pi$

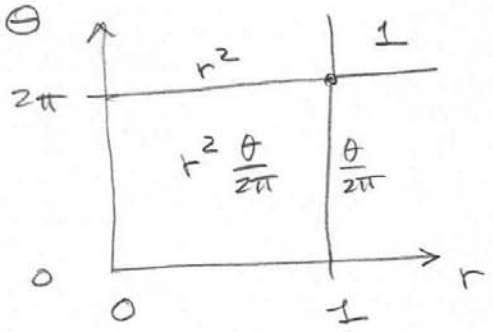


(a) $P[R \le r, \Theta \le \theta]$

$$= \frac{\text{area of pie slice}}{\text{area of unit circle}}$$

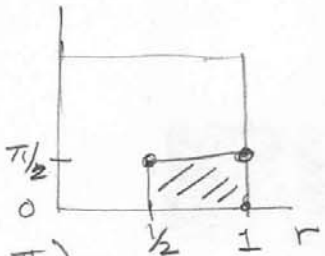
$$= \frac{\pi r^2 \cdot \frac{\theta}{2\pi}}{\pi(1)^2} = r^2 \frac{\theta}{2\pi} \qquad \begin{matrix} 0 \le r \le 1 \\ 0 \le \theta \le 2\pi \end{matrix}$$

(b)



(b) $F_R(r) = r^2 \qquad 0 \le r \le 1$

$$F_\Theta(\theta) = \frac{\theta}{2\pi} \qquad 0 \le \theta \le 2\pi$$



(c) $P[R > \frac{1}{2}, 0 < \Theta < \frac{\pi}{2}]$

$$= F_{R\Theta}(1, \tfrac{\pi}{2}) - F_{R\Theta}(1, 0) - F_{R\Theta}(\tfrac{1}{2}, \tfrac{\pi}{2}) + F_{R\Theta}(\tfrac{1}{2}, 0)$$

$$= \frac{1}{4} - 0 - \frac{1}{4} \cdot \frac{1}{4} + 0 = \frac{3}{16}$$

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5.19

(g) $P[\{\min(X,Y)>0\} \cup \{\max(X,Y)<0\}]$

$= P[\{\min(X,Y)>0\}] + P[\{\max(X,Y)<0\}]$

$= P[X>0, Y>0] + P[X<0, Y<0]$

$= F_{XY}(0^-, 0^-) + 1 - P[\{X>0\}^c \cup \{Y>0\}^c]$

$= F_{XY}(0^-, 0^-) + 1 - P[X\le 0] - P[Y\le 0]$
$+ P[X\le 0, Y\le 0]$

$= F_{XY}(0^-, 0^-) + 1 - F_{XY}(0, \infty)$
$- F_{XY}(\infty, 0) + F_{XY}(0, 0)$

(h) $P[\{X<0, Y>0\} \cup \{X>0, Y<0\}]$

$= P[\{X<0, Y>0\}] + P[\{X>0, Y<0\}]$

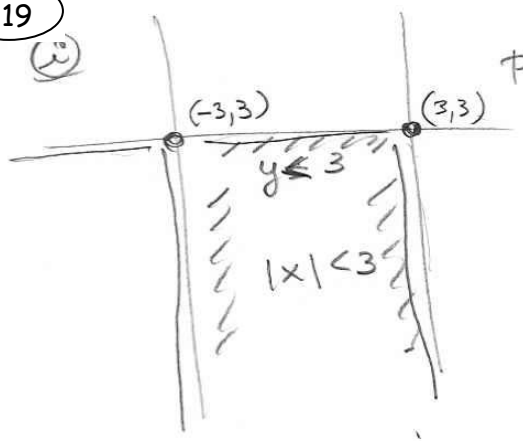
$= P[\{X<0\}] - P[X<0, Y\le 0]$
$+ P[\{Y<0\}] - P[X\le 0, Y<0]$

$= F_{XY}(0^-, \infty) - F_{XY}(0^-, 0)$
$+ F_{XY}(\infty, 0^-) - F_{XY}(0, 0^-)$

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5.19

(a)



$$P[-3 < X \leq 3,\ Y \leq 3] = F_{XY}(3^-, 3^-) - F_{XY}(-3, 3^-)$$

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5.20 (b) $F_X(x) = F_{XY}(x, \infty)$

$$= \begin{cases} 1 - 1/x^2 & , x > 1 \\ 0 & , \text{otherwise} \end{cases}$$

$$F_Y(y) = F_{XY}(\infty, y)$$

$$= \begin{cases} 1 - 1/y^2 & , y > 1 \\ 0 & , \text{otherwise} \end{cases}$$

(c) $P\{X < 3, Y \leq 5\}$

$= F_{XY}(3,5)$

$= (1 - \frac{1}{9})(1 - \frac{1}{25})$

$= \frac{64}{75}$

$P\{X > 4, Y > 3\}$

$= 1 - F_{XY}(4, \infty) - F_{XY}(\infty, 3) + F_{XY}(4, 3)$

$= 1 - (1 - \frac{1}{16}) - (1 - \frac{1}{9}) + (1 - \frac{1}{16})(1 - \frac{1}{9})$

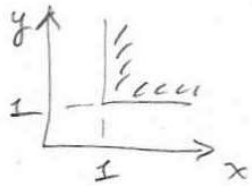
$= 1 - \frac{15}{16} - \frac{8}{9} + \frac{5}{6}$

$= \frac{1}{144}$

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5.21

$$F_{XY}(x,y) = \begin{cases} 1 - \frac{1}{x^2 y^2} & x>1, y>1 \\ 0 & \text{elsewhere} \end{cases}$$



$$F_X(x) = \lim_{y\to\infty} F_{XY}(x,y) = 1 \quad \text{all } x>1$$

$F_X(x)$ cannot be equal to 1 for all x

so not a valid cdf

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5.22 Properties

(i) $F_{XY}(x_1, y_1) \le F_{XY}(x_2, y_2)$ if $x_1 \le x_2$ and $y_1 \le y_2$

$F_X(x_1) F_Y(y_1) \le F_X(x_2) F_Y(y_2)$

is true since $0 \le F_X(x_1) \le F_X(x_2)$

and $0 \le F_Y(y_1) \le F_Y(y_2)$

(ii) $F_{XY}(x_1, -\infty) = 0$

$F_X(x_1) F_Y(-\infty) = 0$

is true since $F_Y(-\infty) = 0$

$F_{XY}(-\infty, y_1) = 0$

$F_X(-\infty) F_Y(y_1) = 0$

is true since $F_X(-\infty) = 0$

$F_{XY}(\infty, \infty) = 1$

$F_X(\infty) F_Y(\infty) = 1$

is true since $F_X(\infty) = F_Y(\infty) = 1$

(iii) $F_X(x_1) = F_{XY}(x_1, \infty)$

$= F_X(x_1) F_Y(\infty)$

$= F_X(x_1) \cdot 1$

$= F_X(x_1)$

$F_Y(y_1) = F_{XY}(\infty, y_1)$

$= F_X(\infty) F_Y(y_1)$

$= 1 \cdot F_Y(y_1)$

$= F_Y(y_1)$

(iv) $\lim_{x \to a^+} F_{XY}(x, y)$

$= \lim_{x \to a^+} F_X(x) F_Y(y)$

$F_X(x)$ right continuous

$= F_X(a) F_Y(y)$

$= F_{XY}(a, y)$

$\lim_{y \to b^+} F_{XY}(x, y)$

$= \lim_{y \to b^+} F_X(x) F_Y(y)$

$F_Y(y)$ right continuous

$= F_X(x) F_Y(b)$

$= F_{XY}(x, b)$

(v) $P[x_1 < X \le x_2, y_1 < Y \le y_2]$

$= P[x_1 < X \le x_2] P[y_1 < Y \le y_2]$

$= (F_X(x_2) - F_X(x_1))(F_Y(y_2) - F_Y(y_1))$

$= F_X(x_2) F_Y(y_2) - F_X(x_1) F_Y(y_2) - F_X(x_2) F_Y(y_1) + F_X(x_1) F_Y(y_1)$

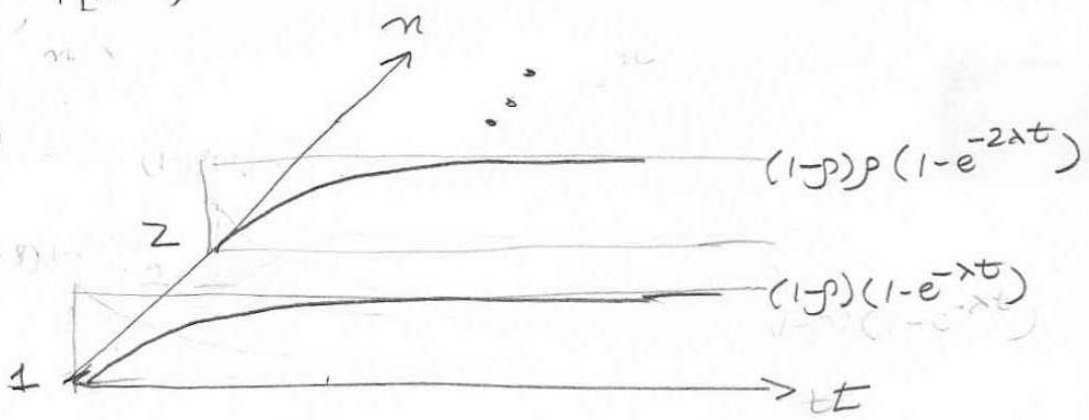
$= F_{XY}(x_2, y_2) - F_X(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1)$

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5.23 N is discrete and X is continuous

$$P[N=n, X\le t] = (1-p)p^{n-1}(1-e^{-n\lambda t}) \qquad n=1,2,\ldots,\ t>0$$

(a)



(b) $$P[N=n] = \lim_{t\to\infty} P[N=n, X\le t] = (1-p)p^{n-1} \qquad n=1,2,\ldots$$ geometric pmf

(c) $$P[X\le t] = \sum_{n=1}^{\infty}(1-p)p^{n-1}(1-e^{-\lambda n t})$$

$$= 1-(1-p)e^{-\lambda t}\sum_{n=1}^{\infty}\underbrace{p^{n-1}e^{-\lambda(n-1)t}}_{(pe^{-\lambda t})^{n-1}}$$

$$= 1-\frac{(1-p)e^{-\lambda t}}{1-pe^{-\lambda t}} = \frac{1-e^{-\lambda t}}{1-pe^{-\lambda t}}$$

(d) $$P\left[N\le 3, X>\frac{3}{\lambda}\right] = P[N\le 3] - P\left[N\le 3, X\le\frac{3}{\lambda}\right]$$

$$= \sum_{n=1}^{3}(1-p)p^{n-1} - \sum_{n=1}^{3}(1-p)p^{n-1}(1-e^{-3n})$$

$$= \sum_{n=1}^{3}(1-p)p^{n-1}e^{-3n}$$

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$$\sum_{n=1}^{3} (1-p)\, p^{n-1} e^{-3n} = (1-p)\, e^{-3} \sum_{n=1}^{3} (pe^{-3})^{n-1}$$

$$= (1-p)e^{-3}\left(1 + pe^{-3} + (pe^{-3})^2\right)$$

$$= (1-p)\, e^{-3}\, \frac{1-(pe^{-3})^3}{1-pe^{-3}}$$

5.24

$$P[N=k] = \binom{n}{k} p^k (1-p)^{n-k}$$

$$P[T \le t | N=k] P[N=k] = (1-e^{-k\alpha t}) \binom{n}{k} p^k (1-p)^{n-k}$$

$$P[T \le t] = \sum_{k=0}^{n} (1-e^{-k\alpha t}) \binom{n}{k} p^k (1-p)^{n-k}$$

$$= 1 - \sum_{k=0}^{n} \binom{n}{k} (pe^{-\alpha t})^k (1-p)^{n-k}$$

$$= 1 - (pe^{-\alpha t} + (1-p))^n$$

$$\lim_{t\to\infty} P[T \le t] = 1 - \underbrace{(1-p)^n}_{\substack{\text{Prob. that all} \\ \text{machines } \underline{\text{not}} \text{ working}}}$$

that is, the time to complete an item is infinite when no machines are available to do the work.

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5.4 The Joint cdf of Two Continuous Random Variables

5.25

(a) for $x>0, y>0$

$$F_{XY}(x,y) = \int_0^x \int_0^y \frac{1}{2} e^{-x/2} 2y e^{-y^2} dx dy$$

$$= (1-e^{-x/2})(1-e^{-y^2})$$

(b)

$$P[X>Y] = \int_0^\infty \int_0^x 2y e^{-y^2} dy \frac{1}{2} e^{-x/2} dx$$

$$= \int_0^\infty (1-e^{-x^2}) \frac{1}{2} e^{-x/2} dx$$

$$(x+\tfrac{1}{4})^2 = \tfrac{1}{16}$$

$$= x^2 + \tfrac{1}{2}x + \tfrac{1}{16} - \tfrac{1}{16}$$

$$= 1 - \frac{1}{2}\int_0^\infty e^{-(x^2+x/2)} dx$$

$$= 1 - \frac{1}{2} e^{1/16} \underbrace{\int_0^\infty e^{-(x+\frac{1}{4})^2} dx}$$

← Gaussian pdf mean 1/4 variance 1/2

$$\sqrt{2\pi \tfrac{1}{2}} = \sqrt{\pi}$$

$$= 1 - \frac{\sqrt{\pi} e^{1/16}}{2}$$

(c)

$$F_X(x) = \lim_{y\to\infty} F_{XY}(x,y) = 1-e^{-x/2} \qquad x>0$$

$$f_X(x) = \frac{1}{2} e^{-x/2}$$

$$F_Y(y) = 1-e^{-y^2}$$

$$f_Y(y) = 2y e^{-y^2}$$

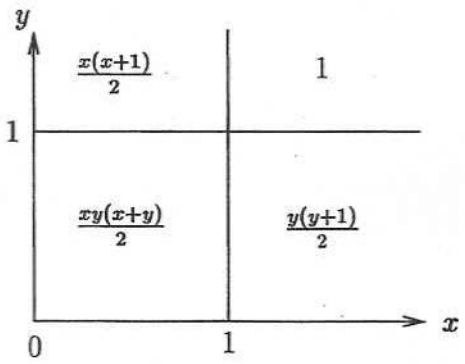
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5.26

$$1 = k\int_0^1\int_0^1 (x+y)dxdy = k\int_0^1 \left(\frac{x^2}{2}+xy\right)_0^1 dy$$
$$= k\int_0^1 \left(\frac{1}{2}+y\right)dy = k\left[\frac{1}{2}y+\frac{y^2}{2}\right]_0^1 = k$$

$\therefore k = 1$

b)



c) $F_X(x) = \lim_{y\to\infty} F_{XY}(x,y) = F_{XY}(x,1) \quad 0 < x < 1$

$$\Rightarrow f_X(x) = \frac{d}{dx}F_X(x) = x + \frac{1}{2}$$

Similarly

$$f_Y(y) = y + \frac{1}{2}$$

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5.27

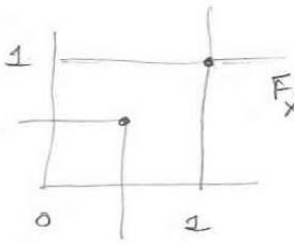
$f_{XY}(x,y) = kx(1-x)y \qquad 0<x<1,\ 0<y<1$

(a)

$$1 = k\int_0^1\int_0^1 x(1-x)y\,dx\,dy = k\left[\frac{x^2}{2}-\frac{x^3}{3}\right]_0^1\left[\frac{y^2}{2}\right]_0^1 = k\underbrace{\left[\frac{1}{2}-\frac{1}{3}\right]}_{\frac{3-2}{6}}\left[\frac{1}{2}\right]$$

$$\Rightarrow k = 12$$

(b)



$0<x<1 \qquad 0<y<1$

$$F_{XY}(x,y) = 12\int_0^x\int_0^y (x'-x'^2)\,y'\,dx'dy'$$

$$= 12\left(\frac{x^2}{2}-\frac{x^3}{3}\right)\left(\frac{y^2}{2}\right)$$

$$= (3x^2-2x^3)y^2$$

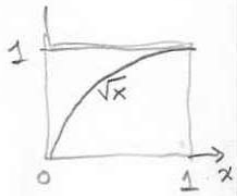
(c)

$$f_X(x) = \int_0^1 f_{XY}(x,y')\,dy' = 12x(1-x)\int_0^1 y\,dy = 6x(1-x)$$

$$f_Y(y) = \int_0^1 f_{XY}(x',y)\,dx' = 12y\int_0^1 (x-x^2)\,dx = 12y\left(\frac{x^2}{2}-\frac{x^3}{3}\right)_0^1$$

$$= 2y$$

(d)



$$P[Y\le\sqrt{X}] = 12\int_0^1 dx\int_0^{\sqrt{x}} (x-x^2)\,y\,dy$$

$$= 12\int_0^1 dx\,(x-x^2)\underbrace{\left[\frac{y^2}{2}\right]_0^{\sqrt{x}}}_{x}$$

$$= 6\int_0^1 dx\,(x^2-x^3) = 6\underbrace{\left[\frac{x^3}{3}-\frac{x^4}{4}\right]}_{\frac{4-3}{12}} = \frac{1}{2}$$

$$P[X<Y] = 12\int_0^1 dy\int_0^y (x-x^2)\,y\,dx$$

$$= 12\int_0^1 y\left[\frac{x^2}{2}-\frac{x^3}{3}\right]_0^y dy = 12\int_0^1\left(\frac{y^3}{2}-\frac{y^4}{3}\right)dy$$

$$= 12\left[\frac{1}{2}\frac{y^4}{4}-\frac{1}{3}\frac{y^5}{5}\right]_0^1 = 12\left[\underbrace{\frac{1}{2}\frac{1}{4}}_{\frac{1}{8}}-\underbrace{\frac{1}{3}\frac{1}{5}}_{\frac{1}{15}}\right] = \frac{7}{10}$$

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5.28

a)
$$k \cdot \pi = 1 \quad \Rightarrow k = \frac{1}{\pi} \quad \text{(i)}$$
$$k \cdot \sqrt{2} \cdot \sqrt{2} = 1 \quad \Rightarrow k = \frac{1}{2} \quad \text{(ii)}$$
$$k \cdot 1^2/2 = 1 \quad \Rightarrow k = 2 \quad \text{(iii)}$$

b) (i) $f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} = \frac{2\sqrt{1-x^2}}{\pi} \qquad -1 < x < 1$

Similarly
$$f_Y(y) = \frac{2\sqrt{1-y^2}}{\pi} \qquad -1 < x < 1$$

(ii) $f_X(x) = \int_{|x|-1}^{1-|x|} \frac{dy}{2} = 1 - |x| \qquad -1 < x < 1$

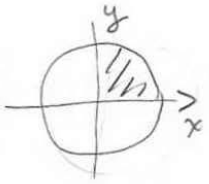
Similarly
$$f_Y(y) = 1 - |y| \qquad -1 < y < 1$$

(iii) $f_X(x) = \int_0^{1-x} 2dy = 2(1-x) \qquad 0 < x < 1$

Similarly
$$f_Y(y) = 2(1-y) \qquad 0 < y < 1$$

5.28c

(i) $P[X>0, Y>0] = \int_0^1 dx \int_0^{\sqrt{1-x^2}} \frac{1}{\pi} dy = \frac{1}{\pi} \int_0^1 \sqrt{1-x^2}\, dx$



$= \frac{1}{\pi} \left[x\sqrt{1-x^2} + \sin^{-1} x \right]_0^1$

$= \frac{1}{\pi} \left[0 + \underbrace{\sin^{-1} 1}_{\pi/4} - 0 + \underbrace{\sin^{-1} 0}_{0} \right] = \frac{1}{\pi} \frac{\pi}{4} = \frac{1}{4}$

(ii) $P[X>0, Y>0] = \int_0^1 dx \int_0^{1-x} \frac{1}{2} dy = \frac{1}{2} \int_0^1 (1-x)\, dx = \frac{1}{2} \left[x - \frac{x^2}{2} \right]_0^1$

$= \frac{1}{2} \left[1 - \frac{1}{2} - 0 + 0 \right] = \frac{1}{4}$ ✓

(iii) $P[X>0, Y>0] = \int_0^1 dx \int_0^{1-x} 2\, dy = 2\left[\frac{1}{2}\right] = 1$ ✓

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5.29

a) For $0 \leq y_0 \leq x_0$ we integrate along the strip indicated below.

b) The marginal cdf's are obtained by taking the appropriate limits of the joint cdf:

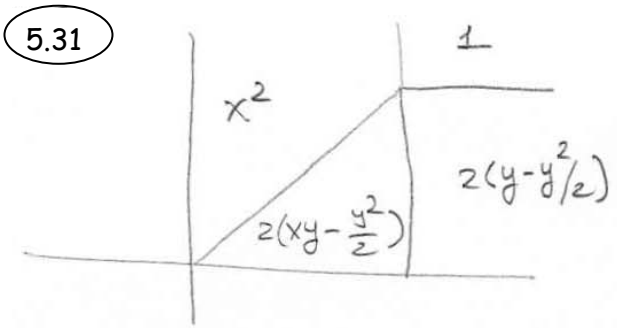
$$F_X(x_0) = \lim_{y_0 \to \infty} F_{XY}(x_0, y_0) = F_{XY}(x_0, x_0) = 1 - 2e^{-x_0} + e^{-2x_0}$$
$$F_Y(y_0) = \lim_{x_0 \to \infty} F_{XY}(x_0, y_0) = 1 - e^{-2y_0}$$

5.30

$$f_X(x) = \int_0^\infty xe^{-x}e^{-xy}dy = xe^{-x}\left(\frac{-1}{x}e^{-xy}\right)_0^\infty = e^{-x}$$
$$f_Y(y) = \int_0^\infty xe^{-x(1+y)}dx = \frac{e^{-x(1+y)}((1+y)x-1)}{(1+y)^2}\Bigg|_0^\infty$$
$$= \frac{1}{(1+y)^2}$$

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5.31

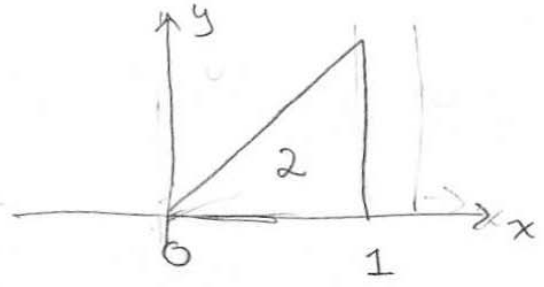


$0 < y < x < 1$

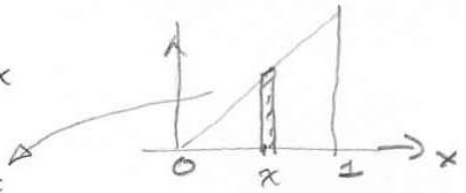
$$f_{XY}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x,y)$$

$\Rightarrow$ only the region $0 < y < x < 1$ has non zero pdf

$$f_{XY}(x,y) = \frac{\partial^2}{\partial x \partial y}\left(2xy - \frac{y^2}{2}\right) = \frac{\partial}{\partial x}(2x - 2y) = 2$$

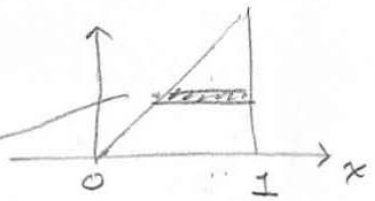


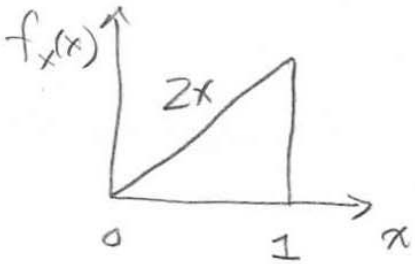
pdf is constant inside the triangle



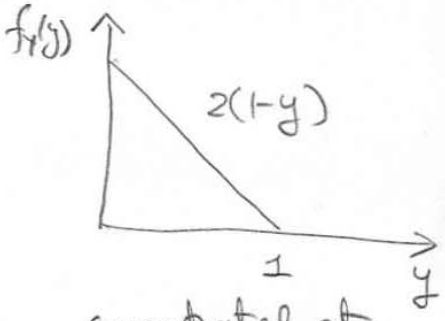
$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dy = \int_0^x 2\,dy = 2x \qquad 0 < x < 1$$

$$f_Y(y) = \int_y^1 2\,dx = 2(1-y)$$





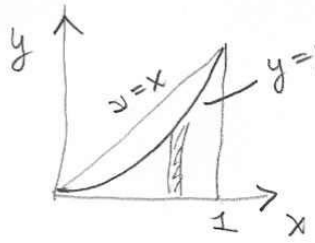
concentrated at higher values



concentrated at lower values

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5.31



5.32

5.33

$$
\begin{aligned}
P[X^2+Y^2<R^2] &= \int\int_{x^2+y^2<R^2} \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2}\,dx\,dy \\
&= \int_0^{2\pi}\int_0^R \frac{1}{2\pi\sigma^2} e^{-r^2/2\sigma^2} r\,dr\,d\theta \\
&\quad \text{where we let } x = r\cos\theta,\ y = r\sin\theta \\
&= \frac{r}{\sigma^2}\int_0^R r e^{-r^2/2\sigma^2}\,dr \\
&= 1 - e^{-R^2/2\sigma^2}
\end{aligned}
$$

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5.34

$$f_X(x) = \int_{-\infty}^{\infty} \frac{\exp\left\{-\frac{\left(\frac{x-m_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-m_1}{\sigma_1}\right)\left(\frac{y-m_2}{\sigma_2}\right) + \left(\frac{y-m_2}{\sigma_2}\right)^2}{2(1-\rho^2)}\right\}}{2\pi\sigma_1\sigma_2(1-\rho^2)^{1/2}} dy$$

Complete the square of the argument in the exponent:

$$\begin{aligned}
&\left(\frac{x-m_1}{\sigma_2}\right)^2 - 2\rho\left(\frac{x-m_1}{\sigma_1}\right)\left(\frac{y-m_2}{\sigma_2}\right) + \rho^2\left(\frac{x-m_1}{\sigma_1}\right)^2 \\
&\quad -\rho^2\left(\frac{x-m_1}{\sigma_1}\right)^2 + \left(\frac{y-m_2}{\sigma_2}\right)^2 \\
&\quad = \left[\left(\frac{y-m_2}{\sigma_2}\right) - \rho\left(\frac{x-m_1}{\sigma_1}\right)\right]^2 + (1-\rho^2)\left(\frac{x-m_1}{\sigma_1}\right)^2
\end{aligned}$$

Thus

$$\begin{aligned}
f_X(x) &= \frac{e^{-(x-m_1)^2/2\sigma_1^2}}{\sqrt{2\pi}\sigma_1} \int_{-\infty}^{\infty} \frac{e^{-\left[y-\left(m_2+\rho\sigma_2\left(\frac{x-m_1}{\sigma_1}\right)\right)\right]^2/2\sigma_2^2(1-\rho^2)}}{\sqrt{2\pi}\sigma_2\sqrt{1-\rho^2}} \\
&= \underbrace{\frac{e^{-(x-m_1)^2/2\sigma_1^2}}{\sqrt{2\pi}\sigma_1}}_{1}
\end{aligned}$$

integral of Gaussian pdf with
mean $m_2 + \rho\sigma_2\left(\frac{x-m_1}{\sigma_1}\right)$
variance $\sigma_2^2(1-\rho^2)$

$f_Y(y)$ has the same form.

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5.35

(a) for $j=-1$:

$$P[X=j, Y\le y]$$

$$= P[X=-1, N-1\le y]$$

$$= P[X=-1, N\le y+1]$$

$$= (1-p)\cdot\frac{1}{0.5\sqrt{2\pi}}\int_{-\infty}^{y+1} e^{-x^2/2(0.25)}\,dx$$

$$= (1-p)\sqrt{\frac{2}{\pi}}\int_{-\infty}^{y+1} e^{-2x^2}\,dx$$

for $j=1$:

$$P[X=1, Y\le y]$$

$$= P[X=1, N+1\le y]$$

$$= P[X=1, N\le y-1]$$

$$= p\sqrt{\frac{2}{\pi}}\int_{-\infty}^{y-1} e^{-2x^2}\,dx$$

(b) $p_X(-1) = 1-p \qquad p_X(1) = p$

$$F_Y(y) = P[Y\le y\mid X=-1]P[X=-1] + P[Y\le y\mid X=1]P[X=1]$$

$$= P[N-1\le y](1-p) + P[N+1\le y](p)$$

$$= \int_{-\infty}^{y}\frac{(1-p)e^{-(y'+1)^2/2(0.25)}}{0.5\sqrt{2\pi}}dy' + \int_{-\infty}^{y}\frac{pe^{-(y'-1)^2/2(0.25)}}{0.5\sqrt{2\pi}}dy'$$

$$f_Y(y) = \frac{d}{dy}F_Y(y) =$$

$$= (1-p)e^{-2(y+1)^2}\cdot\sqrt{\frac{2}{\pi}} + pe^{-2(y-1)^2}\cdot\sqrt{\frac{2}{\pi}}$$

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(c) test for $X=1$:

$$P[X=1 \mid Y>0]$$

$$= \frac{P[X=1, Y>0]}{P[Y>0]}$$

$$= \frac{P[X=1, N+1>0]}{P[Y>0 \mid X=-1]P[X=-1] + P[Y>0 \mid X=1]P[X=1]}$$

$$= \frac{P[X=1, N>-1]}{(1-p)P[N>1] + pP[N>-1]}$$

$$= \frac{pQ\left(\frac{-1}{0.5}\right)}{(1-p)Q\left(\frac{1}{0.5}\right) + pQ\left(\frac{-1}{0.5}\right)}$$

$$= \frac{pQ(-2)}{(1-p)Q(2) + pQ(-2)}$$

$$= \frac{p(1-Q(2))}{(1-p)Q(2) + p(1-Q(2))}$$

from table:

$$= \frac{0.9772p}{0.0228 - 0.0228p + 0.9772p}$$

$$= \frac{0.9772p}{0.0228 + 0.9544p}$$

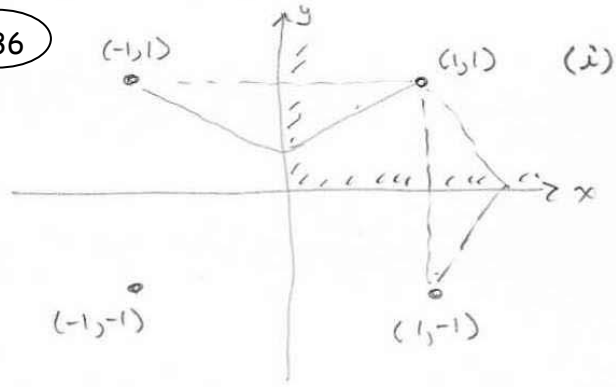
$P[X=1 \mid Y>0] > \frac{1}{2}$ when $0.0228 < p \leq 1$

and $< \frac{1}{2}$ when $0 \leq p \leq 0.0228$

$\therefore X=1$ more likely when $p \in [0.0228, 1]$ //

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5.36



(i) The y axis is perpendicular to the line between (-1,1) and (1,1). All points on the y axis are equidistant to (-1,1) and to (1,1). All pts to the right of the y axis are closer to (1,1) than to (-1,1).

(ii) Similarly all points above the x-axis are closer to (1,1) than to (1,-1). So the first quadrant is the set of points closer to (1,1) than to the other points.

$$P[\underline{Y} \text{ in 1st quadrant} \mid \underline{X}=(1,1)]$$

$$= \int_0^\infty \int_0^\infty \frac{1}{2\pi\sigma^2} e^{-\{(x_1-1)^2+(x_2-1)^2\}/2\sigma^2} dx_1 dx_2$$

$$= \int_0^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-(x_1-1)^2/2\sigma^2} dx_1 \int_0^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-(x_2-1)^2/2\sigma^2} dx_2$$

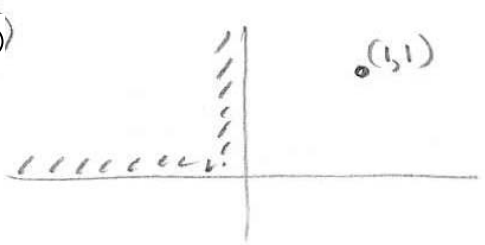
$$= \int_{-\frac{1}{\sigma}}^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du \int_{-\frac{1}{\sigma}}^\infty \frac{1}{\sqrt{2\pi}} e^{-v^2/2} dv$$

$$= Q(-\tfrac{1}{\sigma})\, Q(-\tfrac{1}{\sigma})$$

$$= (1-Q(\tfrac{1}{\sigma}))^2$$

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5.36b



Given $\underline{X} = (1,1)$
find prob of 2nd quadrant

$$P[\underline{Y} \text{ in 2nd Quad} \mid \underline{X} = (1,1)]$$

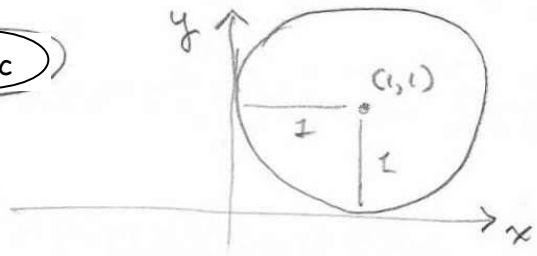
$$= \int_{-\infty}^{0} dn_1 \int_{0}^{\infty} dn_2 \frac{1}{2\pi\sigma^2} e^{-\{(n_1-1)^2+(n_2-1)^2\}/2\sigma^2}$$

$$= \int_{-\infty}^{-\frac{1}{\sigma}} \frac{e^{-u^2/2}}{\sqrt{2\pi}} du \int_{-1/\sigma}^{\infty} \frac{e^{-v^2/2}}{\sqrt{2\pi}} dv$$

$$= \left(1 - Q\left(-\tfrac{1}{\sigma}\right)\right) Q\left(-\tfrac{1}{\sigma}\right)$$

$$= Q\left(\tfrac{1}{\sigma}\right)\left(1 - Q\left(\tfrac{1}{\sigma}\right)\right)$$

5.36c



$$\{(x,y): \sqrt{(x-1)^2+(y-1)^2} > 1\} \supset [\underbrace{\{x \le 0, y \ge 0\}}_{\text{quad 2}} \cup \underbrace{\{x<0, y<0\}}_{\text{quad 3}} \cup \underbrace{\{x>0, y<0\}}_{\text{quad 4}}]$$

$$P\left[\sqrt{(X-1)^2+(Y-1)^2} \le 1\right] = \iint_{\sqrt{(x-1)^2+(y-1)^2} \le 1} \frac{1}{2\pi\sigma^2} e^{-\{(n_1-1)^2+(n_2-1)^2\}/2\sigma^2} dn_1\, dn_2$$

$$= \iint_{\sqrt{x^2+y^2} \le 1} \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2} dx\, dy$$

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5.36c — continued — change to polar coordinates

$$P\left[\sqrt{(X-1)^2+(Y-1)^2} \le 1\right] = \int_0^{2\pi}\int_0^1 \frac{1}{2\pi\sigma^2} e^{-r^2/2\sigma^2} r\,dr\,d\theta$$

$$= \int_0^1 \frac{r e^{-r^2/2\sigma^2}}{\sigma^2} dr = e^{-r^2/2\sigma^2}\Big|_0^1$$

$$= 1 - e^{-1/2\sigma^2}$$

$$\therefore P\left[\sqrt{(X-1)^2+(Y-1)^2} \ge 1\right] = e^{-1/2\sigma^2}$$

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5.5 Independence of Two Random Variables

5.37

N	1	2	3	4	5	6	outcome of toss
X	0	1	1	2	2	3	full pairs
Y	1	0	1	0	1	0	remainder
$p(x,y)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	

x \ y	0	1	
0		$\frac{1}{6}$	$\frac{1}{6}$
1	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{2}{6}$
2	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{2}{6}$
3	$\frac{1}{6}$		$\frac{1}{6}$
	$\frac{1}{2}$	$\frac{1}{2}$	

$P[Y=0] = \frac{1}{2} = P[Y=1]$

$P[X=0] = \frac{1}{6} = P[X=3]$

$P[X=1] = \frac{2}{6} = P[X=2]$

$p(x,y) \neq p(x)\,p(y)$ all x, y

$\Rightarrow$ X and Y not independent

5.38 from 5.16b

$0 \le k \le n$

$0 \le j \le n$

$P[X=(2k-n)\Delta, Y=(2j-n)\Delta]$

$= P[X=(2k-n)\Delta, Y=(2j-n)\Delta]$

$\Rightarrow$ X and Y are independent RVs

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5.39

(a) $p_{XY}(r,r) = 0$

but $p_X(r) = 1/8$

and $p_Y(r) = 1/8$

∴ Since $p_{XY}(r,r) \neq p_X(r) \cdot p_Y(r)$

X and Y are <u>not</u> independent.

(b) joint pmf:

X \ Y	$-r$	$-\frac{r}{\sqrt{2}}$	0	$\frac{r}{\sqrt{2}}$	r
$-r$	0	0	1/6	0	0
$-\frac{r}{\sqrt{2}}$	0	1/12	0	1/12	0
0	1/6	0	0	0	1/6
$\frac{r}{\sqrt{2}}$	0	1/12	0	1/12	0
r	0	0	1/6	0	0

marginal pmf:

$p_X(-r) = \frac{1}{6}$ $p_X(-\frac{r}{\sqrt{2}}) = \frac{1}{6}$ $p_X(0) = \frac{1}{3}$ $p_X(\frac{r}{\sqrt{2}}) = \frac{1}{6}$ $p_X(r) = \frac{1}{6}$

$p_Y(-r) = \frac{1}{6}$ $p_Y(-\frac{r}{\sqrt{2}}) = \frac{1}{6}$ $p_Y(0) = \frac{1}{3}$ $p_Y(\frac{r}{\sqrt{2}}) = \frac{1}{6}$ $p_Y(r) = \frac{1}{6}$

$p_{XY}(0,0) = 0$

but

$p_X(0)\, p_Y(0) = 1/9$

∴ X and Y are <u>not</u> independent.

5.40

Only the joint pmf for 5.11(ii) corresponds to a pair of independent RV's.

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5.41

Let M represent Michael's arrival time (minutes after 7:00)
Let B represent the arrival time of the bus (minutes after 7:00)

M, B uniform RVs

$$f_M(m) = \frac{1}{15} \quad , 25 \le m \le 40$$
$$f_B(b) = \frac{1}{10} \quad , 27 \le b \le 37$$

(a) $f_{MB}(m,b) = f_M(m) f_B(b)$ since M, B independent

$$= \frac{1}{150} \quad , 25 \le m \le 40, \ 27 \le b \le 37$$
$$0 \quad , \text{otherwise}$$

$$P[M+5 < B] = \int_{30}^{37} \int_{25}^{b-5} \left(\frac{1}{150}\right) dm\,db$$
$$= \frac{1}{150} \int_{30}^{37} (b-30)\,db$$
$$= \frac{1}{150} \left[\frac{b^2}{2} - 30b\right]_{30}^{37}$$
$$= 0.163$$

(b) $$P[M > B] = \int_{27}^{37} \int_{b}^{40} \frac{1}{150}\, dm\,db$$
$$= \frac{1}{150} \int_{27}^{37} (40-b)\,db$$
$$= \frac{1}{150} \left[40b - \frac{b^2}{2}\right]_{27}^{37}$$
$$= \frac{8}{15}$$
$$\approx 0.533$$

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5.42 From marginal cdfs in 5.19bc

$$F_{R,\Theta}(r,\theta) = P[R \le r, \Theta \le \theta] = F_R(r) F_\Theta(\theta) \quad \text{all } r, \theta$$

$\Rightarrow$ R and Θ are ndep RVs

5.43 $F_{XY}(x,y) = F_X(x) F_Y(y)$ all x, y

$\therefore$ X and Y are independent

5.44

$$F_{XY}(x,y) = (1-e^{-x/2})(1-e^{-y^2})$$

$$= F_X(x)\, F_Y(y) \quad \text{all } x, y$$

$\Rightarrow$ X and Y are ndependent

5.45

$$F_{XY}(x,y) = \frac{xy(x+y)}{2} \qquad \begin{matrix} 0 \le x \le 1 \\ 0 \le y \le 1 \end{matrix}$$

$$F_X(x) = x + \tfrac{1}{2} \qquad 0 \le x \le 1$$

$$F_Y(y) = y + \tfrac{1}{2} \qquad 0 \le y \le 1$$

$$F_{XY}(x,y) \neq F_X(x) F_Y(y)$$

$\Rightarrow$ X and Y are not ndependent.

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5.46

$$f_{XY}(x,y) = 12x(1-x)y$$

$$f_X(x) = 6x(1-x)$$

$$f_Y(y) = 2y$$

$\therefore$ X and Y independent since $f_{XY}(x,y) = f_X(x) f_Y(y)$

5.47

~~4.25~~ a) $P[a < X \leq b, Y \leq d] = P[a < X \leq b]P[Y \leq d]$
$= (F_X(b) - F_X(a))F_Y(d)$

b) $P[a \leq X \leq b, c \leq Y \leq a] = (F_X(b) - F_X(a^-))(F_Y(d) - F_X(c^-))$

c) $P[|X| > a, c \leq Y \leq d] = (1 - F_X(a) + F_X(a^-))(F_Y(d) - F_X(c^-))$

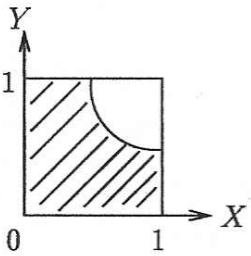
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5.48

a) $P[X^2 < \frac{1}{2}, |Y| < \frac{1}{2}] = P[X^2 < \frac{1}{2}]P[|Y| < \frac{1}{2}]$
$= P[X < \frac{1}{\sqrt{2}}]P[Y < \frac{1}{2}] = \frac{1}{2}\frac{1}{\sqrt{2}}$

b) $P[X < 1, Y < 0] = P[X < \frac{1}{4}]P[Y < 0] = (\frac{1}{4}) \cdot 0 = 0$

c)



$$f(x,y) = f(x)f(y) = 1$$

$$\begin{aligned} P[XY < \frac{1}{2}] &= \frac{1}{2} + \int_{\frac{1}{2}}^{1} \int_{0}^{\frac{1}{2x}} 1 \cdot dydx \\ &= \frac{1}{2} + \int_{\frac{1}{2}}^{1} \frac{1}{2x} dx \\ &= \frac{1}{2} + \frac{1}{2} \ln |x| \,|_{\frac{1}{2}}^{1} \\ &= 0.85 \end{aligned}$$

d) $P[\min(X,Y) > \frac{1}{3}] == P[X > \frac{1}{3}]P[Y > \frac{1}{3}] = (\frac{2}{3})^2 \ \frac{4}{9}$

5.49

a)

X \ Y	-1	0	1
	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$
-1 $\frac{1}{4}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{16}$
0 $\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$
1 $\frac{1}{4}$	$\frac{1}{16}$	$\frac{1}{6}$	$\frac{1}{16}$

X^2 \ Y^2	-1	0
	$\frac{1}{2}$	$\frac{1}{2}$
0 $\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
1 $\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

b)

X \ Y	-1	0	1
	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$
-1 $\frac{3}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
0 $\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$
1 $\frac{1}{2}$	0	$\frac{1}{8}$	0

X^2 \ Y^2	0	1
	$\frac{1}{2}$	$\frac{1}{2}$
0 $\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
1 $\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

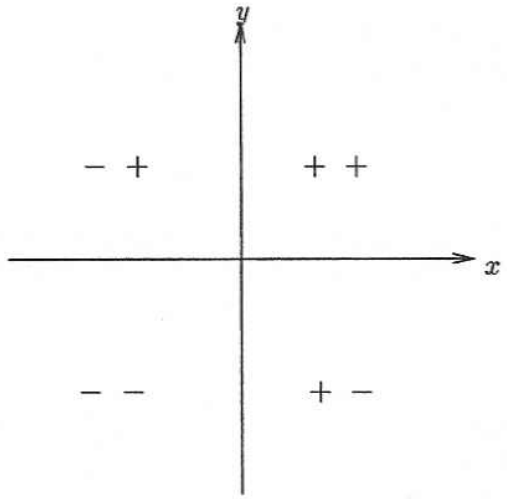
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5.50

4.25 a) If $\rho = 0$ in Problem 35, we have

$$\begin{aligned} f_{XY}(x,y) &= \frac{1}{2\pi\sigma_1\sigma_2} e^{-\left[\left(\frac{x-m_1}{\sigma_1}\right)^2+\left(\frac{y-m_2}{\sigma_2}\right)^2\right]/2} \\ &= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-m_1)^2}{2\sigma_1^2}} \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(y-m_2)^2}{2\sigma_2^2}} \\ &= f_X(x) f_Y(y) \quad \text{for all } x, y \\ &\Rightarrow X, Y \text{ indep. RV's} \end{aligned}$$

b) If $\rho = 0$ then



$$\begin{aligned} P[XY > 0] &= P[X \text{ and } Y \text{ have same sign}] \\ &= \underbrace{\int\int_{++}}_{\text{quadrant}} f_{XY}(x,y)dxdy + \underbrace{\int\int_{--}}_{\text{quadrant}} f_{XY}(x,y)dxdy \\ &= \int_0^\infty f_X(x)dx \int_0^\infty f_Y(y)dy + \int_{-\infty}^0 f_X(x)dx \int_{-\infty}^0 f_Y(y)dy \end{aligned}$$

but

$$\int_0^\infty \frac{1}{\sqrt{2\pi}\sigma_1} e^{-(x-m_1)^2/2\sigma_1^2} dx = \int_{-\frac{m_1}{\sigma_1}}^\infty \frac{e^{-t^2/2}}{\sqrt{2\pi}} dt = Q\left(-\frac{m_1}{\sigma_1}\right)$$

and similarly for other integrals, thus

$$P[XY > 0] = Q\left(-\frac{m_1}{\sigma_1}\right) Q\left(-\frac{m_2}{\sigma_2}\right) + \left(1 - Q\left(-\frac{m_1}{\sigma_1}\right)\right)\left(1 - Q\left(-\frac{m_2}{\sigma_2}\right)\right)$$

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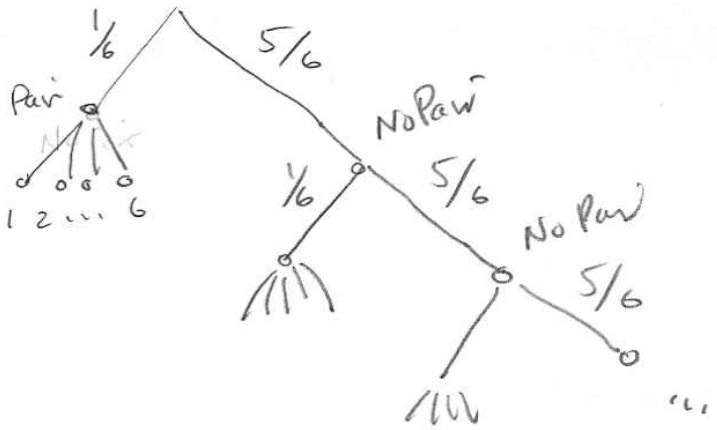
5.51

$$p_{KX}(k,x) = \left(\tfrac{5}{6}\right)^{k-1}\left(\tfrac{1}{6}\right)\left(\tfrac{1}{6}\right) \qquad 1 \le x \le 6, \quad k > 0$$

$$p_K(k) = \left(\tfrac{5}{6}\right)^{k-1}\left(\tfrac{1}{6}\right), \quad k > 0$$

$$p_X(x) = \tfrac{1}{6}, \quad 1 \le x \le 6$$

$\therefore$ K and X are independent since $p_{KX}(k,x) = p_K(k)\,p_X(x)$



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5.52

L is geometric dist p. # devices produced in 1 day

N = # working devices

M = # defective devices

$$P[L=k] = (1-p)^k p \qquad k=0,1,\ldots$$

$$P[N=n \mid L=k] = \binom{k}{n} \alpha^n (1-\alpha)^{k-n} \qquad n=0,\ldots,k \quad 0 \le n \le k$$

$$P[N=n, M=m \mid L=n+m] = \binom{n+j}{n} \alpha^n (1-\alpha)^m$$

$$P[N=n, M=m] = \binom{n+j}{n} \alpha^n (1-\alpha)^m (1-p)^{n+m} p$$

$$P[N=n] = \sum_{k=n}^{\infty} \binom{k}{n} \alpha^n (1-\alpha)^{k-n} (1-p)^k p$$

* Note k must be greater than n in order for N=n to be possible.

$$= \alpha^n (1-p)^n p \sum_{k=n}^{\infty} \binom{n}{k} \left[(1-\alpha)(1-p)\right]^{k-n}$$

$$\frac{1}{(1-(1-\alpha)(1-p))^{n+1}}$$

$$= \frac{\alpha^n (1-p)^n p}{(1-(1-\alpha)(1-p))^{n+1}} = \frac{(\alpha\bar{p})^n p}{(1-\bar{\alpha}\bar{p})^{n+1}} = \left(\frac{\alpha\bar{p}}{1-\bar{\alpha}\bar{p}}\right)^n \frac{p}{1-\bar{\alpha}\bar{p}}$$

Similarly

$$P[M=m] = \frac{(\bar{\alpha}\bar{p})^m p}{(1-\alpha\bar{p})^{n+1}} = \left(\frac{\bar{\alpha}\bar{p}}{1-\alpha\bar{p}}\right)^n \left(\frac{p}{1-\alpha\bar{p}}\right)$$

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5.52 — continued —

$$P[N=n, M=m] = \binom{n+j}{n} \alpha^n (1-\alpha)^m (1-p)^{n+m} p$$

only one way for n & m at the same time

$$\neq P[N=n]P[M=m]$$

$\Rightarrow$ many ways for n and m to occur separately

$\Rightarrow$ N & M are not independent

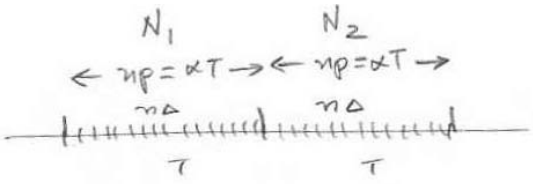
Also

$$P[N=n \mid L=k] = \binom{k}{n} \alpha^n (1-\alpha)^{k-n}$$

$$\neq P[N=n]$$

$\therefore$ L and N are not independent

5.53



In problem 13 we found that N_1 and N_2 are independent Bernoulli RV's

$$P[N_1=k, N_2=j] = \binom{n}{k} p^k (1-p)^{n-k} \binom{n}{j} p^j (1-p)^{n-j}$$

Allow the intervals Δ to decrease while keeping $np=\alpha T$ then each Bernoulli approaches a Poisson pmf

$$P[N_1=k, N_2=j] = \frac{(\alpha T)^k}{k!} e^{-\alpha T} \frac{(\alpha T)^j}{j!} e^{-\alpha T}.$$

The intervals need not be of the same length T.

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5.54

4.27 a) Without loss of generality, let event A, be $x_1 < X \leq x_2$, event A_2 be $y_1 < Y \leq y_2$

$$\begin{aligned} P[A] &= P[A_1 \cap A_2] \\ &= F_{X,Y}(x_2, y_2) - F_{X,Y}(x_2, y_1) - F_{X,Y}(x_1, y_2) + F_{X,Y}(x_1, y_2) \\ &= F_X(x_2)F_Y(y_1) - F_X(x_2)F_Y(y_1) - F_X(x_1)F_Y(y_2) + F_X(x_1)F_Y(y_2) \\ &= [F_X(x_2) - F_X(x_1)][F_Y(y_2) - F_Y(y_1)] \\ &= P[A_1]P[A_2] \end{aligned}$$

b) Let event A be $\{-\infty < X < x,\ -\infty < Y < y\}$, A_1 be $\{-\infty < X < x\}$, A_2 be $\{-\infty < Y < y\}$, then

$$\begin{aligned} P[A] &= F_{X,Y}(x, y) \\ &= P[A_1]P[A_2] \\ &= F_X(x)F_Y(y) \end{aligned}$$

5.55 a) (4.20) $\Rightarrow$ (4.21)

$$\begin{aligned} F_{X,Y}(x, y) &= F_X(x)F_Y(y) \\ \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y} &= \frac{\partial^2}{\partial x \partial y}(F_X(x)F_Y(y)) \\ f_{X,Y}(x, y) &= \frac{\partial}{\partial x}(F_X(x)f_Y(y)) \\ &= f_X(x)f_Y(y) \end{aligned}$$

b) (4.21) $\Rightarrow$ (4.20)

$$\begin{aligned} f_{X,Y}(x', y') &= f_X(x')f_Y(y') \\ \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(x', y')dx'dy' &= \int_{-\infty}^{x}\int_{-\infty}^{y} f_X(x')f_Y(y')dx'dy' \\ \text{i.e.} \quad F_{X,Y}(x, y) &= \int_{-\infty}^{x} f_X(x')F_Y(y)dx' \\ &= F_X(x)F_Y(y) \end{aligned}$$

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5.6 Joint Moments and Expected Value of a Function of Two Random Variables

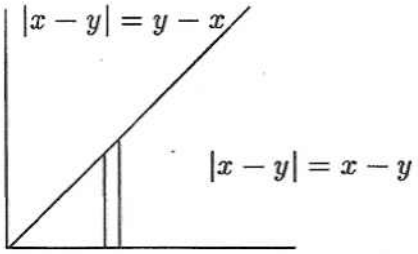
5.56

4.59 a) $\mathcal{E}[(X+Y)^2] = \mathcal{E}[X^2 + 2XY + Y^2] = \mathcal{E}[X^2] + 2\mathcal{E}[XY] + \mathcal{E}[Y^2]$

b)
$$\begin{aligned} VAR[X+Y] &= \mathcal{E}[(X+Y)^2] - \mathcal{E}[X+Y]^2 \\ &= \mathcal{E}[X^2] + 2\mathcal{E}[XY] + \mathcal{E}[Y^2] - \mathcal{E}[X]^2 \\ &\quad -2\mathcal{E}[X]\mathcal{E}[Y] - \mathcal{E}[Y]^2 \\ &= VAR[X] + VAR[Y] + 2[\mathcal{E}[XY] - \mathcal{E}[X]\mathcal{E}[Y]] \end{aligned}$$

c) $VAR[X+Y] = VAR[X] + VAR[Y]$ if $\mathcal{E}[XY] = \mathcal{E}[X]\mathcal{E}[Y]$ that is, if X and Y are uncorrelated.

5.57



$$\begin{aligned} \mathcal{E}[|X-Y|] &= \int_0^\infty \int_0^\infty |x-y| e^{-(x+y)} dx dy \\ &= 2\int_0^\infty \int_0^x (x-y) e^{-x} e^{-y} dy dx \\ &= 2\int_0^\infty e^{-x} [x(1-e^{-x}) - \underbrace{\int_0^x y e^{-y} dy}_{1-(1+x)e^{-x}}] \\ &\qquad \underbrace{\qquad\qquad\qquad\qquad}_{x+e^{-x}-1} \\ &= 2\int_0^\infty (x e^{-x} + e^{-2x} - e^{-x}) dx \\ &= 2[1 + \tfrac{1}{2} - 1] \end{aligned}$$

$2\left[\frac{1}{2} + 2\cdot\frac{1}{3} - 1\right] = 1 + \frac{4}{3} - 2 = \frac{1}{3}$

5.58

b) $E[X^2 e^Y] = E[X^2]E[e^Y] = 1 \cdot E[e^Y] = \frac{1}{3}(e^3 - 1)$

a) $E[X^2 Y] = \mathcal{E}[X^2]\mathcal{E}[Y] = 1(1) = 1$

$\frac{1}{3}\int_0^3 e^y dy = \frac{1}{3} e^y \Big|_0^3 = \frac{1}{3}(e^3 - 1)$

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5.59 from problem 5.9:

$$E[X] = 0\cdot\frac{7}{16} + 1\cdot\frac{8}{16} + 2\cdot\frac{1}{16} = \frac{10}{16}$$

$$E[Y] = 0\cdot\frac{1}{16} + 1\cdot\frac{8}{16} + 2\cdot\frac{7}{16} = \frac{22}{16}$$

$$E[XY] = \underbrace{0\cdot\left(0\cdot\tfrac{1}{16} + 1\cdot\tfrac{4}{16} + 2\cdot\tfrac{2}{16}\right)}_{0} + 1\cdot\left(1\cdot\tfrac{4}{16} + 2\cdot\tfrac{4}{16}\right) + 2\cdot 2\cdot\frac{1}{16}$$

$$= \frac{4}{16} + \frac{8}{16} + \frac{4}{16} = 1 \Rightarrow \text{not orthogonal}$$

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

$$= 1 - \frac{10}{16}\cdot\frac{22}{16} = \frac{36}{256} = \frac{9}{64} \Rightarrow \text{not uncorrelated}$$

$\Rightarrow$ not independent

5.60 From Problem 5.10

$$E[X] = -2\cdot\frac{1}{16} - 1\cdot\frac{4}{16} + 0\cdot\frac{6}{16} + 1\cdot\frac{4}{16} + 2\cdot\frac{1}{16} = 0$$

$E[Y] = 0$ same pmf

$$E[XY] = -2(2)\cdot\frac{1}{16} - 1\left(1\cdot\frac{2}{16} + 3\cdot\frac{2}{16}\right) + 0(\quad)$$

$$+ 1\cdot\left(1\cdot\frac{2}{16} + 3\cdot\frac{2}{16}\right) + 2(2)\frac{1}{16}$$

$= 0 \Rightarrow$ X and Y are orthogonal

$$Cov(X,Y) = E[XY] - E[X]E[Y] = 0$$

$\Rightarrow$ X and Y are uncorrelated

however $P[X=i, Y=j] \neq P[X=i]P[Y=j]$

$\Rightarrow$ not independent

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5.61

(i) $E[X] = -1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = 0$

$E[Y] = 0$ same pmf

$E[XY] = (-1)(-1)\frac{1}{6} + (-1)(1)\frac{1}{6} = 0$

$\Rightarrow$ X and Y are orthogonal

$Cov(X, Y) = E[XY] - E[X]E[Y] = 0$

$\Rightarrow$ X and Y are uncorrelated

However X and Y are not independent

since $P[X=i, Y=j] \neq P[X=i]P[Y=j]$

(ii) $E[X] = E[Y] = 0$ as before

$E[XY] = (-1)(-1)\frac{1}{9} + (-1)(1)\frac{1}{9} + (1)(-1)\frac{1}{9} + (1)(1)\frac{1}{9}$

$= 0$

$\therefore$ X and Y uncorrelated and orthogonal

However $P[X=i, Y=j] = P[X=i]P[Y=j]$

all i, j

$\Rightarrow$ X and Y indep.

(iii) $E[X] = E[Y] = 0$

$E[XY] = (-1)(-1)\frac{1}{3} + (1)(1)\frac{1}{3} = \frac{2}{3}$

$\Rightarrow$ not orthogonal and not uncorrelated

Also $P[X=i, Y=j] \neq P[X=i]P[Y=j]$

$\Rightarrow$ not indep.

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5.62

$$P_{N_1,N_2}(n_1,n_2) = \binom{100}{n_1}(0.05)^{n_1}(0.95)^{100-n_1}\binom{100}{n_2}(0.05)^{n_2}(0.95)^{100-n_2}$$

correlation

$$E[N_1N_2] = \sum_{n_1=0}^{100}\sum_{n_2=0}^{100}\binom{100}{n_1}(0.05)^{n_1}(0.95)^{100-n_1}\binom{100}{n_2}(0.05)^{n_2}(0.95)^{100-n_2}\, n_1 n_2$$

$$= \sum_{n_1=0}^{100}\binom{100}{n_1}(0.05)^{n_1}(0.95)^{100-n_1} n_1 \sum_{n_2=0}^{100}\binom{100}{n_2}(0.05)^{n_2}(0.95)^{100-n_2} n_2$$

$$= E[N_1]E[N_2]$$

$$= 100(0.05)\cdot 100(0.05)$$

$$= 25$$

covariance

$$E[N_1N_2] - E[N_1]E[N_2]$$

$$= 25-25$$

$$= 0$$

∴ Independent: yes since $COV(N_1,N_2) = 0$

Orthogonal: no since $E[N_1N_2] \neq 0$

Uncorrelated: yes since N_1 and N_2 independent.

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5.63

N_1 = # req in 1st n

N_2 = total # req in 1st n and 2nd n

Let M = # requests in 2nd n, then

$N_2 = N_1 + M$ where N_1 & M are indep (for Problem 5.62)

$E[N_1] = np \quad E[N_2] = 2np$

$E[N_1 N_2] = E[N_1(N_1+M)] = E[N_1^2] + E[N_1]E[M]$

$= npq + (np)^2 + (np)(np)$

$= npq + 2(np)^2 \Rightarrow$ not orthogonal

$COV(N_1, N_2) = npq + 2(np)^2 - np(2np)$

$= npq \Rightarrow$ not uncorrelated

Also N_1 and N_2 are not independent (N_2 always $\geq N_1$)

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5.64

$$P[N=n, X \le t] = (1-p)p^{n-1}(1-e^{-n\lambda t}) \qquad n=1,2,\ldots \quad t>0$$

$$E[NX] = \sum_{n=1}^{\infty} \int_0^{\infty} n\, P[N=n, X>t]\, dt$$

$$= \sum_{n=1}^{\infty} (1-p)p^{n-1} n \underbrace{\int_0^{\infty} e^{-n\lambda t}\, dt}$$

$$-\frac{e^{-n\lambda t}}{n\lambda}\Big|_0^{\infty}$$

$$= \sum_{n=1}^{\infty} (1-p)p^{n-1} \cdot n \cdot \frac{1}{n\lambda}$$

$$= \frac{1}{\lambda} \Rightarrow \text{not orthogonal}$$

$$E[N] = \sum_{n=1}^{\infty} n(1-p)p^{n-1} = \frac{1}{1-p}$$

$$E[X] = \int_0^{\infty} P[X>t]\, dt = \int_0^{\infty} \frac{(1-p)e^{-\lambda t}}{1-pe^{-\lambda t}}\, dt$$

$$= \frac{(1-p)}{p}\left[\frac{1}{p}\ln(1-pe^{-\lambda t})\right]_0^{\infty}$$

$$= \frac{1-p}{p}\left[0 - \ln(1-p)\right]$$

$$= \frac{(1-p)}{p}\left(\ln\frac{1}{1-p}\right)$$

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5.64 – continued –

$$\text{COV}(N,X) = E[NX] - E[N]E[X]$$

$$= \frac{1}{\lambda} - \frac{1}{1-p}\,\frac{1-p}{p}\ln\frac{1}{1-p}$$

$$= \frac{1}{\lambda} + \frac{\ln(1-p)}{p}$$

$\Rightarrow$ correlated

$\Rightarrow$ not independent

5.65

$$\begin{aligned}
f(x,y) &= x+y \quad 0<x<1 \quad 0<y<1 \\
f_X(x) &= x+\frac{1}{2} \quad 0<x<1 \quad f_Y(y) = y+\frac{1}{2} \quad 0<y<1 \\
\mathcal{E}[X] &= \int_0^1 x\left(x+\frac{1}{2}\right)dx = \frac{7}{12} = \mathcal{E}[Y] \\
\mathcal{E}[X^2] &= \int_0^1 x^2\left(x+\frac{1}{2}\right)dx = \frac{5}{12} = \mathcal{E}[Y^2] \\
\Rightarrow VAR[X] &= \frac{5}{12} - \left(\frac{7}{12}\right)^2 = \frac{11}{144} = VAR[Y] \\
\mathcal{E}[XY] &= \int_0^1\int_0^1 xy(x+y)dxdy = 2\int_0^1\int_0^1 x^2y\,dxdy = \frac{1}{3} \\
\rho &= \frac{\frac{1}{3}-\left(\frac{7}{12}\right)^2}{\frac{11}{144}} = -\frac{1}{11}
\end{aligned}$$

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5.66 $f_{XY}(x,y) = 12x(1-x)y \quad 0<x<1, \quad 0<y<1$

correlation $E[XY] = \int_0^1 \int_0^1 12x(1-x)y \cdot x \cdot dy dx$

$= \int_0^1 \int_0^1 12x^2(1-x)y^2 \, dy dx$

$= 12 \int_0^1 \left[\frac{y^3}{3} x^2(1-x)\right]_0^1 dx$

$= \frac{12}{3} \int_0^1 x^2(1-x) dx$

$= 4\left[\frac{1}{3}x^3 - \frac{1}{4}x^4\right]_0^1$

$= \frac{4}{12}$

$= \frac{1}{3}$ //

covariance $E[XY] - E[X]E[Y]$

$E[X] = \int_0^1 6x(1-x) \cdot x dx$

$= 6 \int_0^1 (x^2 - x^3) dx$

$= 6\left[\frac{1}{3}x^3 - \frac{1}{4}x^4\right]_0^1$

$= \frac{1}{2}$

$E[Y] = \int_0^1 2y \cdot y dy$

$= 2\left[\frac{1}{3}y^3\right]$

$= \frac{2}{3}$

$cov(X,Y) = \frac{1}{3} - \frac{1}{2} \cdot \frac{2}{3}$

$= 0$

∴ orthogonal: no since $E[XY] \neq 0$
independent: yes since $cov(X,Y) = 0$
uncorrelated yes since X,Y independent.

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5.67

i) $\mathcal{E}[X] = \mathcal{E}[Y] = 0$

$$\mathcal{E}[XY] = \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} x\frac{1}{x}dydx = 0$$

$\Rightarrow \rho = 0$ orthogonal & uncorrelated

ii) $\mathcal{E}[X] = \mathcal{E}[Y] = 0$

$$\mathcal{E}[XY] = \int_{-1}^{1}\int_{-(1-|x|}^{1-|x|} xydydx = 0$$

$\Rightarrow \rho = 0$ orthogonal & uncorrelated

iii) $$\mathcal{E}[X] = \int_0^1 2x(1-x)dx = \frac{1}{3} = \mathcal{E}[Y]$$

$$\mathcal{E}[X^2] = \int_0^1 2x^2(1-x)dx = \frac{1}{6}$$

$$VAR[X] = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{18}$$

$$\mathcal{E}[XY] = \int_0^1\int_0^{1-x} 2xydydx = \int_0^1 x(1-x)^2dx = \frac{1}{12}$$

$$\rho = \frac{\frac{1}{12} - \left(\frac{1}{3}\right)^2}{\frac{1}{18}} = -\frac{1}{2}$$

5.68

$$\rho = \frac{COV(X,Y)}{\sigma_x\sigma_Y} = \frac{\mathcal{E}[X(aX+b) - \mathcal{E}(X)\mathcal{E}(aX+b)]}{\sqrt{\mathcal{E}[X^2] - \mathcal{E}[X]^2}\sqrt{\mathcal{E}[(aX+b)^2 - \mathcal{E}[aX+b]^2]}}$$

$$= \frac{a\mathcal{E}[X^2] + b\mathcal{E}[X] - a\mathcal{E}[X]^2 - b\mathcal{E}[X]}{\sqrt{\mathcal{E}[X^2] - \mathcal{E}[X]^2}\sqrt{a\mathcal{E}[X^2] - a\mathcal{E}[X]^2}}$$

$$= \frac{a}{|a|} = \begin{cases} 1 & a > 0 \\ -1 & a < 0 \end{cases}$$

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5.69 The following expression suggests an approach to estimate the covariance:

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

Let $(X_1,Y_2), (X_2,Y_2), \ldots, (X_n,Y_n)$ be n sample pairs then

$E[X]$ is estimated by $\frac{1}{n}\sum_{i=1}^{n} Y_i$

$E[Y]$ " $\frac{1}{n}\sum_{i=1}^{n} Y_i$

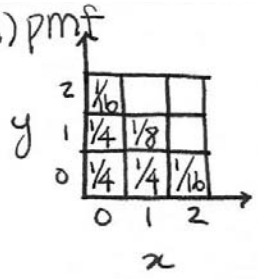
$E[XY]$ " $\frac{1}{n}\sum_{i=1}^{n} X_i Y_i$

$Cov(X,Y)$ is estimated by

$$\frac{1}{n}\sum_{i=1}^{n} X_i Y_i - \frac{1}{n^2}\sum_{i=1}^{n} X_i \sum_{j=1}^{n} Y_j$$

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5.70 (a) pmf



$$\rho_{XY} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y}$$

$$E[XY] = \sum_{x=0}^{2}\sum_{y=0}^{2} p_{XY}(x,y)\cdot x\cdot y$$

$$= \frac{1}{8}(1)(1)$$

$$= \frac{1}{8}$$

$$E[X] = \frac{6}{16}(1) + \frac{1}{16}(2)$$

$$= \frac{1}{2}$$

$$E[Y] = \frac{6}{16}(1) + \frac{1}{16}(2)$$

$$= \frac{1}{2}$$

$$\text{cov}(X,Y) = E[XY] - E[X]E[Y]$$

$$= \frac{1}{8} - \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)$$

$$= -\frac{1}{8}$$

$$E[X^2] = \frac{6}{16}(1) + \frac{1}{16}(4)$$

$$= 5/8$$

$$\sigma_Y = \sigma_X = (\text{VAR}[X])^{1/2}$$

$$= (E[X^2] - E[X]^2)^{1/2}$$

$$= (5/8 - 1/4)^{1/2}$$

$$= \sqrt{3/8}$$

$$\rho_{XY} = \frac{-\frac{1}{8}}{(\sqrt{3/8})(\sqrt{3/8})}$$

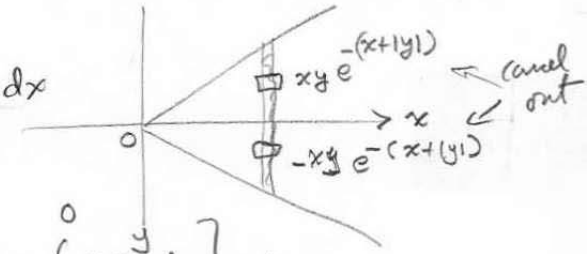
$$= -\frac{1}{3} \;//$$

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5.70 (b)

$$f_{XY}(x,y) = e^{-(x+|y|)} \qquad x>0, \quad -x<y<x$$

$$E[XY] = \int_0^\infty \int_{-x}^{x} xy\, e^{-(x+|y|)}\, dy\, dx$$



$$= \int_0^\infty x e^{-x} \left[\int_0^x y e^{-y}\, dy + \int_{-x}^{0} y e^{y}\, dy \right] dx$$

$$\int_{-x}^{0} y e^{y}\, dy = -\int_0^x y' e^{-y'}\, dy' \qquad y' = -y, \quad dy' = -dy$$

$$= 0$$

X and Y are orthogonal.

$$E[X] = \int_0^\infty x e^{-x} \int_{-x}^{x} e^{-|y|}\, dy\, dx = \int_0^\infty x e^{-x} \left[2\int_0^x e^{-y}\, dy \right] dx$$

$$= 2\int_0^\infty x e^{-x}(1-e^{-x})\, dx = 2\int_0^\infty x\,(e^{-x} - e^{-2x})\, dx$$

$$= 2\left[1 - \tfrac{1}{2}\right] = 1$$

$E[Y] = 0$ from above

$$Cov(X,Y) = E[XY] - E[X]E[Y] = 0$$

$\Rightarrow$ X and Y are uncorrelated.

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5.71

(a) $\rho_{XY} = \dfrac{COV(X,Y)}{\sigma_X \sigma_Y}$

$COV(X,Y) = E[XY] - \cancel{E[X]^0}\,E[Y]$

$= E[XY]$

$E[XY]$

$= E[X(X+N)]$

$= E[X^2] + E[XN]$

$E[X^2] = VAR[X] + \cancel{E[X]^0}^2$

$= \sigma_X^2$

independence

$E[XN] = E[X]\underbrace{E[N]}_{0} = 0$

$COV(X,Y) = \sigma_X^2$

$\rho_{XY} = \dfrac{\sigma_X^2}{\sigma_X \sigma_Y}$

$= \dfrac{\sigma_X}{\sigma_Y}$, $\sigma_Y = \sqrt{\sigma_X^2 + \sigma_N^2}$

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(b)

$$E[(X-aY)^2]$$

$$= E[X^2 - 2aXY + a^2Y^2]$$

$$= E[X^2] - 2aE[XY] + a^2E[Y^2]$$

$$= \sigma_X^2 - 2a\sigma_X^2 + a^2(\sigma_X^2 + \sigma_N^2)$$

Minimize:

$$\frac{d}{da}MSE = 0$$

$$-2\sigma_X^2 + 2a\sigma_X^2 + 2a\sigma_N^2 = 0$$

$$2a(\sigma_X^2 + \sigma_N^2) = 2\sigma_X^2$$

$$a = \frac{\sigma_X^2}{\sigma_X^2 + \sigma_N^2}$$

(c)

$$MSE = \sigma_X^2 - 2\left(\frac{\sigma_X^2}{\sigma_X^2+\sigma_N^2}\right)\sigma_X^2 + \left(\frac{\sigma_X^2}{\sigma_X^2+\sigma_N^2}\right)^2(\sigma_X^2+\sigma_N^2)$$

$$= \sigma_X^2 - \frac{2\sigma_X^4}{\sigma_X^2+\sigma_N^2} + \frac{\sigma_X^4}{\sigma_X^2+\sigma_N^2}$$

$$= \sigma_X^2 - \frac{\sigma_X^4}{\sigma_X^2+\sigma_N^2}$$

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5.72

$$E[XY] = \int_0^{2\pi} \cos\frac{\theta}{4}\sin\frac{\theta}{4}\frac{d\theta}{2\pi}$$

$$= \frac{1}{4\pi}\int_0^{2\pi} \sin\frac{\theta}{2}\,d\theta = \frac{1}{4\pi}\left[-2\cos\frac{\theta}{2}\right]_0^{2\pi}$$

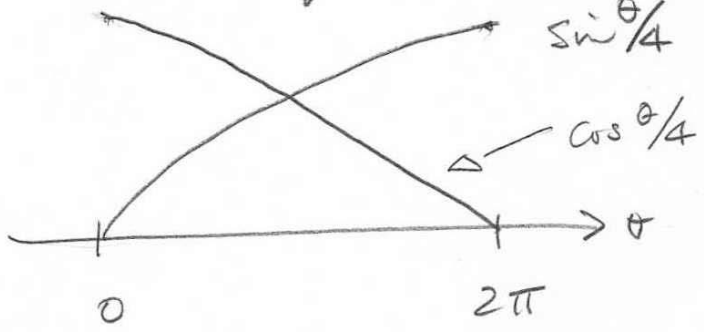
$$= \frac{2}{4\pi}\left[\cos 0 - \cos\frac{2\pi}{2}\right]$$

($\cos 0 = 1$, $\cos\frac{2\pi}{2} = -1$)

$$= \frac{4}{4\pi} = \frac{1}{\pi}$$

X and Y are correlated.

This is evident from the following



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5.73

(a) $\text{cov}(X, E[Y|X]) = \underbrace{E[X\,E[Y|X]]}_{E[XY]} - E[X]\underbrace{E[E[Y|X]]}_{E[Y]}$

$$= E[XY] - E[X]E[Y]$$

$$= \text{cov}(X,Y).$$

(b) If $E[Y|X=x] = E[Y]$ for all x

$$\Rightarrow E[Y|X] = E[Y]$$

$$\therefore \quad \text{cov}(X, E[Y|X]) = \underbrace{E[X\,E[Y]]}_{E[X]E[Y]} - E[X]E[Y]$$

$$= 0$$

$\Rightarrow$ uncorrelated

5.74

$$0 \le E[(tX+Y)^2] = t^2 E[X^2] + 2t\,E[XY] + E[Y^2]$$

View as quadratic equation in t, then equation has at most a double real root

$\therefore$ the discriminant is non-positive

$$(2E[XY])^2 - 4E[X^2]E[Y^2] \le 0$$

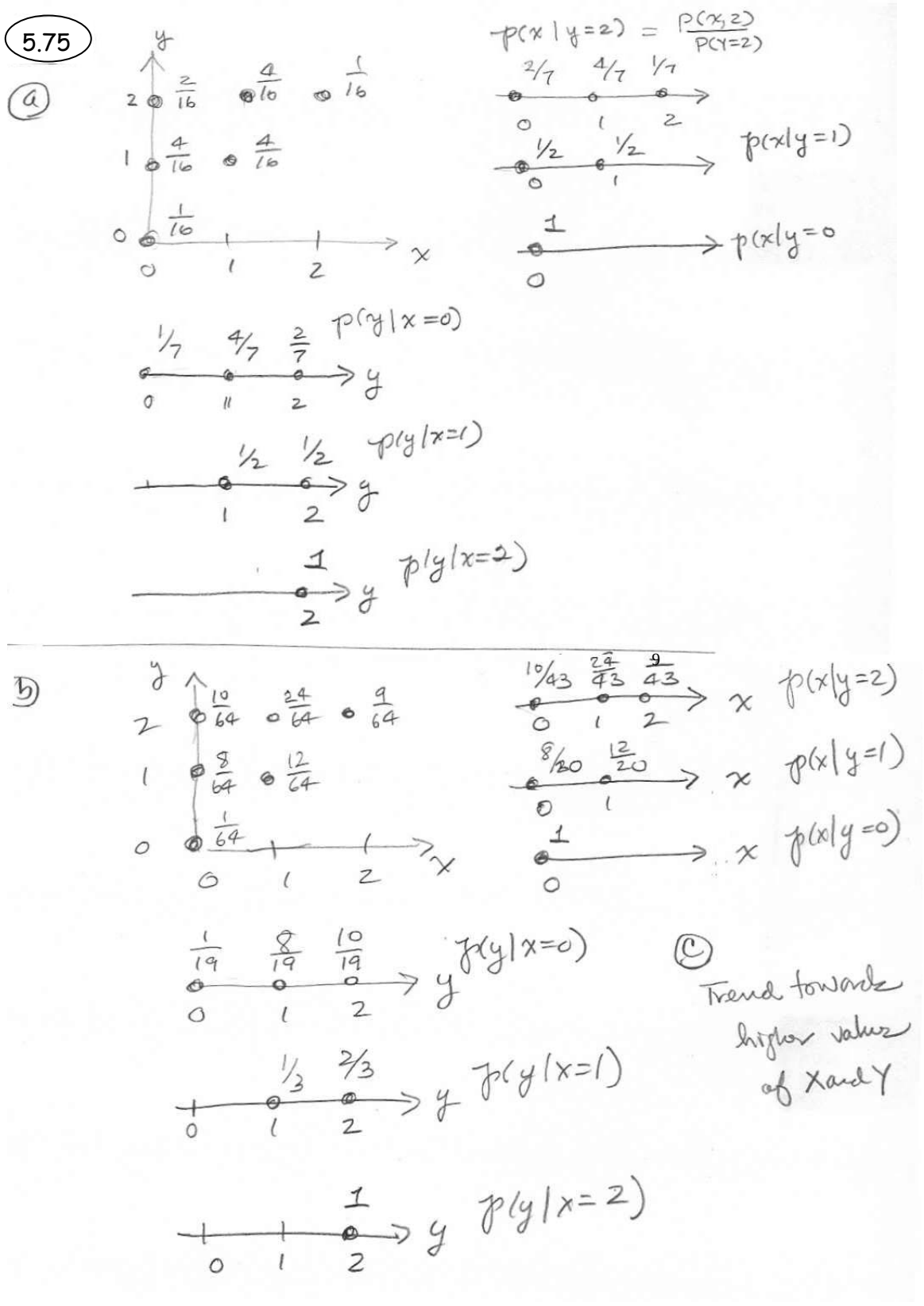
$$\Rightarrow \quad E[XY]^2 \le E[X^2]\,E[Y^2]$$

$$|E[XY]| \le \sqrt{E[X^2]\,E[Y^2]}$$

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5.7 Conditional Probability and Conditional Expectation

5.75



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5.75

(a) $E[X|y=2] = 0\cdot\frac{2}{7} + 1\cdot\frac{4}{7} + 2\cdot\frac{1}{7} = \frac{6}{7}$

$E[X|y=1] = 0\cdot\frac{1}{2} + \frac{1}{2}\cdot 1 = \frac{1}{2}$

$E[X|y=0] = 0$

$E[X] = 0\cdot\frac{1}{16} + \frac{1}{2}\cdot\frac{8}{16} + \frac{6}{7}\frac{7}{16} = \frac{5}{8}$

$E[Y|x=0] = 0\cdot\frac{1}{7} + 1\cdot\frac{4}{7} + 2\cdot\frac{2}{7} = \frac{8}{7}$

$E[Y|x=1] = 1\cdot\frac{1}{2} + 2\cdot\frac{1}{2} = \frac{3}{2}$

$E[Y|x=2] = 1$

$E[Y] = \frac{15}{8}\cdot\frac{7}{16} + \frac{3}{2}\frac{8}{16} + 1\cdot\frac{1}{16} = \frac{35+96+8}{128}$

$= \frac{139}{128}$

$E[X|y=2] = 1\cdot\frac{24}{43} + 2\cdot\frac{9}{43} = \frac{42}{43}$

$E[X|y=1] = \frac{12}{20}\cdot 1 = \frac{12}{20}$

$E[X|y=0] = 0$

$E[X] = \frac{42}{43}\cdot\frac{43}{64} + \frac{12}{20}\cdot\frac{20}{64}$

$= \frac{54}{64}$

$E[Y|x=0] = 1\cdot\frac{8}{19} + 2\cdot\frac{10}{19} = \frac{28}{19}$

$E[Y|x=1] = 1\cdot\frac{1}{3} + 2\cdot\frac{2}{3} = \frac{5}{3}$

$E[Y|x=2] = 1$

$E[Y] = \frac{28}{19}\cdot\frac{19}{64} + \frac{5}{3}\cdot\frac{36}{64} + 1\cdot\frac{9}{64}$

$= \frac{28+60+9}{64} = \frac{97}{64}$

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5.76

(a) $p_X(x|y) = \dfrac{p_{XY}(x,y)}{p_Y(y)}$

$$p_X(-1|-1) = \frac{\frac{1}{4}(1-p-p_e)}{\frac{1}{4}(1-p-p_e)+\frac{3}{4}p} \qquad p_X(1|-1) = \frac{\frac{3}{4}p}{\frac{1}{4}(1-p-p_e)+\frac{3}{4}p}$$

$$p_X(-1|0) = \frac{\frac{1}{4}p_e}{p_e} = \frac{1}{4} \qquad p_X(1|0) = \frac{\frac{3}{4}p_e}{p_e} = \frac{3}{4}$$

$$p_X(-1|1) = \frac{\frac{1}{4}p}{\frac{3}{4}(1-p-p_e)+\frac{1}{4}p} \qquad p_X(1|1) = \frac{\frac{3}{4}(1-p-p_e)}{\frac{3}{4}(1-p-p_e)+\frac{1}{4}p}$$

(b) for $y=0$, $p_X(x|y)$ maximum for $x=1$

for $y=-1$, $p_X(x|y)$ maximum for $x=1$ assuming $p > \frac{1}{4} - \frac{1}{4}p_e$

for $y=1$, $p_X(x|y)$ maximum for $x=1$ assuming $p < \frac{3}{4} - \frac{3}{4}p_e$

(c) $P_{error} = 1 - p_{XY}(1,1) - p_{XY}(-1,-1)$

$= 1 - \frac{1}{4}(1-p-p_e) - \frac{3}{4}(1-p-p_e)$

$= p + p_e$

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5.77

(a)

$p(y\|-1)$	$\frac{1}{2}$	$\frac{1}{2}$	0
$p(y\|0)$	0	0	$\frac{1}{3}$
$p(y\|1)$	$\frac{1}{2}$	$\frac{1}{2}$	0

⇐ this pmf provides most info about Y

(b) The conditional pmf's in (ii) are equiprobable so are most random;
The conditional pmf's in (iii) have one value with probability 1 and hence remove all uncertainty about the outcome

(c) (i) $E[X|-1] = -1 \cdot \frac{1}{2} = -\frac{1}{2}$
$E[X|0] = 1 \cdot \frac{1}{3} = \frac{1}{3}$
$E[X|+1] = -1 \cdot \frac{1}{2}$

$E[X] = \frac{1}{3}\left[-\frac{1}{2} + \frac{1}{3} - \frac{1}{2}\right] = 0$

$E[X|-1] = -\frac{1}{2} + \frac{1}{2} = 0$
$E[X|0] = -\frac{1}{2} + \frac{1}{2} = 0$
$E[X|1] = 0$

$E[X] = 0.$

(ii) $E[Y|i] = 0$ all i
$E[X|i] = 0$ all i

$E[X] = E[Y] = 0$

(iii) $E[Y|-1] = -1$
$E[Y|0] = 0$
$E[Y|1] = 1$

$E[Y] = \frac{1}{3}[-1 + 1] = 0$

$E[X|-1] = -1$
$E[X|0] = 0$
$E[X|1] = 1$

$E[X] = 0.$

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5.77 (d)

i) $E[Y^2|-1] = \frac{1}{2}$
$E[Y^2|0] = 1$
$E[Y^2|+1] = \frac{1}{2}$

$E[Y^2] = \frac{1}{3}[\frac{1}{2}+1+\frac{1}{2}] = \frac{2}{3}$

$VAR[Y] = \frac{2}{3} - 0^2 = \frac{2}{3}$

$E[X^2|-1] = \frac{1}{2}+\frac{1}{2} = 1$
$E[X^2|0] = \frac{1}{2}+\frac{1}{2} = 1$
$E[X^2|1] = 0$

$E[X^2] = \frac{1}{3}[1+1] = \frac{2}{3}$

$VAR(X) = \frac{2}{3}$

(ii) $E[Y^2|i] = ((-1)^2+0+1^2)\frac{1}{3} = \frac{2}{3}$ all i

$\Rightarrow E[Y^2] = \frac{2}{3} = VAR[Y]$

Similarly $E[X^2|i] = \frac{2}{3} = E[X^2] = VAR[X]$.

(iii) $E[Y^2|-1] = 1$
$E[Y^2|0] = 0$
$E[Y^2|+1] = 1$

$E[Y^2] = \frac{1}{3}(1+1) = \frac{2}{3} = VAR[Y]$

$E[X^2|-1] = 1$
$E[X^2|0] = 0$
$E[X^2|+1] = 1$

$E[X^2] = \frac{2}{3} = VAR[X]$

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5.78

for $\ell \geq k$ M = # heads in second n trials

$$P[N_1=k, N_2=\ell] = P[N_1=k]P[M=\ell-k]$$

(a) $$P[N_1=k \mid N_2=\ell] = \frac{P[N_2=\ell \mid N_1=k]P[N_1=k]}{P[N_2=\ell]}$$

$$= \frac{\binom{n}{\ell-k} p^{\ell-k}(1-p)^{n-\ell+k} \binom{n}{k} p^k (1-p)^{n-k}}{\binom{2n}{\ell} p^\ell (1-p)^{2n-\ell}}$$

$$= \frac{\binom{n}{\ell-k}\binom{n}{k}}{\binom{2n}{\ell}}$$

this is the hypergeometric probability we saw in chapter 3

(b) $$P[N_1=k \mid N_2=2k] = \frac{\binom{n}{k}\binom{n}{k}}{\binom{2n}{2k}}$$

Use Stirling's formula $n! \sim \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \approx \frac{\sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}}{\sqrt{2\pi}\, k^{k+\frac{1}{2}} e^{-k} \sqrt{2\pi}\, (n-k)^{n-k+\frac{1}{2}} e^{-(n-k)}}$$

$$= \frac{n^{n+\frac{1}{2}}}{\sqrt{2\pi}\, k^{k+\frac{1}{2}} (n-k)^{n-k+\frac{1}{2}}}$$

$$\frac{\binom{n}{k}\binom{n}{k}}{\binom{2n}{2k}} \approx \frac{n^{2n+1}}{2\pi k^{2k+1} (n-k)^{2(n-k)+1}} \frac{\sqrt{2\pi}\,(2k)^{2k+\frac{1}{2}} (2n-2k)^{2n-2k+\frac{1}{2}}}{(2n)^{2n+1/2}}$$

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5.78 — continued —

$$\frac{\binom{n}{k}\binom{n}{k}}{\binom{2n}{2k}} \approx \frac{n^{1/2}\, 2^{2k+\frac{1}{2}}\, 2^{2\cdot(n-k)+\frac{1}{2}}}{\sqrt{2\pi}\, 2^{2n+1/2}\, k^{1/2}\,(n-k)^{1/2}} = \frac{n^{1/2}\, 2^{1/2}}{\sqrt{2\pi}\, k^{1/2}(n-k)^{1/2}}$$

$$= \sqrt{\frac{2n}{2\pi k(n-k)}}$$

$n=100$ $\quad k=5$ $\quad k=10$ $\quad k=20$

$$\sqrt{\frac{200}{2\pi 5(95)}} \quad \sqrt{\frac{200}{2\pi 10(90)}} \quad \sqrt{\frac{200}{2\pi 20(80)}}$$

$$= \quad = \quad =$$

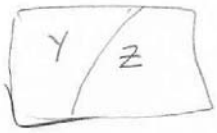
$$0.2589 \quad 0.1880 \quad 0.1404$$

(c) $$E[N_1 | N_2 = \ell] = \sum_{k=0}^{\ell} k \frac{\binom{n}{\ell-k}\binom{n}{k}}{\binom{2n}{\ell}} = \frac{\ell}{2}$$

$$E[N_1] = E\left[\tfrac{1}{2}N_2\right] = \tfrac{1}{2}E[N_2] = \tfrac{1}{2}2np = np$$

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5.79



$X = Y + Z$ X ~ Poisson

$$P[Y=j, Z=l] = P[Y=j \mid X=j+l]\, P[X=j+l]$$

$$= \binom{j+l}{j} p^j (1-p)^{l-j} \frac{\alpha^{j+l}}{(j+l)!} e^{-\alpha}$$

$$P[Z=l \mid Y=j] = \frac{P[Y=j, Z=l]}{P[Y=j]}$$

$$= \frac{\binom{j+l}{j} p^j (1-p)^l \alpha^{j+l} e^{-\alpha}}{(j+l)! \; \frac{(\alpha p)^j e^{-\alpha p}}{j!}}$$

$$= \frac{(1-p)^l \alpha^l}{l!} e^{-\alpha(1-p)}$$

$$= \frac{((1-p)\alpha)^l}{l!} e^{-\alpha(1-p)}$$

Poisson and indep of $Y=j$.

Result is "intuitive" if we view defect process as aggregation of multiplicity of Bernoulli trials over different regions of the device

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5.80

(a) $$f_Y(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{x+y}{x+\frac{1}{2}} \qquad 0<x<1$$

(b) $$P[Y>X|x] = \int_x^1 \frac{x+y}{x+\frac{1}{2}}\,dy = \frac{1}{x+\frac{1}{2}}\int_x^1 (x+y)\,dy$$

$$= \frac{1}{x+\frac{1}{2}}\left[xy\Big|_x^1 + \frac{y^2}{2}\Big|_x^1\right]$$

$$= \frac{1}{x+\frac{1}{2}}\left[x(1-x) + \frac{1}{2} - \frac{x^2}{2}\right]$$

$$= \frac{1}{x+\frac{1}{2}}\left[x + \frac{1}{2} - \frac{3}{2}x^2\right]$$

(c) $$P[Y>X] = \int_0^1 P[Y>X|x] f_X(x)\,dx$$

$$= \int_0^1 \frac{x+\frac{1}{2}-\frac{3}{2}x^2}{x+\frac{1}{2}}\left(x+\frac{1}{2}\right)dx$$

$$= \int_0^1 \left(x+\frac{1}{2}-\frac{3}{2}x^2\right)dx$$

$$= \frac{1}{2} + \frac{1}{2} - \frac{3}{2}\frac{1}{3} = \frac{1}{2} \checkmark$$

(d) $$E[Y|X=x] = \int_0^1 y\,\frac{x+y}{x+\frac{1}{2}}\,dy = \frac{1}{x+\frac{1}{2}}\int_0^1 (xy+y^2)\,dy$$

$$= \frac{\frac{1}{2}x + \frac{1}{3}}{x+\frac{1}{2}}$$

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5.81

(a) (i) $f_Y(y|x) = \dfrac{f_{XY}(x,y)}{f_X(x)} = \dfrac{\pi}{2\sqrt{1-x^2}}$ $\quad -1 \le x \le 1$, $-\sqrt{1-x^2} \le y \le \sqrt{1-x^2}$

(ii) $f_Y(y|x) = \dfrac{1}{1-|x|}$ $\quad -1 \le x \le 1$, $-(1-|x|) \le y \le 1-|x|$

(iii) $f_Y(y|x) = \dfrac{1}{2(1-x)}$ $\quad 0 \le x \le 1$, $0 < y < 1-x$

(b) (i) $E[Y|x] = \dfrac{\pi}{\sqrt{1-x^2}} \displaystyle\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} y\,dy = 0 \Rightarrow E[Y] = 0$

(ii) $E[Y|x] = \dfrac{1}{1-|x|} \displaystyle\int_{-(1-|x|)}^{1-|x|} y\,dy = 0 \Rightarrow E[Y] = 0$

(iii) $E[Y|x] = \dfrac{1}{2(1-x)} \displaystyle\int_0^{1-x} y\,dy = \dfrac{1}{2(1-x)} \cdot \dfrac{(1-x)^2}{2} = \dfrac{1-x}{2}$

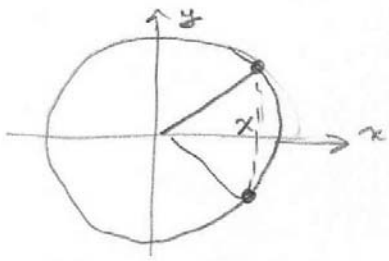
$$E[Y] = \int_0^1 \frac{1-x}{2} \cdot 2(1-x)\,dx = \int_0^1 (1-x)^2\,dx$$

$$= \int_0^1 (1-2x+x^2)\,dx$$

$$= \left[1 - 1 + \frac{1}{3}\right] = \frac{1}{3}. \checkmark$$

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5.82 (a)



$X = \cos\Theta$

$Y = \sin\Theta$

Θ uniform on $(0, 2\pi)$

$X^2 + Y^2 = 1$

$$f_Y(y|x) = \frac{1}{2}\delta(y - \sqrt{1-x^2}) + \frac{1}{2}\delta(y + \sqrt{1-x^2})$$

(a)
$$E[XY|X] = \frac{x}{2}\int_{-\sqrt{1-x^2}}^{\sqrt{1+x^2}} y\{\delta(y-\sqrt{\ }) + \delta(y+\sqrt{\ })\}\,dy$$

$$= \frac{x}{2}\left\{\sqrt{1+x^2} - \sqrt{1+x^2}\right\}$$

$$= 0$$

(b) $E[XY] = E[E[XY|X]] = 0$

(b)
$$E[Y|x] = \frac{1}{2}\int_{-\sqrt{1-x^2}}^{\sqrt{1+x^2}} y\{\delta(y-\sqrt{\ }) + \delta(y+\sqrt{\ })\}\,dy$$

$$= \frac{1}{2}\left\{\sqrt{1+x^2} - \sqrt{1+x^2}\right\} = 0$$

(c) $E[Y] = 0$

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5.83

$$f_Y(y|x) = \frac{\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{\left(\frac{x-m_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-m_1}{\sigma_1}\right)\left(\frac{y-m_2}{\sigma_2}\right) + \left(\frac{y-m_2}{\sigma_2}\right)^2}{2(1-\rho^2)}\right\}}{\frac{1}{\sqrt{2\pi}\sigma_1} \exp\left\{-\frac{(x-m_1)^2}{2\sigma_1^2}\right\}}$$

$$\begin{aligned} f_Y(y|x) &= \frac{\exp\left\{-\frac{\left(\frac{y-m_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{y-m_2}{\sigma_2}\right)\left(\frac{x-m_1}{\sigma_1}\right) + \rho^2\left(\frac{x-m_1}{\sigma_1}\right)^2}{2(1-\rho^2)}\right\}}{\sqrt{2\pi\sigma_2^2(1-\rho^2)}} \\ &= \frac{\exp\left\{-\frac{\left\{\left(\frac{y-m_2}{\sigma_2}\right) - \rho\left(\frac{x-m_1}{\sigma_1}\right)\right\}^2}{2(1-\rho^2)}\right\}}{\sqrt{2\pi\sigma_2^2(1-\rho^2)}} \\ &= \frac{\exp\left\{-\frac{(y-m_2-\rho\frac{\sigma_2}{\sigma_1}(x-m_1))^2}{2\sigma_2^2(1-\rho^2)}\right\}}{\sqrt{2\pi\sigma_2^2(1-\rho^2)}} \\ &= \text{Gaussian pdf with mean } m_2 + \rho\tfrac{\sigma_2}{\sigma_1}(x-m_1) \\ &\quad \text{and variance } \sigma_2^2(1-\rho^2) \end{aligned}$$

Similarly $f_X(x|y)$ is a Gaussian pdf with mean $m_1 + \rho\frac{\sigma_1}{\sigma_2}(y-m_2)$ and variance $\sigma_1^2(1-\rho^2)$.

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5.84

(a) $f_X(x|N=n) = \frac{d}{dx} F_X(x|N)$

$$F_X(x|N) = \frac{P[X \le x, N=n]}{P[N=n]}$$

$$= \frac{(1-p)p^{n-1}(1-e^{-n\lambda x})}{(1-p)p^{n-1}}$$

$$= 1 - e^{-n\lambda x}$$

$$f_X(x|N) = \frac{d}{dx}(1-e^{-n\lambda x})$$

$$= n\lambda e^{-n\lambda x} \qquad x > 0$$

(b) $E[X|N=n]$

$$= \int f_X(x|n)\, x\, dx$$

$$= \int_0^\infty n\lambda e^{-n\lambda x} \cdot x\, dx$$

integration by parts...

$$= \frac{-e^{-n\lambda x}}{n\lambda}\Big]_0^\infty$$

$$= 1/n\lambda$$

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(5.84) continued

$$P[N=n \mid x<T<t+dt] = \frac{P[N=n,\ t<T<t+dt]}{P[t<T<t+dt]}$$

$$P[N=n,\ t<T<t+dt] = P[N=n,\ T<t+dt] - P[N=n,\ T<t]$$

$$= (1-p)p^{n-1}\left[e^{-n\lambda t} - e^{-n\lambda(t+dt)}\right]$$

$$= (1-p)p^{n-1}e^{-n\lambda t}\left[1 - e^{-n\lambda dt}\right]$$

$$\approx (1-p)p^{n-1}e^{-n\lambda t}\left[n\lambda dt + o(n\lambda dt)\right]$$

From 5.24©

$$P[T\le t] = \frac{1-e^{-\lambda t}}{1-pe^{-\lambda t}} \Rightarrow f_T(t) = \frac{\lambda(1-p)e^{-\lambda t}}{(1-pe^{-\lambda t})^2}$$

© $$P[N=n \mid t<T<t+dt] = \frac{(1-p)p^{n-1}e^{-n\lambda t}\, n\lambda\, dt}{f_T(t)\, dt}$$

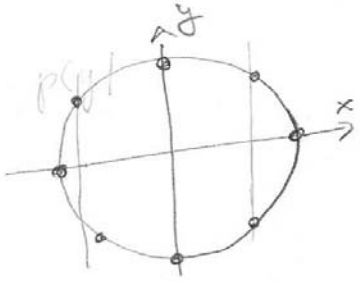
$$= A(t)\underbrace{(pe^{-\lambda t})^n}_{}\, n$$

To find the value of n that maximizes this probability we consider the ratio

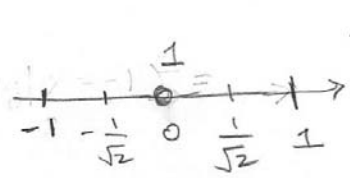
$$1 = \frac{\beta^{n+1}(n+1)}{\beta^n n} = \beta\left(1+\frac{1}{n}\right) \Rightarrow n^* \approx \frac{\beta}{\beta-1} = \frac{pe^{-\lambda t}}{1-pe^{-\lambda t}}$$

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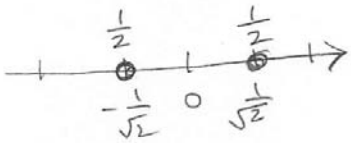
5.85



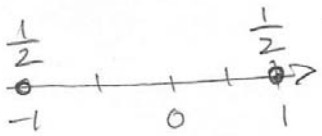
(a)



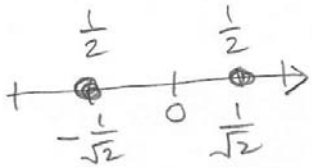
$p(y|x=-1)$



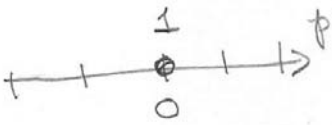
$p(y|x=-\frac{1}{\sqrt{2}})$



$p(y|x=0)$



$p(y|x=+\frac{1}{\sqrt{2}})$



$p(y|x=1)$

(b) $E[Y|-1] = 0$

$E[Y|-\frac{1}{\sqrt{2}}] = 0$

$E[Y|0] = 0$

$E[Y|\frac{1}{\sqrt{2}}] = 0$

$E[Y|1] = 0$

(c) $E[XY|X] = \sum_j jx\, p_Y(j|x) = x\, E[Y|X]$

$\therefore E[XY|X] = 0$ all x

$\Rightarrow E[XY] = E[E[XY|X]] = 0$

$\Rightarrow$ X and Y are uncorrelated.

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5.86

$$\alpha = \frac{\alpha x_m}{\alpha - 1} = \frac{2.5}{1.5} x_m$$

$$f_X(t) = \sum_{k=1}^{3} f_X(t|k)\, p_k$$

$$= \frac{1}{3}\left[\delta(t-2) + \frac{1}{2} e^{-t/2} + 2.5 \frac{\left(\frac{3}{2.5}\right)^{2.5}}{t^{3.5}} u\left(t - \frac{3}{2.5}\right) \right]$$

$$E[X] = \sum_{k=1}^{3} \underbrace{E[X|k]}_{2}\, p_k = 2$$

$$VAR[X] = E[(X-2)^2]$$

$$= \int_0^\infty (t-2)^2 f_X(t)\, dt = \sum_{k=1}^{3} \int_0^\infty (t-2)^2 f_X(t|k)\, dt\, p_k$$

$$= \frac{1}{3}\left\{ 0 + \left(\frac{1}{1/2}\right)^2 + \frac{(3/2.5)^2 2.5}{(1.5)^2 (.5)^2} \right\} = \frac{52}{15}$$

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5.87

$$P[K=k|N=n] = \binom{n}{k} p^k (1-p)^{n-k}$$

$$P[K=k] = \sum_{n=\max(k,1)}^{\infty} P[K=k|N=n]P[N=n]$$

can have $k > 0$ new messages only if $N \geq k$

$$\begin{aligned} P[K=0] &= \sum_{n=1}^{\infty}(1-p)^n(1-a)a^{n-1} \\ &= (1-p)(1-a)\sum_{n=1}^{\infty}[(1-p)a]^{n-1} \\ &= \frac{(1-p)(1-a)}{1-(1-p)a} \end{aligned}$$

For $k \geq 1$

$$\begin{aligned} P[K=k] &= \sum_{n=k}^{\infty}\binom{n}{k}p^k(1-p)^{n-k}(1-a)a^{n-1} \\ &= \frac{(1-a)p^k a^k}{a}\sum_{n=k}^{\infty}\binom{n}{k}[(1-p)a]^{n-k} \\ &= \frac{(1-a)p^k a^k}{a(1-(1-p)a)^{k+1}} \\ &= \frac{(1-a)}{a(1-(1-p)a)}\left(\frac{pa}{1-(1-p)a}\right)^k \qquad k=1,2,... \end{aligned}$$

K is geometric-like for $k \geq 1$, but the $P[K=0]$ is not consistent with the probability of success.

b)
$$\begin{aligned} \mathcal{E}[K] &= \mathcal{E}[\mathcal{E}[K|N]] = \sum_{n=1}^{\infty}\mathcal{E}[K|n](1-\alpha)a^{n-1} \\ &= \sum_{n=1}^{\infty} n - p(1-a)a^{n-1} = p\mathcal{E}[N] = \frac{p}{1-a} \\ \mathcal{E}[K^2] &= \sum_{n=1}^{\infty}\mathcal{E}[K^2|n](1-a)a^{n-1} \\ &= \sum_{n=1}^{\infty}(npq+(np)^2)(1-a)a^{n-1} \\ &= pq\mathcal{E}[N]+p^2\mathcal{E}[N^2] \\ VAR[K] &= \mathcal{E}[K^2]-\mathcal{E}[K]^2 \\ &= pq\mathcal{E}[N]+p^2\mathcal{E}[N^2]-p^2\mathcal{E}[N]^2 \\ &= pq\mathcal{E}[N]+p^2VAR[N] \\ &= \frac{pq}{1-a}+\frac{p^2a}{(1-a)^2} \end{aligned}$$

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5.88

$$\begin{aligned}
P[N=k] &= \int_0^\infty P[N=k|R=r]f_R(r)dr \\
&= \int_0^\infty \frac{r^k}{k!}e^{-r}\frac{\lambda(\lambda r)^{\alpha-1}}{\Gamma(\alpha)}e^{-\lambda r}dr \\
&= \frac{\lambda^\alpha}{k!\Gamma(\alpha)}\int_0^\infty r^{k+\alpha-1}e^{-(1+\lambda)r}dr \qquad \text{let } t=(1+\lambda)r \\
&= \frac{\lambda^\alpha}{k!\Gamma(\alpha)}\frac{1}{(1+\lambda)^{k+\alpha}}\underbrace{\int_0^\infty t^{k+\alpha-1}e^{-t}dt}_{\Gamma(k+\alpha)} \\
&= \underbrace{\frac{\Gamma(k+\alpha)}{\Gamma(\alpha)k!}}_{\substack{\text{generalization}\\ \text{of Binomial}\\ \text{Coefficient}}}\quad \underbrace{\left(\frac{\lambda}{1+\lambda}\right)^\alpha\left(\frac{1}{1+\lambda}\right)^k}_{\substack{\text{form of}\\ \text{Binomial}\\ \text{dist.}}}
\end{aligned}$$

N is called the generalized Binomial RV.

$$\begin{aligned}
E[N] &= \int_0^\infty \mathcal{E}[N|r]f_R(r)dr = \int_0^\infty r f_R(r)dr = \mathcal{E}[R] = \frac{\alpha}{\lambda} \\
\mathcal{E}[N^2] &= \int_0^\infty \mathcal{E}[N^2|r]f_R(r)dr = \int_0^\infty (r+r^2)f_R(r) = \mathcal{E}[R]+\mathcal{E}[R^2] \\
VAR[N] &= \mathcal{E}[R^2]+\mathcal{E}[R] = \mathcal{E}[R]^2 = VAR[R]+\mathcal{E}[R] \\
&= \frac{\alpha}{\lambda^2}+\frac{\alpha}{\lambda}
\end{aligned}$$

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5.89

(a)

$$P[X=+1 \mid y<Y\le y+dy] = \frac{P[X=+1,\ y<Y\le y+dy]}{f_Y(y)\,dy}$$

$$= \frac{f_Y(y|+1)\,p\,dy}{f_Y(y)\,dy} = p\,\frac{f_Y(y|+1)}{f_Y(y)}$$

$\sigma^2 = \frac{1}{4}$

$$= p\,\frac{e^{-(y-1)^2/2\sigma^2}}{p\,e^{-(y-1)^2/2\sigma^2} + (1-p)\,e^{-(y+1)^2/2\sigma^2}}$$

$$= \frac{p}{p + (1-p)\,e^{-2[(y+1)^2-(y-1)^2]}}$$

$$= \frac{p}{p + q\,e^{-8y}}$$

(b) +1 is more probable if

$$\frac{p}{p+q\,e^{-8y}} > \frac{1}{2} \iff 1 + \frac{q}{p}e^{-8y} < 2$$

$$\iff y > -\frac{1}{8}\ln\frac{p}{q} = \gamma$$

(c)

$$P[\text{error}] = P[\text{error}\mid +1]\,p + P[\text{error}\mid -1]\,(1-p)$$

$$= p\int_{-\infty}^{\gamma} f_Y(y|+1)\,dy + q\int_{\gamma}^{\infty} f_Y(y|-1)\,dy$$

$$= p\left(1 - Q\left(\frac{\gamma-1}{\sigma}\right)\right) + q\,Q\left(\frac{\gamma+1}{\sigma}\right)$$

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5.8 Functions of Two Random Variables

5.90

X_1 exponential $\frac{1}{\lambda} = 100$

X_2 Rayleigh $E[X_2] = \alpha\sqrt{\pi/2} = 100$

$T = \min(X_1, X_2)$

(a) $P[T > t] = P[X > t]P[Y > t] = e^{-\lambda t} e^{-\alpha t^2}$

$P[X \le t] = 1 - e^{-\lambda t} e^{-\alpha t^2}$

(b) $P[T - t_0 > t \mid T > t_0] = \dfrac{P[T > t + t_0]}{P[T > t_0]}$ $t_0 = 100$ where $T - t_0$ is the additional time

$= P[T > t + t_0 \mid T > t_0] = \dfrac{e^{-\lambda(t+t_0) - \alpha(t+t_0)^2}}{e^{-\lambda t_0 - \alpha t_0^2}} = e^{-\lambda t - \alpha(t+t_0)^2 + \alpha t_0^2}$ $t > 0$

(c) $P[T > t + t_0 \mid T > t_0]$ given above where T is the total time

5.91

(a) $T = \max(X_1, X_2)$

$P[T \le t] = P[X_1 \le t]P[X_2 \le t] = (1 - e^{-\lambda t})(1 - e^{-\alpha t^2})$

$t_0 = 100$

(b) $P[T > t + t_0 \mid T > t_0] = \dfrac{P[T > t + t_0]}{P[T > t_0]}$

$= \dfrac{1 - (1 - e^{-\lambda(t+t_0)})(1 - e^{-\alpha(t+t_0)^2})}{1 - (1 - e^{-\lambda t_0})(1 - e^{-\alpha t_0^2})}$

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5.92

(a) $p_M(m)$

$= P[M=m]$

$= P[K+N=m]$

$= \sum_{n=0}^{m} P[N=n]\,P[K=m-n]$

$= \sum_{n=0}^{m} p_N(n)\,p_K(m-n)$

(b) $p_M(m) = \sum_{t=0}^{m} p_N(t)\,p_K(m-t)$

$= \sum_{t=0}^{m} \binom{n}{t} p^t (1-p)^{n-t} \binom{k}{m-t} p^{m-t} (1-p)^{k-m+t}$

$= \sum_{t=0}^{m} \binom{n}{t}\binom{k}{m-t} p^m (1-p)^{k+n-m} = p^m(1-p)^{k+n-m} \sum_{t=0}^{m} \binom{n}{t}\binom{k}{m-t}$

$= \binom{n+k}{m} p^m (1-p)^{k+n-m}$ Also binomial

$\sum_{t=0}^{m} \binom{n}{t}\binom{k}{m-t} = \binom{n+k}{m}$

(c) $p_M(m) = \sum_{t=0}^{m} p_N(t)\,p_K(m-t)$

$= \sum_{t=0}^{m} \frac{\alpha_1^t}{t!} e^{-\alpha_1} \cdot \frac{\alpha_2^{m-t}}{(m-t)!} e^{-\alpha_2}$

$= \frac{\alpha_2^m e^{-(\alpha_1+\alpha_2)}}{m!} \sum_{t=0}^{m} \frac{m!}{t!(m-t)!} \left(\frac{\alpha_1}{\alpha_2}\right)^t$

$\sum_{t=0}^{m} \frac{m!}{t!(m-t)!} \left(\frac{\alpha_1}{\alpha_2}\right)^t = \left(1+\frac{\alpha_1}{\alpha_2}\right)^m$

$= \frac{(\alpha_1+\alpha_2)^m}{m!} e^{-(\alpha_1+\alpha_2)}$ Also Poisson

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5.93

$$E[X] = \frac{1-p}{p} = 2 \qquad E[Y] = \frac{1-q}{q} = 4$$

$$p = \frac{1}{3} \qquad q = \frac{1}{5}$$

(a) $p_Z(z) = P[Z = z]$

$= P[X - Y = z]$

for $z \geq 0$:

$$p_Z(z) = \sum_{k=0}^{\infty} \sum_{l=0}^{k-z} \frac{1}{3}\left(\frac{2}{3}\right)^k \cdot \frac{1}{5}\left(\frac{4}{5}\right)^l$$

$$= \begin{cases} \sum_{k=0}^{\infty} \sum_{l=0}^{k-z} \frac{1}{15}\left(\frac{2}{3}\right)^k \left(\frac{4}{5}\right)^l, & z \geq 0 \\ \sum_{k=0}^{\infty} \sum_{l=0}^{k-z} \frac{1}{15}\left(\frac{4}{5}\right)^k \left(\frac{2}{3}\right)^l, & z < 0 \end{cases}$$

(b) P["Bulldogs beat Flames"]

$= P[X > Y]$

$$= \sum_{k=1}^{\infty} \sum_{l=0}^{k-1} \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^k \left(\frac{1}{5}\right)\left(\frac{4}{5}\right)^l$$

$$= \sum_{k=1}^{\infty} \frac{1}{15}\left(\frac{2}{3}\right)^k \left(\frac{1 - \frac{4}{5}^k}{1/5}\right)$$

$$= \frac{1}{3}\left(\sum_{k=1}^{\infty}\left(\frac{2}{3}\right)^k - \sum_{k=1}^{\infty}\left(\frac{8}{15}\right)^k\right)$$

$$= \frac{1}{3}\left(2 - \frac{8}{7}\right)$$

$$= \frac{2}{7}$$

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P5.93 cont

$$P[\text{"tie"}] = P[X=Y]$$

$$= \sum_{k=0}^{\infty} \frac{1}{3}\left(\frac{2}{3}\right)^k \left(\frac{1}{5}\right)\left(\frac{4}{5}\right)^k$$

$$= \frac{1}{15} \sum_{k=0}^{\infty} \left(\frac{8}{15}\right)^k$$

$$= \frac{1}{15}\left(\frac{15}{7}\right)$$

$$= \frac{1}{7}$$

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5.94

(a) $p_T(t) = P[T=t]$

$$= \begin{cases} \binom{t}{2} p^2 (1-p)^{t-2} & , t \geq 2 \\ 0 & , \text{otherwise} \end{cases}$$

(b) $p_W(w) = (1-p)^{w-1} p \quad , w \geq 1$

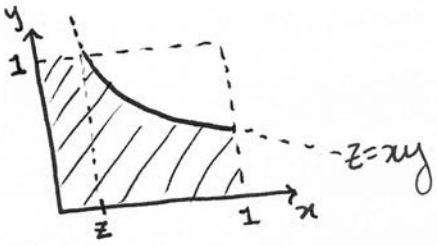
5.95

$Z = XY \quad f_{XY}(x,y) = 1 \quad 0 \leq x \leq 1, \; 0 \leq y \leq 1$

$F_Z(z) = P[Z \leq z]$

$= P[XY \leq z]$

$= P[Y \leq z/X]$



$= f_{XY}(x,y) \times$ shaded area

$$= z + \int_z^1 \int_0^{z/x} dy\,dx$$

$$= z + \int_z^1 z/x \, dx$$

$$= z + [z \ln x]_z^1$$

$$= z - z \ln z$$

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

$$= 1 - \ln z - \frac{z}{z}$$

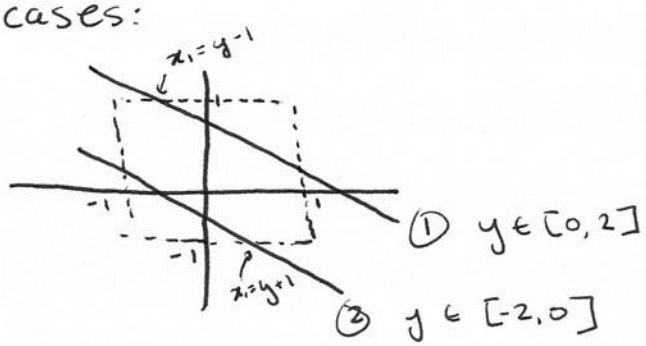
$$= \begin{cases} -\ln z & , 0 \leq z \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

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5.96

(a) $f_{X_1X_2}(x_1, x_2) = \frac{1}{4}$ $-1 \le x_1 \le 1$, $-1 \le x_2 \le 1$

2 cases:



case ①:

$$F_Y(y) = P[X_1 + X_2 \le y]$$
$$= P[X_2 \le y - X_1]$$
$$= \frac{1}{4}(2y) + \int_{y-1}^{1} \int_{-1}^{y-x_1} \frac{1}{4}\, dx_2\, dx_1$$
$$= \frac{y}{2} + \frac{1}{4}\int_{y-1}^{1} (y - x_1 + 1)\, dx_1$$
$$= \frac{y}{2} + \frac{1}{4}\left[yx_1 - \frac{1}{2}x_1^2 + x_1\right]_{y-1}^{1}$$
$$= \frac{y}{2} + \frac{y}{4} + \frac{3}{8} - \frac{y^2}{4} + \frac{y^2}{8} - \frac{y}{4} + \frac{1}{8}$$
$$= -\frac{y^2}{8} + \frac{y}{2} + \frac{1}{2}$$

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case ②

$$F_Y(y) = \frac{1}{4}\int_{-1}^{y+1}\int_{-1}^{y-x_1} dx_2\,dx_1$$

$$= \frac{1}{4}\int_{-1}^{y+1}(y-x_1+1)\,dx_1$$

$$= \frac{1}{4}\left[x_1 y - \frac{1}{2}x_1^2 + x_1\right]_{-1}^{y+1}$$

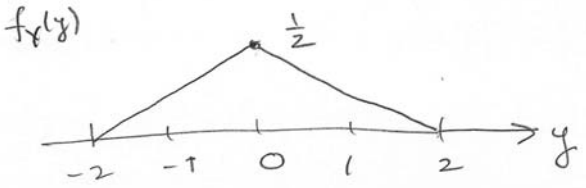
$$= \frac{1}{4}\left(y(y+1) - \frac{1}{2}(y+1)^2 + y + 1 + y + \frac{1}{2} + 1\right)$$

$$= \frac{1}{4}\left(y^2 + y - \frac{1}{2}y^2 - y - \frac{1}{2} + y + 1 + y + \frac{1}{2} + 1\right)$$

$$= \frac{1}{8}y^2 + \frac{1}{2}y + \frac{1}{2}$$

$$\therefore F_Y(y) = \begin{cases} -\frac{1}{8}y^2 + \frac{1}{2}y + \frac{1}{2} & , 0 \le y \le 2 \\ \frac{1}{8}y^2 + \frac{1}{2}y + \frac{1}{2} & , -2 \le y < 0 \\ 1 & , y > 2 \\ 0 & , y < -2 \end{cases}$$

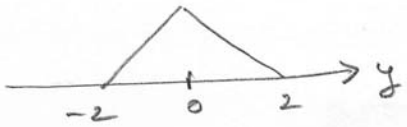
$$f_Y(y) = \frac{d}{dy}F_Y(y) = \begin{cases} -\frac{1}{4}y + \frac{1}{2} & , 0 \le y \le 2 \\ \frac{1}{4}y + \frac{1}{2} & , -2 \le y < 0 \\ 0 & , \text{otherwise} \end{cases}$$

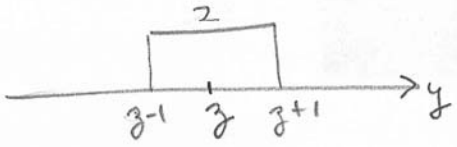


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(b) Use the convolution method

$$f_Z(z) = \int_{-\infty}^{\infty} f_Y(y) f_{X_3}(z-y)\,dy$$





for $z+1<-2$ $\quad f_Z(z)=0$

for $z+1<0$

$$f_Z(z) = \int_{-2}^{z+1} \left(\frac{1}{2}+\frac{1}{4}y\right) dy = \left[\frac{1}{2}y + \frac{1}{4}\frac{y^2}{2}\right]_{-2}^{z+1}$$

$$= \frac{1}{2}(z+1) + 1 + \frac{1}{8}(z+1)^2 - \frac{1}{2} = \frac{1}{8}(z+1)^2 + \frac{1}{2}(z+1) + \frac{1}{2}$$

for $0 < z+1 < 2$

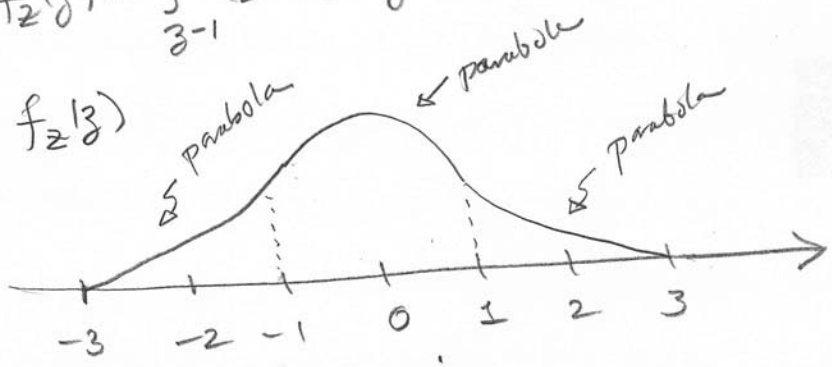
$$f_Z(z) = \int_{z-1}^{0} \left(\frac{1}{2}+\frac{1}{4}y\right) dy + \int_{0}^{z+1} \left(\frac{1}{2}-\frac{1}{4}y\right) dy$$

$$= -\frac{1}{2}(z-1) - \frac{1}{4}\frac{(z-1)^2}{2} + \frac{1}{2}(z+1) - \frac{1}{4}\frac{(z+1)^2}{2}$$

$$= 1 - \frac{1}{8}(z-1)^2 - \frac{1}{8}(z+1)^2 = 1 - \frac{1}{4}(1+z^2) = \frac{3}{4} - \frac{1}{4}z^2$$

for $2 < z+1 < 4$

$$f_Z(z) = \int_{z-1}^{2} \left(\frac{1}{2}-\frac{1}{4}y\right) dy = \frac{1}{2}(2-z+1) - \frac{1}{4}\frac{1}{2}\left(4-(z-1)^2\right)$$



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5.97

$$f_X(x) = \frac{\lambda^{\alpha_1} x^{\alpha_1 - 1}}{\Gamma(\alpha_1)} e^{-\lambda x} \qquad f_Y(y) = \frac{\lambda^{\alpha_2} x^{\alpha_2 - 1}}{\Gamma(\alpha_2)} e^{-\lambda x}$$

$$Z = X + Y$$

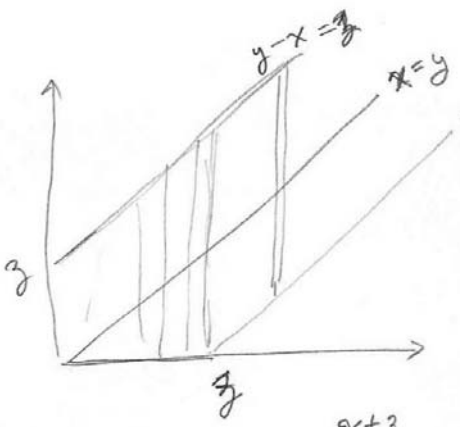
$$f_Z(z) = \underbrace{\frac{\lambda^{\alpha_1+\alpha_2}}{\Gamma(\alpha_1)\Gamma(\alpha_2)}}_{c} \int_0^z x^{\alpha_1 - 1} e^{-\lambda x} (z-x)^{\alpha_2 - 1} e^{-\lambda(z-x)} dx$$

$$= c e^{-\lambda z} \underbrace{\int_0^z x^{\alpha_1 - 1} (z-x)^{\alpha_2 - 1} dx}_{\frac{z^{\alpha_1+\alpha_2-1}\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1+\alpha_2-1)}} = \frac{\lambda^{\alpha_1+\alpha_2} z^{\alpha_1+\alpha_2-1}}{\Gamma(\alpha_1+\alpha_2-1)}$$

Gamma pdf

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5.98



$$Z = |X-Y| = \begin{cases} X-Y & X>Y \\ Y-X & X<Y \end{cases}$$

(a)
$$P[Z \le z] = \int_0^z dx\, e^{-x} \int_0^{x+z} e^{-y}\, dy + \int_z^\infty dx\, e^{-x} \int_{x-z}^{x+z} e^{-y}\, dy$$

$$= \int_0^z dx\, e^{-x}\underbrace{\left(1-e^{-(x+z)}\right)}_{e^{-x}-e^{-2x-z}} + \int_z^\infty dx\, e^{-x}\underbrace{\left(e^{-(x-z)}-e^{-(x+z)}\right)}_{e^{-2x+z}-e^{-2x-z}}$$

$$= \int_0^z e^{-x}\,dx - \int_0^\infty e^{-2x-z}\,dx + \int_z^\infty e^{-2x+z}\,dx$$

$$= 1-e^{-z} - \tfrac{1}{2}e^{-z} + \tfrac{1}{2}e^{z}e^{-2z} = 1-e^{-z}$$

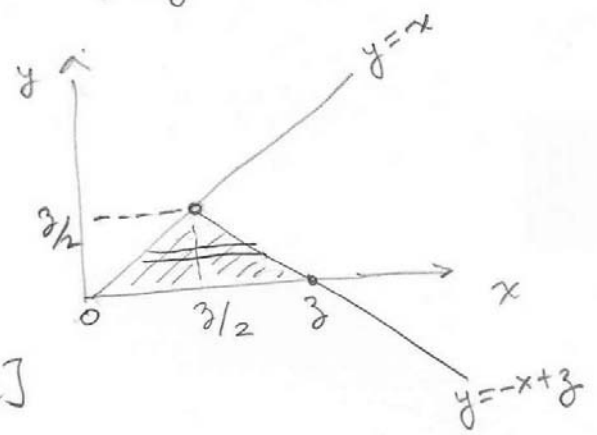
(b)
$$E[Z] = \int_0^\infty P[Z>z]\,dz = \int_0^\infty e^{-z}\,dz = 1 \checkmark$$

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5.99

$f_{XY}(x,y) = e^{-(x+y)} \quad 0 < y < x < 1$

$Z = X + Y$



$$P[Z \le z] = P[X+Y \le z]$$

$$= P[Y \le -X+z]$$

$$= 2\int_0^{z/2} dy\, e^{-y} \int_y^{z-y} e^{-x}\, dx = \int_0^{z/2} dy\, e^{-y}\left[-e^{-x}\right]_y^{z-y}$$

$$= 2\int_0^{z/2} dy\, e^{-y}\left[e^{-y} - e^{y-z}\right]$$

$$= 2\int_0^{z/2} \left[e^{-2y} - e^{-z}\right] dy$$

$$= 2\left\{-\frac{e^{-2y}}{2}\Big|_0^{z/2} - e^{-z}[y]_0^{z/2}\right\}$$

$$= 2\left\{\frac{1}{2} - \frac{1}{2}e^{-z} - e^{-z}\frac{z}{2}\right\}$$

$$= 1 - e^{-z} - ze^{-z}$$

$$f_Z(z) = e^{-z} - e^{-z} + ze^{-z} = ze^{-z} \qquad z > 0$$

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5.100

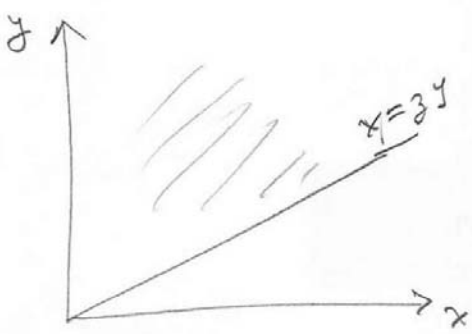
X, Y Rayleigh r.v. $\alpha = \beta = 1$

$Z = X/Y$

for $z > 0$

$$P[Z \le z] = P[X/Y \le z]$$

$$= P[X \le zY]$$



$$P[Z \le z] = \int_0^\infty dx\, x e^{-x^2/2} \int_{x/z}^\infty y e^{-y^2/2}\, dy$$

$$= \int_0^\infty dx\, x e^{-x^2/2} \left[e^{-(x/z)^2/2} \right]$$

$$= \int_0^\infty dx\, x\, e^{-\frac{1}{2}\left(1+\frac{1}{z^2}\right)x^2}$$

$$= \frac{1}{\left(1+\frac{1}{z^2}\right)} = \frac{z^2}{z^2+1}$$

$$f_Z(z) = \frac{2z}{(1+z^2)^2}$$

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5.101

$$f_{XY}(x,y) = \frac{1}{2\pi} e^{-\left(\frac{x^2}{2}+\frac{y^2}{2}\right)}$$

$Z = X/Y$

$$f_Z(z) = \int_{-\infty}^{\infty} |y| f(zy, y)\, dy$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} |y| e^{-\left(\frac{y^2}{2}+\frac{z^2y^2}{2}\right)} dy$$

$$= \frac{1}{\pi} \int_{0}^{\infty} y e^{-y^2\left(\frac{1}{2}+\frac{1}{2}z^2\right)} dy$$

but $$\int_0^{\infty} y e^{-ay^2} dy = \left[-\frac{1}{2a} e^{-ay^2}\right]_0^{\infty} = \frac{1}{2a}(0-(-1)) = \frac{1}{2a}$$

So

$$f_Z(z) = \frac{1}{\pi} \cdot \frac{1}{2\left(\frac{1}{2}+\frac{1}{2}z^2\right)}$$

$$= \frac{1}{\pi(1+z^2)}$$

$\therefore$ Z is a Cauchy RV w/ $\alpha = 1$ //

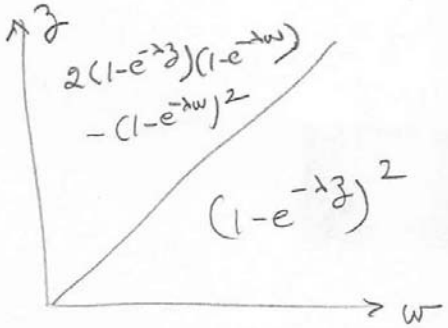
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5.103

$$F_{XY}(x,y) = (1-e^{-\lambda x})(1-e^{-\lambda y}) \quad x>0 \; y>0$$

If $z < w$ then

$$F_{WZ}(w,z) = F_{XY}(z,z) = (1-e^{-\lambda z})^2$$



If $z > w$ then from Ex. 5.43

$$F_{WZ}(w,z) = F_{XY}(w,z) + F_{XY}(z,w) - F_{XY}(w,w)$$

$$= 2(1-e^{-\lambda z})(1-e^{-\lambda w}) - (1-e^{-\lambda w})^2$$

Check: find marginal cdf of $W = \min(X,Y)$.

$$\lim_{z\to\infty} F_{WZ}(w,z) = 2(1-e^{-\lambda w}) - (1-e^{-\lambda w})^2$$

$$= 1 - e^{-2\lambda w} \qquad w > 0$$

exponential w. rate 2λ

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5.104

$$F_{XY}(x,y) = \left(1-\left(\frac{x_m}{x}\right)^{\alpha}\right)\left(1-\left(\frac{x_m}{y}\right)^{\alpha}\right) \qquad x > x_m,\ y > x_m$$

If $z < w$ then

$$F_{WZ}(w,z) = F_{XY}(z,z) = \left(1-\left(\frac{x_m}{z}\right)^{\alpha}\right)^2$$

If $z > w$ then

$$F_{WZ}(w,z) = 2\left(1-\left(\frac{x_m}{z}\right)^{\alpha}\right)\left(1-\left(\frac{x_m}{w}\right)^{\alpha}\right) - \left(1-\left(\frac{x_m}{z}\right)^{\alpha}\right)^2$$

Let's look at marginals of W and Z

$$F_W(w) = \lim_{z\to\infty} F_{WZ}(w,z) = 2\left(1-\left(\frac{x_m}{w}\right)^{\alpha}\right) - 1$$

$$= 1-\left(\frac{x_m}{w}\right)^{\alpha}$$

the minimum is also Pareto with the same parameter!

The marginal for the max is

$$\lim_{w\to\infty} F_{WZ}(w,z) = \left(1-\left(\frac{x_m}{z}\right)^{\alpha}\right)^2$$

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5.105

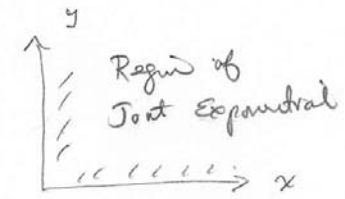
(a) $\begin{bmatrix} W \\ Z \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} w \\ z \end{bmatrix}$$

$$x = \frac{w+z}{2} \qquad y = \frac{w-z}{2}$$

$$\therefore f_{WZ}(w,z) = f_{XY}\left(\frac{w+z}{2}, \frac{w-z}{2}\right)$$



(b) $f_{WZ}(w,z) = f_X\left(\frac{w+z}{2}\right) f_Y\left(\frac{w-z}{2}\right)$

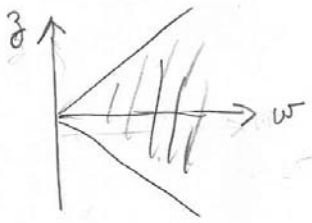
$$= e^{-\frac{(w+z)}{2}} \cdot e^{-\frac{(w-z)}{2}}$$

$$= e^{-w}$$

$w > z$

$w > -z$

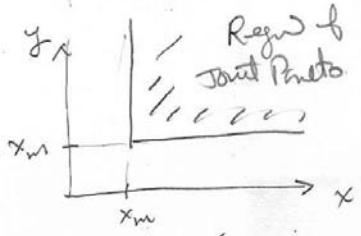
for $w > 0$ $\quad -w < z < w$



(c) $f_{WZ}(w,z) = f_X\left(\frac{w+z}{2}\right) f_Y\left(\frac{w-z}{2}\right)$

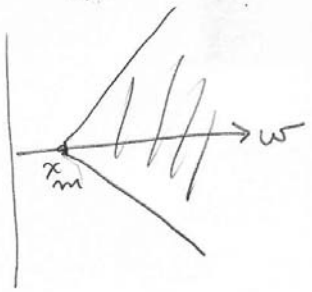
$$= k\frac{x_m^k}{\left(\frac{w+z}{2}\right)^{k+1}} \cdot k\frac{x_m^k}{\left(\frac{w-z}{2}\right)^{k+1}}$$

$$= \frac{k^2 x_m^{2k}}{\left(\frac{w^2-z^2}{4}\right)^{k+1}}$$



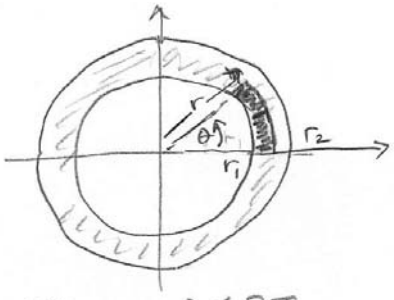
$w > x_m$

$w + z > x_m \qquad w - z > x_m$



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5.106



$r_1 < r < r_2 \quad 0 < \theta < 2\pi$

$$P[R \le r, \Theta \le \theta] = \frac{(\pi r^2 - \pi r_1^2)}{\pi r_2^2 - \pi r_1^2} \frac{\theta}{2\pi}$$

$$\frac{\partial^2}{\partial r \partial \theta} F_{R,\Theta}(r,\theta) = \frac{\partial}{\partial r} \frac{1}{2\pi} \left(\frac{\pi r^2 - \pi r_1^2}{\pi r_2^2 - \pi r_1^2} \right)$$

$$= \frac{1}{2\pi} \frac{2r}{r_2^2 - r_1^2} \qquad r_1 < r < r_2 \quad 0 < \theta < 2\pi$$

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5.107

(a) $\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} a & b \\ c & e \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$

then, $\begin{bmatrix} X \\ Y \end{bmatrix} = \frac{1}{|ae-bc|} \begin{bmatrix} e & -b \\ -c & a \end{bmatrix} \begin{bmatrix} V \\ W \end{bmatrix}$

$$f_{VW} = f_{XY}\left(\frac{ev-bw}{|ae-bc|}, \frac{-cv+aw}{|ae-bc|}\right)$$

$$= f_X\left(\frac{ev-bw}{|ae-bc|}\right) f_Y\left(\frac{-cv+aw}{|ae-bc|}\right)$$

where f_X and f_Y Gaussian pdf with $\mu = 0$ and $\sigma = 1$.

(b) The matrix A will not invertible if its rows are linearly dependent, that is,

$V = aX + bY$

$W = kcX + kbY$

which implies that $W = kV$

∴ W is simply a constant multiple of V

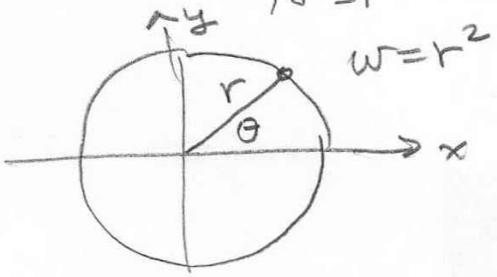
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5.108

X, Y indep, jointly Gaussian RV's $m=0, \sigma^2=1$

$W = X^2 + Y^2$

$\Theta = \tan^{-1} Y/X$



$$P[W \le w, \Theta \le \theta_0] = \iint \frac{1}{2\pi} e^{-(x^2+y^2)/2} \, dx\,dy$$

$$\{(x,y): x^2+y^2 \le r, \text{ & } \tan^{-1}(\tfrac{y}{x}) < \theta_0\}$$

$$= \int_0^{\sqrt{w}} \int_0^{\theta_0} \frac{e^{-r^2/2}}{2\pi} \, r\,dr\,d\theta$$

$$= \frac{\theta_0}{2\pi}\left(1 - e^{-w/2}\right)$$

$0 < \theta_0 < 2\pi$

$0 < r_0 < \infty$

$$F_W(w) = 1 - e^{-w/2} \qquad w > 0$$

$$F_\Theta(\theta) = \frac{1}{2\pi} \qquad 0 < \theta \le 2\pi$$

W is an exponential RV with mean 2

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5.109

$$X = \left(2\ln\frac{2\pi}{\xi_1}\right)^{1/2}\cos\xi_2 \qquad Y = \left(2\ln\frac{2\pi}{\xi_1}\right)^{1/2}\sin\xi_2$$

$$R = \sqrt{X^2+Y^2} \qquad \Theta = \tan^{-1} Y/X$$

Consider Θ first:

$$\Theta = \tan^{-1}\frac{Y}{X} = \tan^{-1}\left(\frac{\sin\xi_2}{\cos\xi_2}\right) = \tan^{-1}(\tan\xi_2) = \xi_2$$

But ξ_2 is uniform in $[0, 2\pi)$ $\Rightarrow$ Θ also uniform in $[0, 2\pi)$

Now consider R

$$R = \sqrt{X^2+Y^2} = \sqrt{2\ln\left(\frac{2\pi}{\xi_1}\right)\cos^2\xi_2 + 2\ln\left(\frac{2\pi}{\xi_1}\right)\sin^2\xi_2}$$

$$= \sqrt{2\ln\left(\frac{2\pi}{\xi_1}\right)}$$

ξ_1 is indep of ξ_2 $\Rightarrow$ R and Θ are independent

$$F_R(r) = P[R<r] = P\left[\sqrt{2\ln\frac{2\pi}{\xi_1}} < r\right]$$

$$= P\left[\frac{2\pi}{\xi_1} < e^{r^2/2}\right] = P\left[2\pi e^{-r^2/2} < \xi_1\right]$$

$$= 1 - \frac{2\pi e^{-r^2/2}}{2\pi} = 1 - e^{-r^2/2}$$

R is Rayleigh, Θ uniform $(0, 2\pi)$ as in Ex. 5.44

$\therefore$ X and Y are indep jointly Gaussian RVs with zero mean + unit variance

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5.9 Pairs of Jointly Gaussian Random Variables

5.110

$$f_{XY}(x,y) = \frac{e^{-(2x^2+y^2/2)}}{2\pi c}$$

The sol'n involves matching the coefficients of the polynomial in the exponent of the Gaussian pdf.

coeff. of $x^2 \Rightarrow \frac{1}{2(1-\rho^2)\sigma_1^2} = 2$.

coeff. of $y^2 \Rightarrow \frac{1}{2(1-\rho^2)\sigma_2^2} = \frac{1}{2}$

coeff of $xy \Rightarrow \frac{-2\rho}{2(1-\rho^2)\sigma_1\sigma_2} = 0 \quad \therefore \rho = 0$

$$\sigma_1^2 = \frac{1}{2(1-0^2)\cdot 2} = \frac{1}{4} \;//$$

$$\sigma_2^2 = \frac{1}{2(1-0^2)\cdot \frac{1}{2}} = 1 \;//$$

$\therefore$ $Cov(X,Y) = \rho_{XY}\sigma_1\sigma_2 = 0$

$VAR[X] = \frac{1}{4}$

$VAR[Y] = 1$

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5.111

$$x^2 + 4y^2 - 3y(x-1) - 2x + 1 =$$

$$= (x-1)^2 - 3(x-1)y + 4y^2$$

$$= \left(\frac{x-1}{1}\right)^2 - 2\left(\frac{3}{4}\right)\left(\frac{x-1}{1}\right)\left(\frac{y}{1/2}\right) + \left(\frac{y}{1/2}\right)^2$$

$m_1 = 1$ $\quad \rho = \frac{3}{4}$ $\quad m_2 = 0$

$\sigma_1 = 1$ $\quad\quad\quad\quad\quad\quad \sigma_2 = 1/2$

5.112 $m_Y = 0 \quad \sigma_1 = 1 \quad \sigma_2 = 2$

$$E[X|Y] = Y/4 + 1$$

$$E[X|Y] = Y/4 + 1 = \rho\frac{\sigma_1}{\sigma_2}(y - m_2) + m_1$$

$$\frac{1}{2} \qquad 0 \qquad +1$$

$$\Rightarrow \rho = \frac{1}{2}$$

$$f_{XY}(x,y) = \frac{\exp\left\{\frac{-1}{2(3/4)}\left[\left(\frac{x-1}{1}\right)^2 - 2\,\frac{1}{2}\left(\frac{x-1}{1}\right)\left(\frac{y}{2}\right) + \left(\frac{y}{2}\right)^2\right]\right\}}{4\pi\sqrt{3/4}}$$

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5.113
~~4.78 a)~~

$$\begin{aligned} P[\sqrt{X^2+Y^2} \le r] &= \int\int_{x^2+y^2 \le r^2} \frac{e^{-(x^2+y^2)/2}}{2\pi} dxdy \\ &= \int_0^{2\pi}\int_0^r \frac{e^{-r^2/2}}{2\pi} r\,dr\,d\theta \\ &\quad \text{let } x = r\cos\theta,\ y = r\sin\theta \\ &= 1 - e^{-r^2/2} = \frac{1}{2} \\ &\quad \text{Note: } \sqrt{X^2+Y^2} \text{ has a Rayleigh dist.} \\ \Rightarrow r &= \sqrt{2\ln 2} \end{aligned}$$

b) $P[\sqrt{X^2+Y^2} > r] = e^{-r^2/2}$ from above.

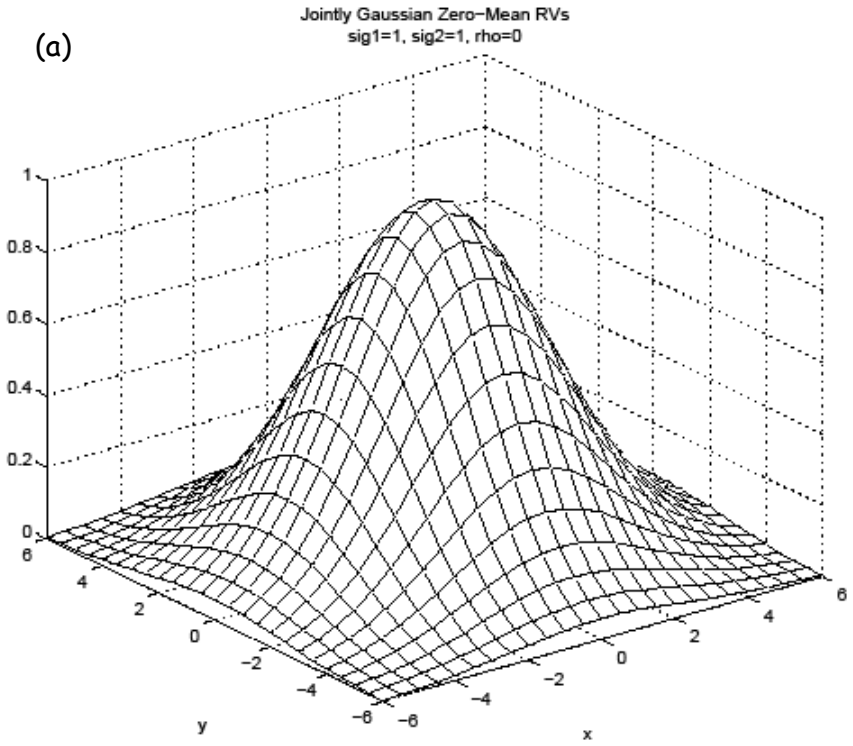
For (x,y) such that $x^2+y^2 > r^2$

$$\begin{aligned} f_{XY}(x,y|R>r)dxdy &= P[x < X \le x+dx, y < Y \le y+dy | R > r] \\ &= \frac{f_{XY}(x,y)dxdy}{P[R>r]} \\ \Rightarrow f_{XY}(x,y|R>r) &= \frac{f_{XY}(x,y)}{P[R>r]} \\ &= \frac{e^{-(x^2+y^2-r^2)/2}}{2\pi} \end{aligned}$$

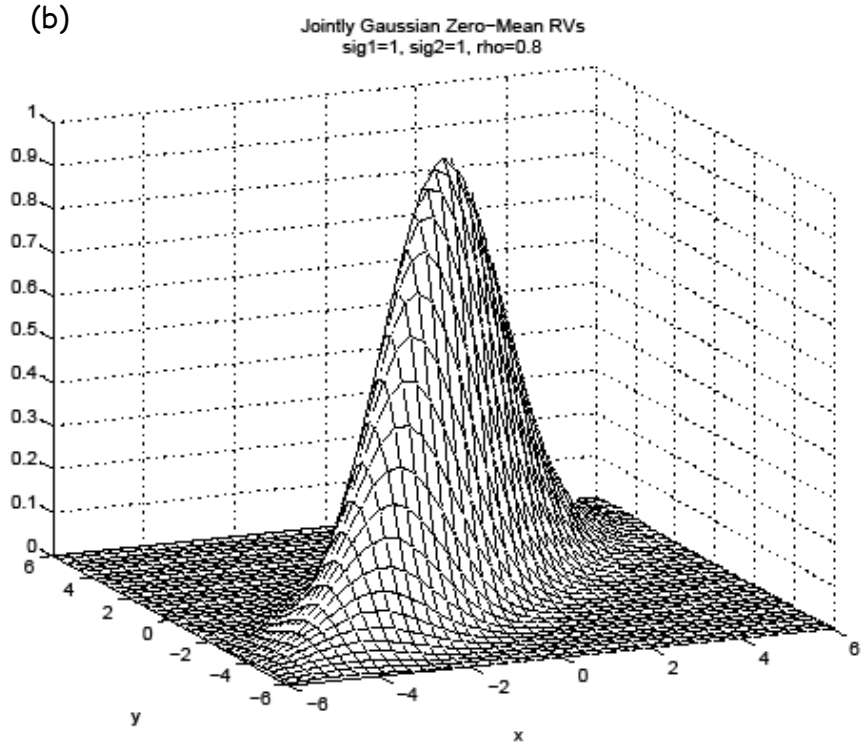
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5.114

(a)

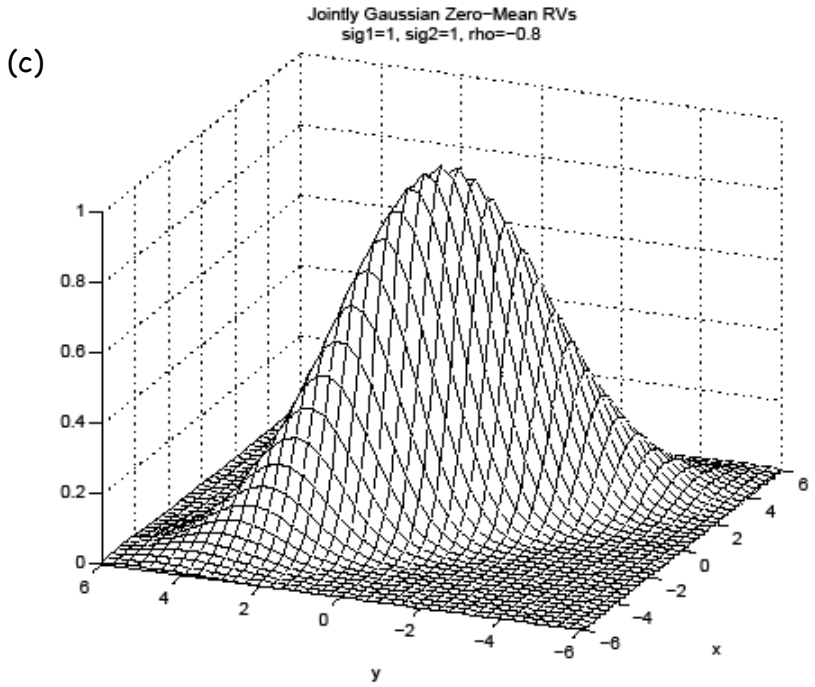


(b)

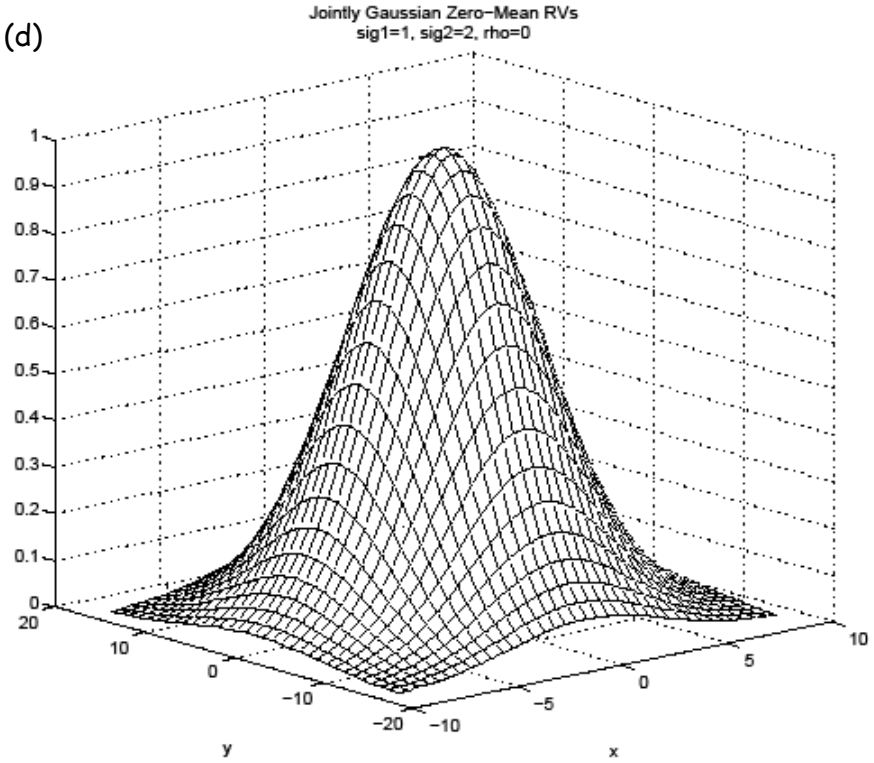


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(c)

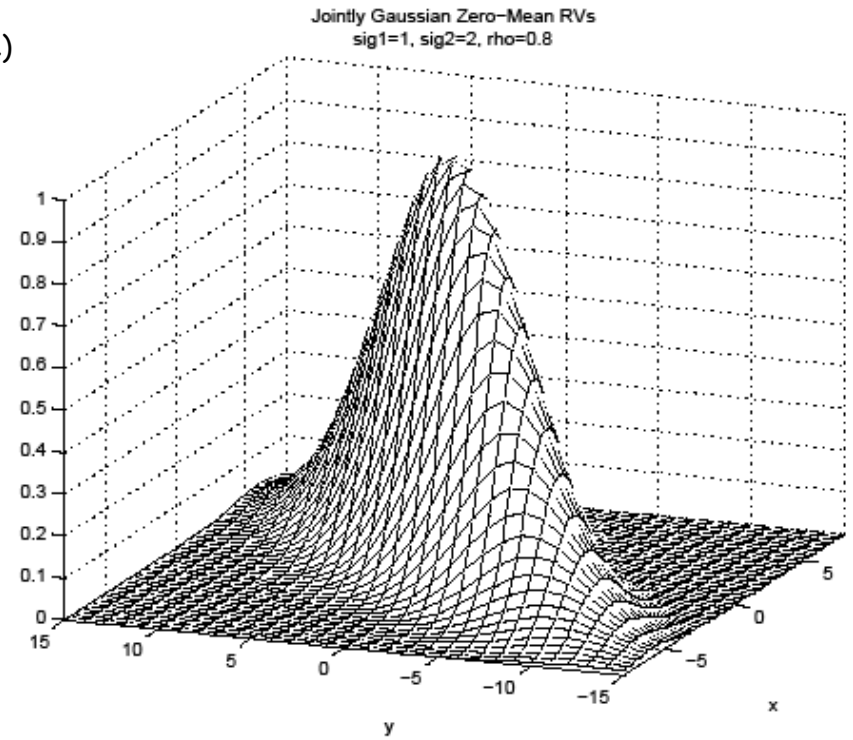


(d)

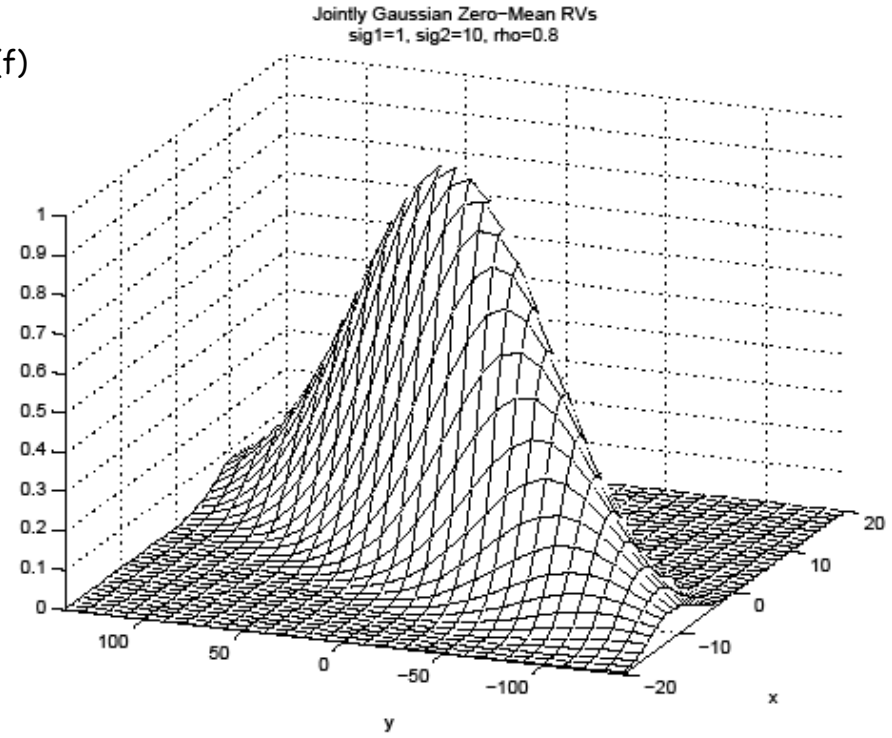


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(e)

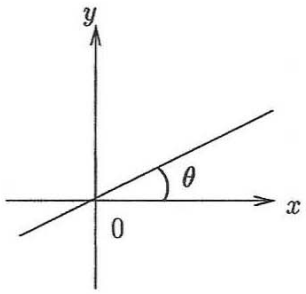


(f)



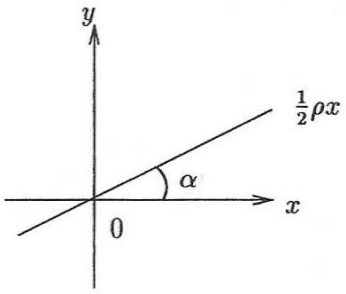
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5.115



$$\theta = \frac{1}{2}\arctan\frac{2\rho\sigma_1\sigma_2}{\sigma_1^2 - \sigma_2^2} = \frac{1}{2}\arctan\frac{4\rho}{3}$$

b)



$$E[Y|X = x] = m_2 + \rho\frac{\sigma_2}{\sigma_1}(x - m_1) = \frac{1}{2}\rho x$$

c) The plots in parts a) and b) are the same only when $\rho = 1$. In this case $E[Y|X = x] = \frac{1}{2}x$, i.e.

$$\tan\alpha = \frac{1}{2}, \qquad \tan 2\alpha = \frac{2\tan\alpha}{1 - \tan^2\alpha} = \frac{4}{3}.$$

$$\therefore \quad \alpha = \frac{1}{2}\arctan\frac{4}{3} = \theta$$

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5.116
~~4.80~~

$$\begin{vmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{vmatrix} = \sigma_X^2\sigma_Y^2(1-\rho^2) = 0 \Rightarrow \rho = \pm 1$$

$$\Rightarrow P[X = \rho Y] = 1 \Rightarrow \text{all probl. mass concentrated along } X = \rho Y \text{ line}$$

Assume $\rho = 1$:

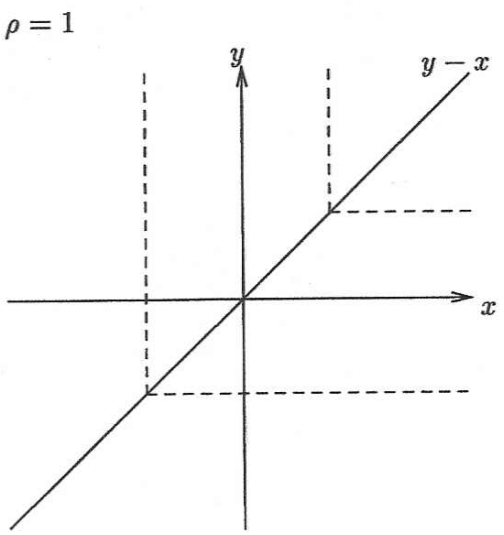
$$\begin{aligned} R_{XY}(x,y) &= P[X \le x, Y \le y] = P[X \le x, X \le y] \\ &= P[\{X \le x\} \cap \{X \le y\}] = P[X \le \min(x,y)] \\ &= \int_{-\infty}^{\min(x,y)} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \end{aligned}$$

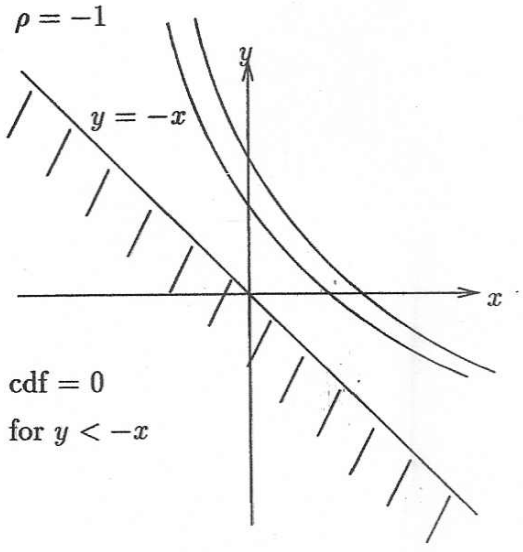
The joint pdf does not exist.

Similarly if $\rho = -1$

$$\begin{aligned} F_{XY}(x,y) &= P[X \le x, -X \le y] = P[X \le x, X \ge -y] \\ &= P[-y \le X \le x] \\ &= \begin{cases} \int_{-y}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt & x \ge -y \Leftrightarrow x + y \ge 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

The joint pdf does not exist.





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5.117 ~~4.81~~

$$h(x,y) = \frac{e^{-(x^2-2\rho_1 xy+y^2)/2(1-\rho_1^2)}}{2\pi\sqrt{1-\rho_1^2}} \qquad g(x,y) = \frac{e^{-(x^2-2\rho_2 xy+y^2)/2(1-\rho_2^2)}}{2\pi\sqrt{1-\rho_2^2}}$$

a) $$f_X(x) = \frac{1}{2}\int_{-\infty}^{\infty} h(x,y)dy + \frac{1}{2}\int_{-\infty}^{\infty} g(x,y)dy = \frac{e^{-x^2/2}}{\sqrt{2\pi}}$$

Similarly

$$f_Y(y) = \frac{e^{-y^2/2}}{\sqrt{2\pi}}$$

$\therefore$ X and Y, individually, are Gaussian RV's.

b) However,

$$f_{XY}(x,y) = \frac{\sqrt{1-\rho_2^2}e^{-(x^2-2\rho_1 xy+y^2)/2(1-\rho_1^2)} + \sqrt{1-\rho_1^2}e^{-(x^2-2\rho_2 xy+y^2)/2(1-\rho_2^2)}}{2\pi\sqrt{1-\rho_1^2}\sqrt{1-\rho_2^2}}$$

does not have the form required for jointly Gaussian RV's.

5.118 ~~4.82~~

$$\begin{aligned}
\mathcal{E}[X^2|y] &= VAR[X|y] + \mathcal{E}[X|y]^2 \quad \text{where we assume } \mathcal{E}[X] = \mathcal{E}[Y] = 0 \\
&= \sigma_X^2(1-\rho^2) + \left(\rho\frac{\sigma_X}{\sigma_Y}y\right)^2 \\
\mathcal{E}[X^2Y^2] &= \mathcal{E}[\mathcal{E}[X^2Y^2|Y]] = \mathcal{E}[Y^2\mathcal{E}[X^2|Y]] \\
&= \mathcal{E}[\sigma_X^2(1-\rho^2)Y^2 + \rho\frac{\sigma_X^2}{\sigma_Y^2}Y^4] \\
&= \sigma_X^2\sigma_Y^2(1-\rho^2) + \rho^2\frac{\sigma_X^2}{\sigma_Y^2}\underbrace{\mathcal{E}[Y^4]}_{3\sigma_Y^4}\} \text{ shown below} \\
&= \sigma_X^2\sigma_Y^2(1+2\rho^2) \\
&= \sigma_X^2\sigma_Y^2 + 2\mathcal{E}[XY] = \mathcal{E}[X^2]\mathcal{E}[Y^2] + 2\mathcal{E}[XY] \\
\mathcal{E}[Y^4] &= \int_{-\infty}^{\infty}\frac{y^4e^{-y^2/2\sigma^2}}{\sqrt{2\pi}\sigma}dy \qquad t = \frac{y}{\sigma} \\
&= \frac{\sigma^5}{\sqrt{2\pi}\sigma}\underbrace{2\int_0^{\infty} t^4e^{-t^2/2}dt}_{\frac{\Gamma\left(\frac{5}{2}\right)}{2\alpha^5}} \qquad \begin{array}{l}\text{from Table in Appendix A} \\ \text{where } \alpha^2 = \frac{1}{2}\end{array} \\
\Gamma\left(\tfrac{5}{2}\right) &= \tfrac{3}{2}\Gamma\left(\tfrac{3}{2}\right) = \tfrac{3}{2}\tfrac{1}{2}\Gamma\left(\tfrac{1}{2}\right) \\
&= \tfrac{3}{4}\sqrt{\pi} \\
&= \frac{\sigma^5}{\sqrt{2\pi}\sigma}\frac{2\frac{3}{4}\sqrt{\pi}}{2\left(\frac{1}{\sqrt{2}}\right)^5} \\
&= 3\sigma^4
\end{aligned}$$

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5.119

$$\begin{aligned} f(x_1,x_2,x_3) &= \exp(-x_1^2 - x_2^2 + \sqrt{2}x_1x_2 - \frac{1}{2}x_3^2)/(2\pi\sqrt{\pi}) \\ &= \frac{\exp(-x_1^2 - x_2^2 + \sqrt{2}x_1x_2)}{\sqrt{2\pi}\cdot\sqrt{\pi}} \frac{\exp(-\frac{1}{2}x_3^2)}{\sqrt{2\pi}} \\ &= \frac{\exp\left\{-\frac{1}{2(1-\frac{1}{2})}[x_1^2 - 2\frac{1}{\sqrt{2}}x_1x_2 + x_2^2]\right\}}{2\pi\sqrt{1-\frac{1}{2}}} \frac{\exp(-\frac{1}{2}x_3^2)}{\sqrt{2\pi}} \end{aligned}$$

We only need to "decorrelate" X_1 and X_2. Try the transform

$$\begin{aligned} A &= \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Note that

$$K = \begin{bmatrix} 1 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Check

$$\begin{aligned} C &= AKA^T \\ &= \begin{bmatrix} 1+\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 1-\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

C is a diagonal matrix

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5.120

$$\hat{X} = \frac{1}{1+\sigma_N^2/\sigma_X^2} Y = cY, \text{ where } Y = X+N$$

$$E[(X-cY)^2] = E[(X-c(X+N))^2]$$

$$= E[((1-c)X - cN)^2]$$

$$= (1-c)^2 E[X^2] - 2(1-c)c\underbrace{E[XN]}_{E[X]E[N]=0} + c^2 E[N^2]$$

$$= (1-c)^2\sigma_X^2 + c^2\sigma_N^2$$

$$= \left(1 - \frac{1}{1+\sigma_N^2/\sigma_X^2}\right)^2 \sigma_X^2 + \left(\frac{1}{1+\frac{\sigma_N^2}{\sigma_X^2}}\right)^2 \sigma_N^2$$

$$= \left(\frac{\sigma_N^2/\sigma_X^2}{1+\sigma_N^2/\sigma_X^2}\right)^2 \sigma_X^2 + \left(\frac{1}{1+\frac{\sigma_N^2}{\sigma_X^2}}\right)^2 \sigma_N^2$$

$$= \frac{\sigma_N^4\sigma_X^2 + \sigma_X^4\sigma_N^2}{(\sigma_X^2+\sigma_N^2)^2}$$

$$= \frac{\sigma_X^2\sigma_N^2}{\sigma_X^2+\sigma_N^2} \rightarrow \begin{cases} \sigma_N^2 \text{ @ Hi SNR} \\ \sigma_X^2 \text{ at low SNR} \end{cases}$$

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5.10 Generating Independent Gaussian Random Variables

5.121 The following Octave code produces the inverse:

```
function z = rayleigh_rnd(s)
     u = rand;
     z = s.*(2.*log(1./(1-u))).^(1/2);
end
```

5.123 The following Octave generates the requested pairs and plot:

```
len = 10000;
X = discrete_rnd([1 -1], [0.5 0.5], 2, len);
N = normal_rnd(0, 1, 2, len);
Y = X + N;
figure;
plot(Y(1,:), Y(2,:), ".");
Xr = sign(Y);
sigerr = (X ~= Xr);
biterr = (sigerr(1,:) | sigerr(2,:));
proberr = sum(biterr)./len
```

5.124 The following Octave generates the requested pairs and plot:

```
X = normal_rnd(0, 2, 1, 1000);
N = normal_rnd(0, 1, 1, 1000);
Y = X + N;
Xr = Y./(1 + 1/2);
err = Xr - X;
figure;
hist(err, [-3:0.25:3], 4);
m = mean(err)
v = var(err)
```

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5.125 The following Octave generates the sequence of X_n and Y_n:

```
X = normal_rnd(0, 1, 1, 1000);
Y = (X + [0 X(1:999)])./2;
figure;
plot(Y(1:999), Y(2:1000), ".");
Z = (X - [0 X(1:999)])./2;
figure;
plot(Z(1:999), Z(2:1000), ".");
```

5.126 The following Octave generates the specified jointly Gaussian random variables:

```
function vw = gaussian_correlate(xy, mu1, mu2, var1, var2, rho)
    sig1 = sqrt(var1);
    sig2 = sqrt(var2);
    K = [var1, rho*sig1*sig2; rho*sig1*sig2, var2];
    [evec eval] = eig(K);
    A = evec * sqrt(eval);
    vw = zeros(size(xy));
    for i = 1:length(xy)
        vw(:,i) = A*xy(:,i) + [mu1; mu2];
    end
end
```

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5.127 The following Octave generates the specified jointly Gaussian random variables and plot:

```
xy = normal_rnd(0, 1, 2, 1000);
vw = gaussian_correlate(xy, 1, -1, 1, 2, -1/2);
plot(vw(1,:), vw(2,:), ".");
```

5.128 The following Octave generates the specified plot:

```
xy = normal_rnd(0, 1, 2, 1000);
muh = 174;
muv = 4.4;
varh = 42.36;
varv = 0.021;
covhv = 0.458;
sigh = sqrt(varh);
sigv = sqrt(varv);
rhohv = covhv/(sigh*sigv);
xy = normal_rnd(0, 1, 2, 1000);
hv = zeros(size(xy));
hv = gaussian_correlate(xy, muh, muv, varh, varv, rhohv);
hw = zeros(size(hv));
hw(1,:) = hv(1,:);
hw(2,:) = e.^hv(2,:);
bmi = hw(2,:)./(hw(1,:).^2);
hist(bmi);
```

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Problems Requiring Cumulative Knowledge

5.129

a)

$$\int_0^{\pi/2}\int_0^{\pi/2} c\sin(x+y)dxdy = 1$$
$$c\int_0^{\pi/2}[-\cos(x+y)]|_0^{\pi/2}dy = 1$$

$$\begin{aligned} 1 &= c\int_0^{\pi/2}(\cos y - \cos(\frac{\pi}{2}+y))dy = c\int_0^{\pi/2}(\cos y + \sin y)dy \\ &= 2c\sin y|_0^{\pi/2} \\ &= 2c \\ c &= \frac{1}{2} \end{aligned}$$

b)

$$\begin{aligned} F_{X,Y}(x,y) &= \int_0^y\int_0^x \frac{1}{2}\sin(u+v)dudv \\ &= \int_0^y \left[-\frac{1}{2}\cos(u+v)\right]\Big|_0^x dv \\ &= \frac{1}{2}\int_0^y(\cos v - \cos(x+v))dv \\ &= \frac{1}{2}(\sin v - \sin(x+v))|_0^y \\ &= \frac{1}{2}(\sin y - \sin(x+y) + \sin x) \end{aligned}$$

c)

$$\begin{aligned} f_X(x) &= \int_0^{\pi/2}\frac{1}{2}\sin(x+y')dy' \\ &= \frac{1}{2}(-\cos(x+y'))|_0^{\pi/2} \\ &= \frac{1}{2}(\cos x + \sin x) \end{aligned}$$

$$f_Y(y) = \frac{1}{2}(\cos y + \sin y)$$

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d)
$$\begin{aligned}
E[X] &= \int_0^{\pi/2} x\frac{1}{2}(\cos x + \sin x)dx \\
&= \frac{1}{2}\int_0^{\pi/2} x d\sin x - \frac{1}{2}\int_0^{\pi/2} x d\cos x \\
&= \frac{1}{2}\left[x\sin x|_0^{\pi/2} - \int_0^{\pi/2}\sin x dx\right] - \frac{1}{2}\left[x\cos x|_0^{\pi/2} - \int_0^{\pi/2}\cos x dx\right] \\
&= \frac{1}{2}\left[\frac{\pi}{2} - 1\right] + \frac{1}{2} \\
&= \pi/4 \\
E[Y] &= \pi/4 \\
E[X^2] &= \int_0^{\pi/2} x^2\frac{1}{2}(\cos x + \sin x)dx \\
&= \frac{1}{2}\int_0^{\pi/2} x^2 d\sin x - \frac{1}{2}\int_0^{\pi/2} x^2 d\cos x \\
&= \frac{1}{2}x^2\sin x|_0^{\pi/2} - \frac{1}{2}\int_0^{\pi/2}\sin x\cdot 2x dx \\
&\quad -\frac{1}{2}x^2\cos x|_0^{\pi/2} + \frac{1}{2}\int_0^{\pi/2}\cos x\cdot 2x dx \\
&= \frac{1}{2}\left(\frac{\pi}{2}\right)^2 - 2\cdot\frac{1}{2} + 2\cdot\frac{1}{2}\left(\frac{\pi}{2} - 1\right) \\
&= \pi^2/8 + \pi/2 \\
VAR[X] &= E[X^2] - E^2[X] \\
&= \pi^2/16 + \pi/2 \\
VAR[Y] &= \pi^2/16 + \pi/2 \\
E[XY] &= \int\int xy\frac{1}{2}\sin(x+y)dxdy \\
&= \frac{1}{2}\int_0^{\pi/2} y dy \int_0^{\pi/2} -x d\cos(x+y) \\
&= -\frac{1}{2}\int_0^{\pi/2} y[x\cos(x+y)|_0^{\pi/2} - \int_0^{\pi/2}\cos(x+y)dx]dy \\
&= -\frac{1}{2}\int_0^{\pi/2} y\cdot\left[-\frac{pi}{2}\sin y - \sin(x+y)|_0^{\pi/2}\right]dy \\
&= \frac{\pi}{2}\int_0^{\pi/2}\frac{1}{2}y\sin y dy + \frac{1}{2}\int_0^{\pi/2} y(\cos y - \sin y)dy \\
&= \left(\frac{\pi}{2} - 1\right)\in_0^{\pi/2}\frac{1}{2}y\sin y dy + \int_0^{\pi/2}\frac{1}{2}y\cos y dy \\
&= \left(\frac{\pi}{2} - 1\right)\frac{1}{2} + \frac{1}{2}\left(\frac{\pi}{2} - 1\right) \\
&= \pi/2 - 1 \\
COV[X,Y] &= E[XY] - E[X]E[Y] \\
&= \pi/2 - 1 - (\pi/4)^2
\end{aligned}$$

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5.130

a) The number of items between consecutive inspections is a geometric random variable with proof.

$$P[M = m] = p(1-p)^{k-1} \qquad k = 1, 2, ...$$

b) The time between inspections is the sum of the M interarrival times:

$$T = \sum_{i=1}^{M} X_i$$

where the X_i are iid exponential random variables with mean 1.

$$f_T(t) = \sum_{j=1}^{\infty} f_T(t|M = j)P[M = j]$$

The sume of j independent exponential random variables is Erlang:

$$f_T(t|M = j) = \frac{\lambda e^{-\lambda x}(\lambda x)^{j-1}}{(j-1)!}$$

Therefore

$$\begin{aligned} f_T(t) &= \sum_{j=1}^{\infty} \frac{\lambda e^{-\lambda x}(\lambda x)^{j-1}}{(j-1)!} p(1-p)^{j-1} \\ &= \lambda p e^{-\lambda x} \sum_{j=1}^{\infty} \frac{(\lambda x(1-p))^{j-1}}{(j-1)!} \\ &= \lambda p e^{-\lambda x} e^{\lambda x(1-p)} \\ &= \lambda p e^{-\lambda p x} \end{aligned}$$

$\therefore T$ is an exponential random variable.

c) Choose p so that

$$\begin{aligned} 0.90 = P[T > t] &= e^{-pt} \\ \Rightarrow \quad p &= \frac{1}{t} \ln \frac{1}{0.90} \end{aligned}$$

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5.131 a)
$$\begin{aligned} f_{X,R}(x,r) &= f_X(x|r)f_R(r) \\ &= re^{-rx}\frac{\lambda(\lambda r)^{\alpha-1}e^{-\lambda r}}{\Gamma(\alpha)} \end{aligned}$$

b)
$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,R}(x,r)dr \\ &= \int_0^{\infty} \frac{(\lambda r)^{\alpha}e^{-(\lambda+X)r}}{\Gamma(\alpha)}dr \\ &= \int_0^{\infty} \frac{\lambda^{\alpha}}{(\lambda+x)^{\alpha}}\frac{[(\lambda+X)r]^{\alpha}e^{-(\lambda+X)r}}{\Gamma(\alpha)}dr \\ &= \frac{\lambda^{\alpha}}{(\lambda+X)^{\alpha}\Gamma(\alpha)}\cdot\frac{\Gamma(\alpha+1)}{\lambda+X} \\ &= \frac{\alpha\lambda^{\alpha}}{(\lambda+x)^{\alpha+1}} \qquad x>0 \end{aligned}$$

c)
$$\begin{aligned} E[X] &= \int_0^{\infty} xf_X(x)dx \\ &= \int_0^{\infty} \frac{\alpha\lambda^{\alpha}x}{(\lambda+x)^{\alpha+1}}dx \\ &= \frac{\lambda^{\alpha}}{-\alpha+1}(\lambda+x)^{-\alpha+1}\Big|_0^{\infty} \\ &= \frac{\lambda}{\alpha-1} \qquad (\alpha>1) \\ E[X^2] &= \int_0^{\infty} x^2f_X(x)dx \\ &= \int_0^{\infty} \frac{\alpha\lambda^{\alpha}x^2}{(\lambda+x)^{\alpha+1}}dx \\ &= -\lambda^{\alpha}\int_0^{\infty} x^2d(\lambda+x)^{-\alpha} \\ &= -\lambda^{\alpha}x^2(\lambda+x)^{-\alpha}\Big|_0^{\infty} + \lambda^{\alpha}\int_0^{\infty}(\lambda+x)^{-\alpha}2xdx \\ &= \frac{2\lambda^{\alpha}}{-\alpha+1}\int_0^{\infty} xd(\lambda+x)^{-\alpha+1} \\ &= \frac{2\lambda^{\alpha}}{-\alpha+1}x(\lambda+x)^{-\alpha+1}\Big|_0^{\infty} + \frac{2\lambda^{\alpha}}{\alpha-1}\int_0^{\infty}(\lambda+x)^{-\alpha+1}dx \\ &= \frac{2\lambda^{\alpha}}{-\alpha-1}\cdot\frac{1}{-\alpha+2}(\lambda+x)^{-\alpha+2}\Big|_0^{\infty} \\ &= \frac{2\lambda^2}{(\alpha-1)(\alpha-2)} \\ VAR[X] &= E[X^2]-E^2[X] = \frac{2\lambda^2}{(\alpha-1)(\alpha-2)} - \frac{\lambda^2}{(\alpha-1)^2} \end{aligned}$$

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5.132 $R^2 = X^2 + Y^2$

a) When signal 0 is present

$$\frac{R^2}{\sigma_0^2} = \left(\frac{X}{\sigma_0}\right)^2 + \left(\frac{Y}{\sigma_0}\right)^2$$

$\frac{X}{\sigma_0}$, $\frac{Y}{\sigma_0}$ are independent, zero-mean, unit-variance RVs. R^2/σ_0^2 is a chi-square RV with 2 degrees of freedom. The pdf of R^2/σ_0^2 is

$$\frac{u^0 e^{-u/2}}{2^1 r(1)} = \frac{e^{-u/2}}{2}$$

The pdf of R_1^2, or $\sigma_0^2 \cdot \frac{R_1^2}{\sigma_0^2}$ is

$$\frac{1}{\sigma_0^2} \cdot \frac{e^{-u/2\sigma_0^2}}{2} = \frac{e^{-T^2/2\sigma_0^2}}{2\sigma_0^2}$$

Similarly, the pdf of R^2 when signal 1 is present

$$f_{R^2}(R^2|1) = \frac{e^{-R^2/2\sigma_1^2}}{2\sigma_0^2}$$

$$\begin{aligned} f_{R^2}(R^2) &= f_{R^2}(R^2|0)p(0) + f_{R^2}(R^2|1)p(1) \\ &= \frac{e^{-R^2/2\sigma_0^2}}{2\sigma_0^2}p + \frac{e^{-R^2/2\sigma_1^2}}{2\sigma_1^2}(1-p) \end{aligned}$$

b)
$$\begin{aligned} p_e &= p[R^2 > T|0]p(0) + p[R^2 < T|]p(1) \\ &= \int_T^\infty \frac{e^{-R^2/2\sigma_0^2}}{2\sigma_0^2} p dR^2 + \int_0^T \frac{e^{-R^2/2\sigma - 1^2}}{2\sigma_1^2}(1-p)dR^2 \end{aligned}$$

c) $\frac{dP_e}{dT} = 0$

$$-p \cdot \frac{e^{-T/2\sigma_0^2}}{2\sigma_0^2} + \frac{e^{-T/2\sigma_1^2}}{2\sigma_1^2}(1-p) = 0$$

$$\ln \frac{p}{2\sigma_0^2} - \frac{T}{2\sigma_0^2} = \ln \frac{1-p}{2\sigma_1^2} - \frac{T}{2\sigma_1^2}$$

$$T = \frac{\ln \frac{p}{2\sigma_0^2} - \ln \frac{1-p}{2\sigma_1^2}}{\frac{1}{2\sigma_0^2} - \frac{1}{2\sigma_1^2}}$$

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5.133 a)

$$X_n = \frac{1}{2}(U_n + U_{n-1})$$
$$X_{n-1} = \frac{1}{2}(U_{n-1} + U_{n-2})$$

or

$$\begin{bmatrix} X_n \\ X_{n-1} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} U_n \\ U_{n-1} \\ U_{n-2} \end{bmatrix}$$

X_n and X_{n-1} are jointly Gaussian

$$\begin{aligned}
E[X_n] &= 0, \quad E[X_{n-1}] = 0 \\
VAR[X_n] &= \frac{1}{4}(VAR[U_n] + VAR[U_{n-1}]) = \frac{1}{2}, \quad VAR[X_{n-1}] = \frac{1}{2} \\
E[X_n X_{n-1}] &= \frac{1}{4}(E[U_n U_{n-1}] + E[U_{n-1}U_{n-1}] + E[U_n U_{n-2}] + E[U_{n-1}U_{n-2}]) \\
&= \frac{1}{4} \\
COV[X_n X_{n-1}] &= E[X_n X_{n-1}] - E[X_n]E[X_{n-1}] = \frac{1}{4} \\
\rho &= \frac{COV[X_n X_{n-1}]}{\sigma_n \sigma_{n-1}} = \frac{1}{2} \\
f(X_n, X_{n-1}) &= \frac{1}{2\pi \cdot \frac{1}{2}\sqrt{1-\frac{1}{4}}} \exp\left\{ \frac{-1}{2(1-\frac{1}{4})} \left[\frac{x_{n-1}^2}{1/2} - \frac{x_n x_{n-1}}{1/2} + \frac{x_{n-1}^2}{1/2} \right] \right\} \\
&= \frac{1}{\pi\sqrt{3/4}} \exp\left\{ -\frac{4}{3}[x^2 - x_n x_{n-1} + x_{n-1}^2] \right\} \\
COV[X_n X_{n+m}] &= 0 \quad \text{for } m > 1 \\
f(X_n X_{n+m}) &= f(X_n) f(X_{n+m}) \\
&= \frac{1}{\pi} \exp[-x_n^2 - x_{n+m}^2], \quad m > 1
\end{aligned}$$

b) In this case, ρ is negative

$$\begin{aligned}
f(y_n, y_{n-1}) &= \frac{1}{\pi\sqrt{3/4}} \exp\left\{ -\frac{4}{3}[y_n^2 + y_n y_{n-1} + y_{n-1}^2] \right\} \\
f(y_n, y_{n+m}) &= \frac{1}{\pi} \exp[-y_n^2 - y_{n+m}^2] < \quad , m > 1
\end{aligned}$$

c) $m = n$

$$\begin{aligned}
E[X_n Y_m] &= \frac{1}{4}E[U_n^2 - U_{n-1}^2] = 0, \quad \rho = 0 \\
f(x_n, y_m) &= \frac{1}{\pi} \exp[-x_n^2 - y_n^2], \quad m = n.
\end{aligned}$$

$m = n + 1$

$$E[X_n Y_m] = \frac{1}{4}E[(U_n + U_{n-1})(U_{n+1} - U_n)] = -\frac{1}{4} = COV[X_n Y_m]$$

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